



Industrial Robot

Communication Guide



Industrial
Automation



New Energy
Vehicle



Intelligent
Elevator



Intelligent
Robot



Digital
Energy



Rail
Transit



Data code 19012876A00

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The product/system described in this documentation may be operated only by personnel qualified for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel can identify the risks of the product/system and prevent possible dangers.

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Proper transportation, storage, assembly, installation, commissioning, operation, and maintenance are required to ensure the safe operation of the product without any problems. The required ambient conditions must be met. All operations must follow the guidelines provided in this documentation.

Preface

Introduction

This guide provides a comprehensive introduction to multi-protocol communication solutions for Inovance industrial robot, covering mainstream industrial communication standards such as Modbus, EtherNet/IP, PROFINET, EtherCAT, and MC protocol. Specific content includes: achieving basic communication between devices via Modbus protocol; enabling data exchange with third-party devices via EtherNet/IP protocol; realizing industrial network integration with Siemens PLCs using PROFINET bus technology; establishing high-speed real-time communication with Omron PLCs via EtherCAT protocol; and utilizing the MC protocol dedicated to Mitsubishi PLCs for efficient control command transmission. This guide offers complete communication technology guidance for industrial automation system integration.

More Data

Data Name	Data Code	Description
IR-TP200 Series Industrial Robot Teach Pendant User Guide	19012798	This guide introduces software-related operations of the robot teach pendant. Contact Inovance technical support to obtain this guide.
Industrial Robot InoRobotLab User Guide	19012861	This guide introduces software-related operations of the robot InoRobotLab software. Contact Inovance technical support to obtain this guide.
Industrial Robot Instruction Guide	19012825	This guide introduces the function descriptions, instruction formats and parameter explanations, usage examples, and application cases for project instructions of Inovance Technology industrial robots.
Industrial Robot Remote I/O Control Function User Guide	19012874	This guide introduces the interface configuration, bus read/write instruction formats, and bus invocation precautions for remote I/O control of Inovance Technology industrial robots.

Data Name	Data Code	Description
Industrial Robot Remote Ethernet Control Function User Guide	19012860	This guide introduces the interface configuration, API call format, API function call examples, common issues and troubleshooting for remote Ethernet control of Inovance Technology industrial robots.
Industrial Robot Controller Software Update User Guide	19012875	This guide introduces the software update procedures and component update procedures for Inovance Technology industrial robots.
Industrial Robot Communication Guide (This guide)	19012876	This guide introduces Inovance industrial robot multi-protocol communication solutions, covering mainstream industrial communication standards such as Modbus, EtherNet/IP, PROFINET, EtherCAT, and MC protocol, providing comprehensive communication technology guidance for industrial automation system integration.

Revision History

Revision Date	Data Version	Revision
2026-02	A00	First issue

Access to the Guide

This guide is not delivered with the product. You can obtain the PDF version in the following ways:

- Do keyword search under Service and Support at <https://www.inovance.com/global>.
- Scan the QR code on the product with your smart phone.
- Scan the QR code below to install My Inovance app, where you can search for and download user guides.



Basic Safety Instructions

Safety Precautions

Safety Disclaimer

- This chapter presents essential safety instructions for a proper use of the equipment. Before operating the equipment, read through the guide and comprehend all the safety instructions. Failure to observe the safety instructions may result in death, severe personal injuries, or equipment damage.
- "DANGER", "WARNING", and "CAUTION" items in this guide do not indicate all safety precautions that need to be followed; instead, they just supplement the safety precautions.
- Use this product according to designated environment requirements. Damage caused by improper use is not covered by warranty.
- Inovance shall take no responsibility for any personal injuries or property damage caused by improper usage.

Safety Levels and Definitions



Indicates that failure to comply with the notice will result in severe personal injuries or even death.



Indicates that failure to comply with the notice may result in severe personal injuries or even death.



Indicates that failure to comply with the notice may result in minor or moderate personal injuries or equipment damage.

Safety Instructions

- Drawings in the guide are sometimes shown without covers or protective guards. Remember to install the covers or protective guards as specified first, and then perform operations in accordance with the instructions.
- The drawings in the guide are shown for illustration only and may be different from the product you purchased.
- Operators must take mechanical precautions to protect personal safety and wear protective equipment, such as anti-smashing shoes, safety clothing, safety glasses, protective gloves, and protective sleeves.

Unpacking



- Do not install the equipment if you find damage, rust, or signs of use on the equipment or accessories upon unpacking.
- Do not install the equipment if you find water seepage or missing or damaged components upon unpacking.
- Do not install the equipment if you find the packing list does not conform to the equipment you received.



- Before unpacking, check whether the outer package is intact, damaged, wet, damped, or deformed.
- Unpack the package in sequence. Do not hit the package violently.
- Check whether there is damage, rust, or injuries on the surface of the equipment and equipment accessories before unpacking.
- Check whether the package contents are consistent with the packing list before unpacking.

Storage and Transportation



- Large-scale or heavy equipment must be transported by qualified professionals using specialized hoisting equipment. Failure to comply may result in personal injuries or equipment damage.
- Before hoisting the equipment, ensure that components such as the front cover and terminal blocks are secured firmly with screws. Loosely-connected components may fall off and result in personal injuries or equipment damage.
- Never stand or stay below the equipment when the equipment is being hoisted by the hoisting equipment.
- When hoisting the equipment with a steel rope, ensure the equipment is hoisted at a constant speed without suffering from vibration or shock. Do not turn the equipment over or let the equipment stay hanging in the air. Failure to comply may result in personal injuries or equipment damage.



- Handle the product with care and mind your steps to prevent personal injuries or product damage.
- When carrying the equipment with bare hands, hold the equipment casing firmly with care to prevent parts from falling. Failure to comply may result in personal injuries.
- Store and transport the equipment based on the storage and transportation requirements. Failure to comply will result in equipment damage.
- Avoid storing or transporting the equipment in environments with water splash, rain, direct sunlight, strong electric field, strong magnetic field, and strong vibration.
- Avoid storing the product for more than 3 months. When the product needs to be stored for an extended period, take more strict protection and necessary inspection.
- Pack the equipment strictly before transportation. Use a sealed box for long-distance transportation.
- Never transport the equipment with other equipment or materials that may harm or have negative impacts on this equipment.

Installation



- The equipment can only be operated by professionals with electrical knowledge. Non-professionals are not allowed to operate on the equipment.



- Read through the guide and safety precautions before installation.
- Do not install the equipment in places with strong electric or magnetic fields.
- Before installation, check that the mechanical strength of the installation site can bear the weight of the equipment. Failure to comply will result in mechanical hazards.
- Do not wear loose clothes or accessories during installation. Failure to comply may result in an electric shock.
- When installing the equipment in a closed environment (such as a cabinet or casing), use a cooling device (such as a fan or air conditioner) to cool the environment down to the required temperature. Failure to comply may result in equipment over-temperature or a fire.
- Do not retrofit the equipment.
- Do not loosen the fixing bolts of the product parts and components, or the bolts marked in red.
- When the equipment is installed in a cabinet or final assembly, a fireproof enclosure providing both electrical and mechanical protections must be provided. The IP rating must meet IEC standards and local laws and regulations.
- Before installing devices with strong electromagnetic interference, such as a transformer, install a shielding device for the equipment to prevent malfunction.
- Install the product on an incombustible object such as metal and do not touch or attach the product to combustibles. Failure to comply can result in fire accident.



- Cover the top of the equipment with a piece of cloth or paper during installation. This is to prevent unwanted objects such as metal shavings, oil, and water from falling into the equipment and causing faults. After installation, remove the cloth or paper on top of the equipment to prevent over-temperature caused by poor ventilation due to blocked ventilation holes.
- Resonance may occur when the equipment operating at a constant speed switches to variable speed operations. In this case, install the vibration-proof rubber under the motor frame or use the vibration suppression function to reduce resonance.

Wiring



- Equipment installation, wiring, maintenance, inspection, or component replacement must be performed only by professionals.
- Before wiring, disconnect all the power supplies of the equipment. Wait for at least the time designated on the equipment warning label before further operations because residual voltage still exists after power-off. Measure the direct voltage of the main circuit and ensure that the voltage is within the safety range. Failure to comply can result in an electric shock.
- Do not perform wiring, remove the equipment cover, or touch the circuit board with power ON. Failure to comply can result in an electric shock.
- Check that the equipment is grounded properly. Failure to comply can result in an electric shock.



- Do not connect the input power supply to the output side of the equipment. Failure to comply can result in equipment damage or even a fire.
- When connecting a drive to the motor, check that the phase sequences of the drive and motor terminals are consistent to prevent reverse motor rotation.
- Cables used for wiring must meet cross sectional area and shielding requirements. The shield of the cable must be reliably grounded at one end.
- Tighten the terminal screws with the tightening torque specified in the user guide. Improper tightening torque may result in overheat, damage to the connecting parts, or a fire.
- After wiring is done, check that all cables are connected properly and no screws, washers or exposed cables are left inside the equipment. Failure to comply may result in electric shock or equipment damage.



CAUTION

- Follow the electrostatic discharge (ESD) procedure and wear an anti-static wrist strap to perform wiring. Failure to comply may result in damage to the equipment or to the internal circuit.
- Use shielded twisted pairs for the control circuit. Connect the shield to the grounding terminal of the equipment for grounding purpose. Failure to comply can result in equipment malfunction.

Power-On



DANGER

- Before power-on, check that the equipment is installed properly with reliable wiring and the motor can be restarted.
- Check that the power supply meets equipment requirements before power-on to prevent equipment damage or a fire.
- Do not open the cabinet door or protective cover of the product, touch any wiring terminal of the product, or remove any part of the product with power on. Failure to comply can result in an electric shock.



WARNING

- After the completion of wiring and parameter setting, perform a robot test run to confirm that the robot can operate safely. Failure to comply may result in personal injuries or equipment damage.
- Before power-on, check that the rated voltage of the equipment is consistent with that of the power supply. Failure to comply may result in fire accident.
- Before power-on, check that no one is near the equipment, motor, or machine. Failure to comply may result in death or personal injuries.

Operation



DANGER

- The equipment must be operated only by professionals. Failure to comply can result in death or personal injuries.
- Do not touch any connecting terminals or disassemble any unit or component of the equipment during operation. Failure to comply can result in an electric shock.



WARNING

- Do not touch the equipment enclosure, fan, or resistor with bare hands. Failure to comply may result in burns.
- Prevent metal or other objects from falling into the equipment during operation. Failure to comply may result in a fire or equipment damage.

Maintenance



- Equipment installation, wiring, maintenance, inspection, or component replacement must be performed only by professionals.
- Do not maintain the equipment with power ON. Failure to comply can result in an electric shock.
- Before maintenance, cut off all the power supplies of the equipment and wait for at least the time designated on the equipment warning label.
- When a permanent magnet motor is used, do not touch the motor terminals immediately after power-off because the motor terminals can generate induced voltage during rotation even after the equipment is powered off. Failure to comply can result in an electric shock.



- Perform routine and regular inspection and maintenance on the equipment in accordance with maintenance requirements and keep a record.

Repair



- Equipment installation, wiring, maintenance, inspection, or component replacement must be performed only by professionals.
- Do not maintain the equipment with power ON. Failure to comply can result in an electric shock.
- Before inspection and repair, cut off all the power supplies of the equipment and wait for at least the time designated on the equipment warning label.



- Submit a request for repair services in accordance with the product warranty agreement.
- When the fuse is blown or the circuit breaker or earth leakage circuit breaker (ELCB) trips, wait for at least the time designated on the equipment warning label before power-on or further operations. Failure to comply may result in death, personal injuries or equipment damage.
- Troubleshooting and repair must be performed by professionals in accordance with the repair instructions, with repair records kept properly.
- Replace quick-wear parts of the equipment in accordance with the replacement instructions.
- Do not operate on damaged equipment. Failure to comply may result in death, personal injuries, or severe equipment damage.
- After the equipment is replaced, check the wiring and set parameters again.

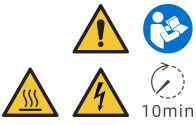
Disposal



- Dispose of retired equipment in accordance with local regulations and standards. Failure to comply may result in death, personal injuries, or property damage.
- Recycle retired equipment in accordance with industry waste disposal standards to avoid environmental pollution.

Safety Label

For safe operation and maintenance, comply with the safety labels on the equipment. Do not damage or remove the safety labels. See the following table for descriptions of the safety labels.

Safety Label	Description
	• Read through the safety instructions before operating the equipment. Failure to comply may result in death, personal injuries, or equipment damage. • Do not touch the terminals or remove the cover with power ON or within 10 min after power-off. Failure to comply can result in an electric shock.

Industrial Information Security

The product provides an interface to connect to the network and transmits data through the network interface. To protect factories, systems, machines, and networks from cyber attacks and ensure safe operation, a proper industrial information security protection mechanism must be implemented.

The customer is responsible for providing and continuously ensuring a secure connection between the product and the customer network or any other network to prevent unauthorized access to its factories, systems, machines, and networks. The system or machine can be connected to the corporate network or the Internet only when secure connections are ensured and appropriate security measures are taken (for example, anti-virus software or firewall installed).

Inovance continuously develops and improves products and solutions to improve safety. It is strongly recommended that you update the product promptly and always use the latest version.



Malware (such as viruses, Trojans, and worms) can put the product in an unsafe running state, which may result in death, serious injuries, and property damage. Observe the following precautions:

- Always use the latest software version. If the product version is no longer supported or the latest version is not used, customers will be exposed to increased risk of cyber attacks.
- Take appropriate protection measures (including but not limited to installing anti-virus software, firewall, WAF, IPS/IDS, and situation awareness system, and applying authentication, data encryption) to prevent files in mobile storage devices from being damaged by malware, and to prevent products, networks, systems, and interfaces from unauthorized access, interference, intrusion, data leakage, or information breach.
- Check all safety-related interfaces and settings after commissioning.

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1 EtherNet/IP Bus Communication

1.1 Communication Introduction

The EtherNet/IP protocol is based on the TCP/IP protocol stack and uses both connection-oriented and connectionless communication methods. It defines two types of communication: Explicit messaging and Implicit messaging.

- Explicit messaging: Transmits data via TCP (Transmission Control Protocol) and is suitable for transferring data with high accuracy and low real-time requirements. Explicit messaging is further divided into UCMM (Unconnected Explicit Messages) and Class 3 (Connected Explicit Messages) based on the connection method.
- Implicit messaging: Transmits data via UDP (User Datagram Protocol) and is suitable for transmitting data with high real-time requirements. There is only one type of implicit messaging, Class 1 "Implicit Message (I/O Messages)".

EtherNet/IP is a high-speed, real-time, flexible, easy-to-integrate, scalable, and secure industrial automation network protocol, widely used in industrial automation control systems.

1.2 Communication Between Inovance PLC and Industrial Robot via EtherNet/IP

1.2.1 Overview

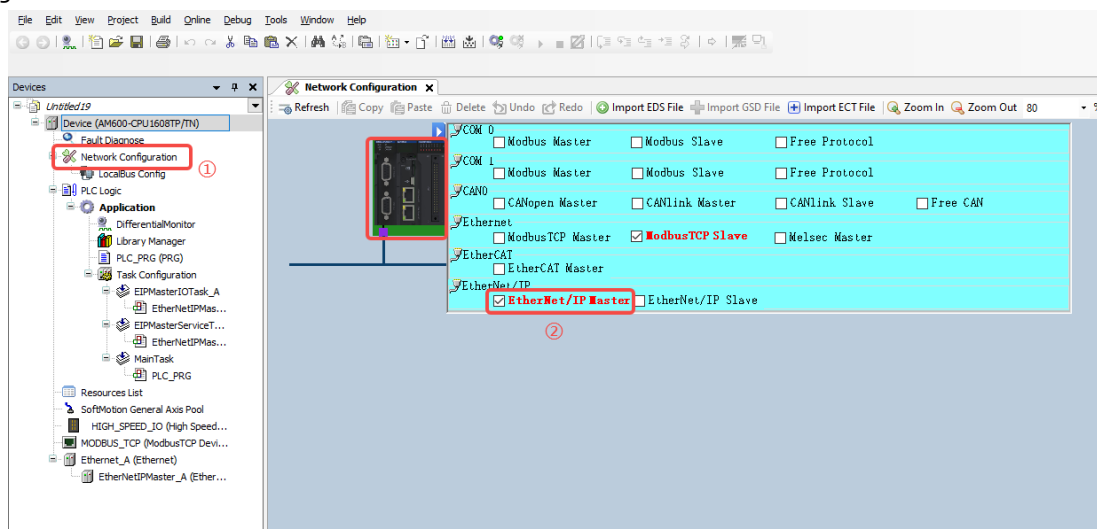
This section applies to the interaction configuration via EDS between Inovance PLCs (currently only for AM600 series) and Inovance robots, as well as the interaction configuration via EDS between robots.

1.2.2 Configuration via InoProShop

1.2.2.1 Configuring the EtherNet/IP Master Function of the PLC

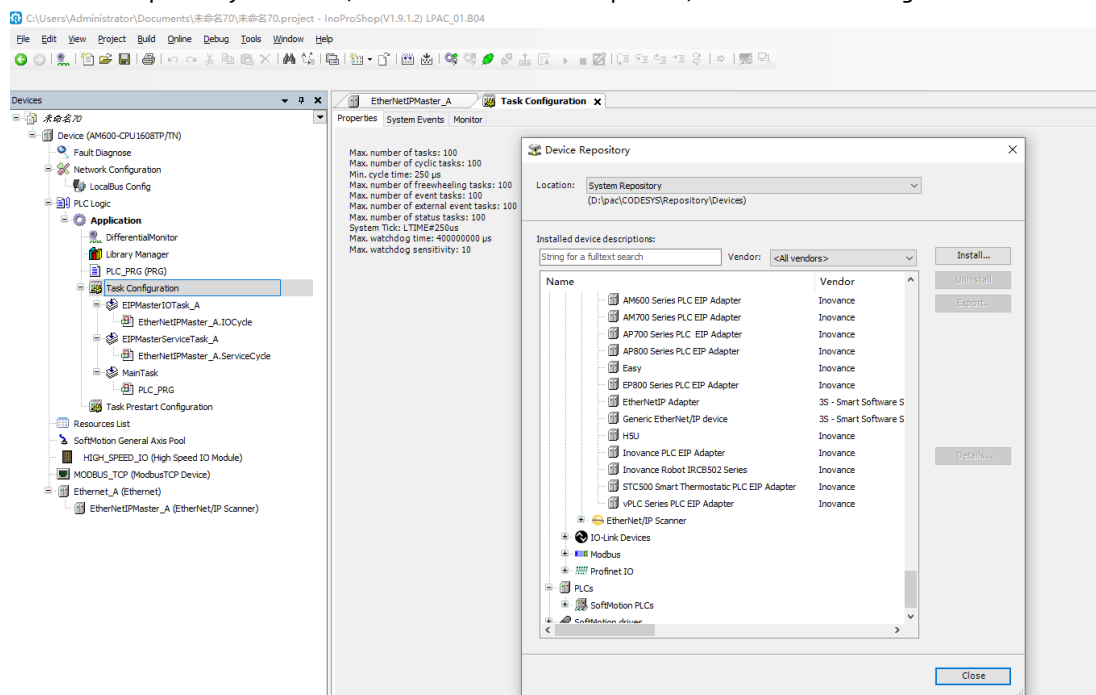
Here we take AM600 as a master with remote EtherNet/IP bus control as an example.

In the left "Devices" pane, double click "Network Configuration" under "Device (AM600-CPU1608TP/TN)", click the AM600 image, then check "EtherNet/IP Master" to enable the EtherNet/IP master function, as shown in the figure below.



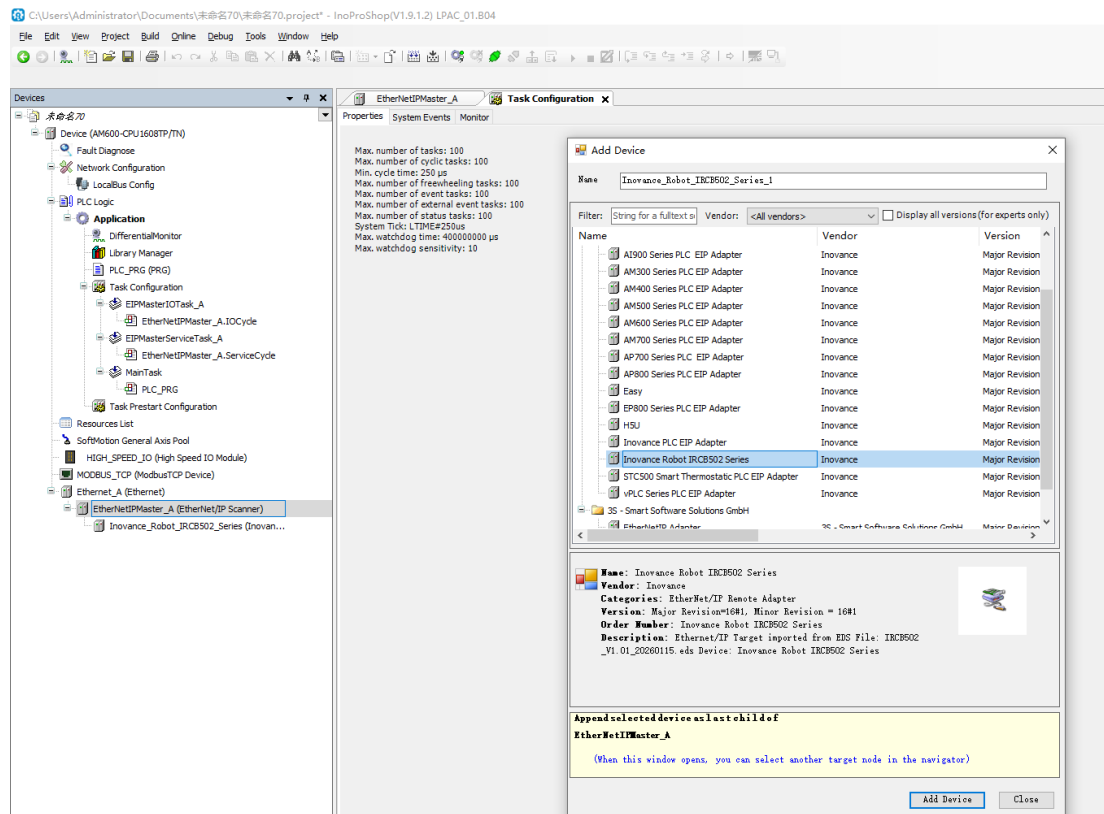
1.2.2.2 Importing the EDS file

1. In the menu bar of the main interface, click "Tools" > "Device Repository" to open the "Device Repository" window.
2. In the "Device Repository" window, select the file to be imported, as shown in the figure below.



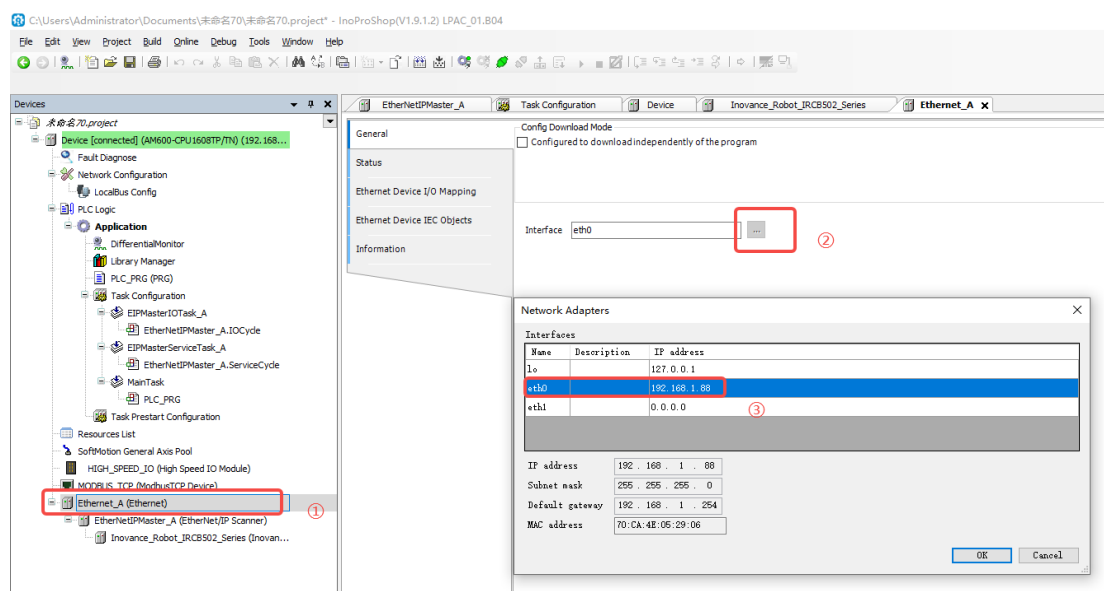
1.2.2.3 Adding a Slave

1. In the left "Devices" pane, right-click "Inovance_Robot IRCB50x_Series" and select "Add Device" to open the "Add Device" dialog box.
2. In the "Add Device" dialog box, select the device to be added, as shown in the figure below.



1.2.2.4 Setting the PLC Network Port

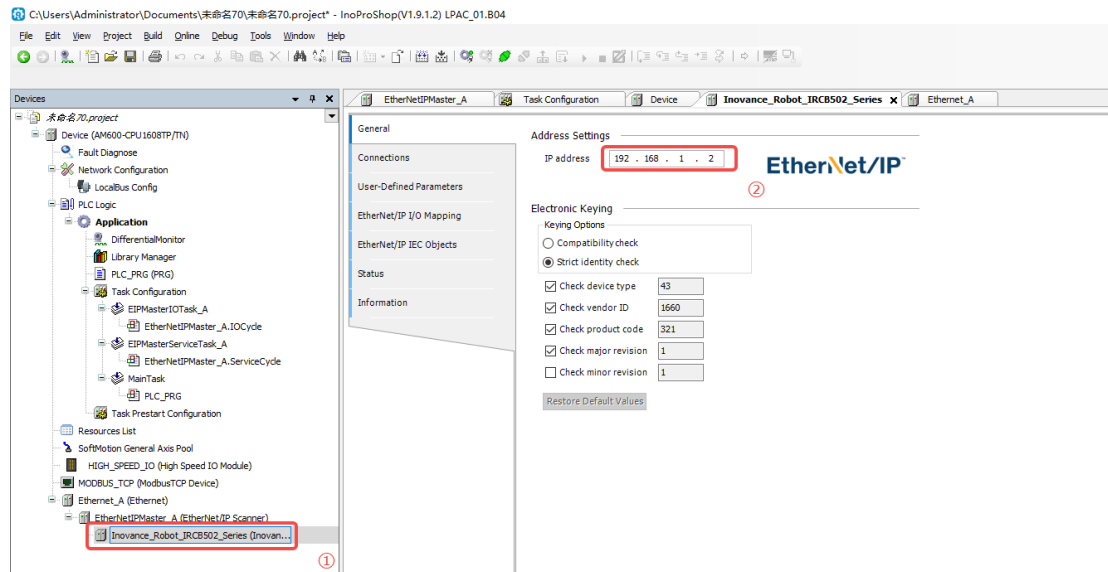
1. In the left "Devices" pane, double-click "Ethernet_A (Ethernet)" to open the "Ethernet_A" interface. In the "Interface" drop-down dialog, click and select "eth0".
2. Click "General", and configure the corresponding IP address in the "Network Adapters" pop-up window.
3. Click "OK" as shown in the figure below.



1.2.2.5 Configuring the Robot Slave

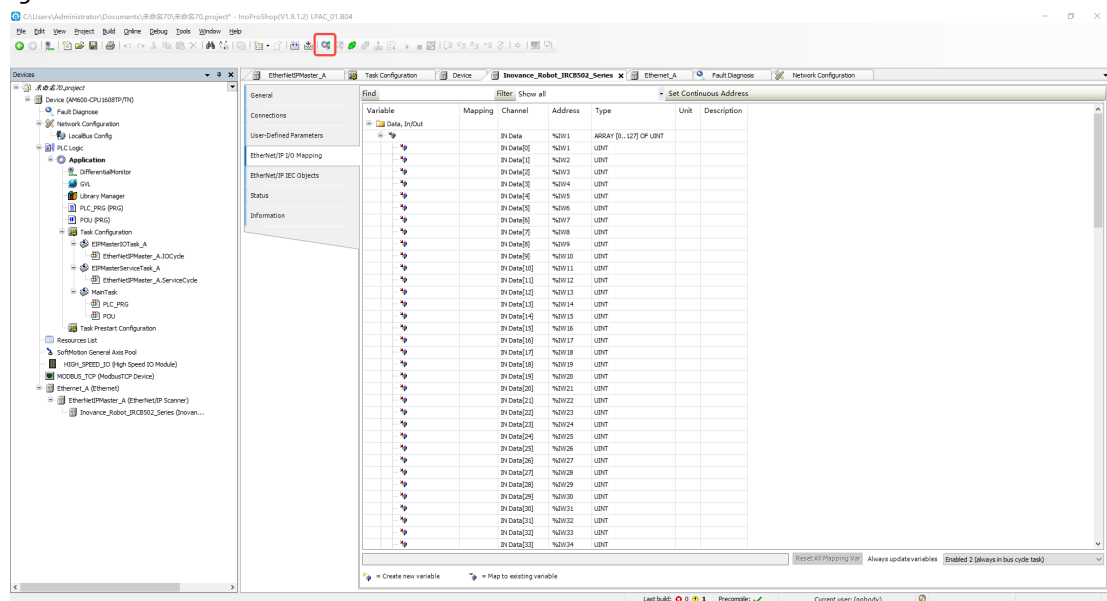
1. In the left "Devices" pane, double-click "Inovance_Robot_IRCB50x_Series" to open the "Inovance_Robot_IRCB50x_Series" interface.
2. On the "Inovance_Robot_IRCB50x_Series" interface, click "EtherNet/IP IEC Objects" to open the "Address Settings" interface.

- On the "Address Settings" interface, set the corresponding IP address to complete the robot slave configuration.

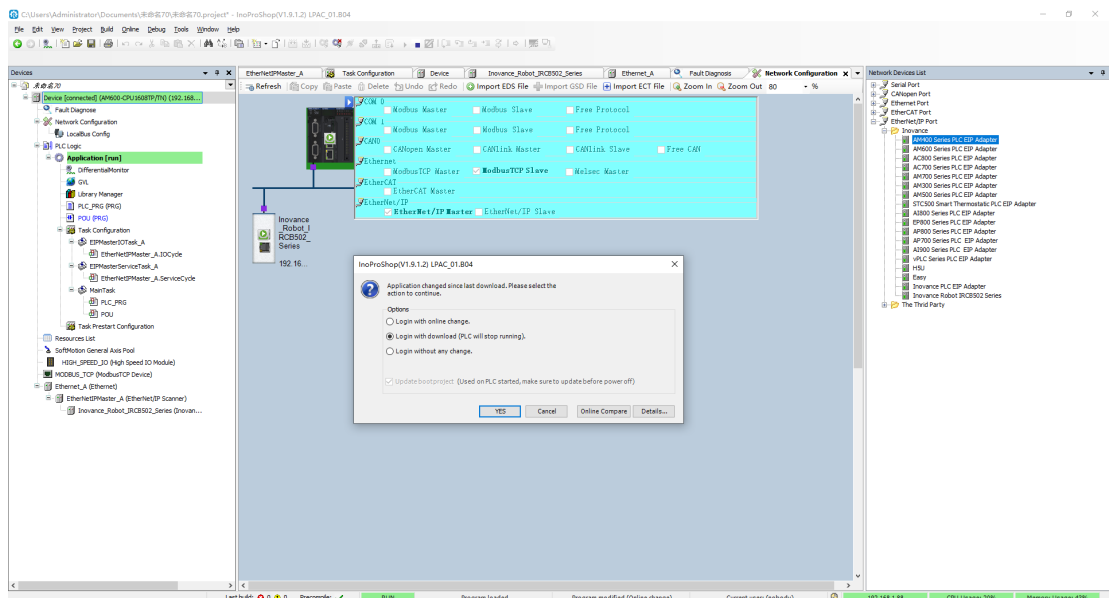


1.2.2.6 Checking the Communication Status

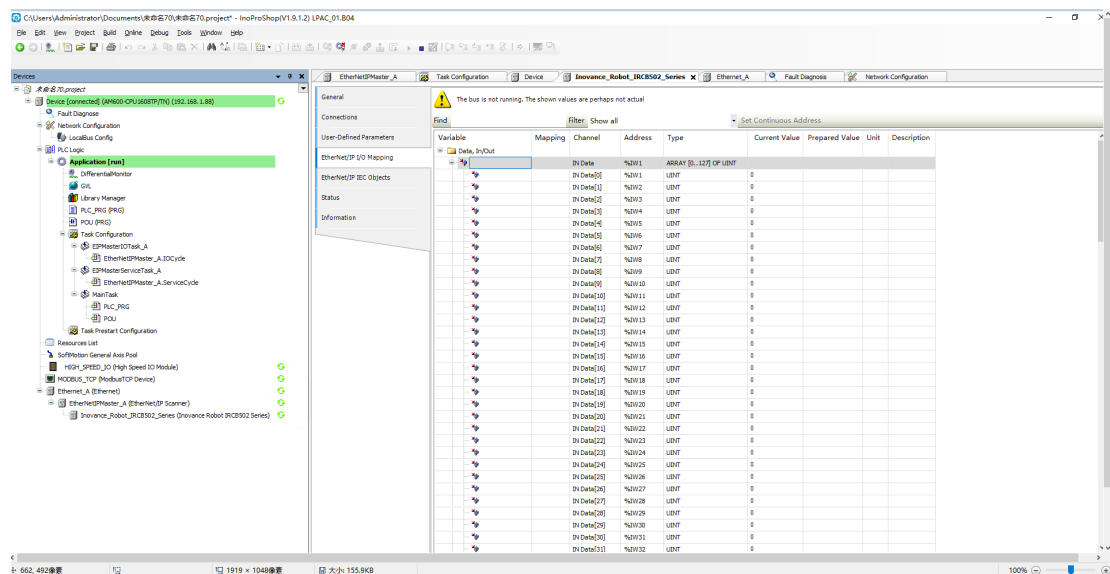
- In the menu navigation bar of InoProShop main interface, click "Login" to jump to the "Hardware Configuration" interface.



- On the "Hardware Configuration" interface, click "YES" to download.



3. After clicking the "Start" button, if both "Device [Connected]" and "Application [run]" in the left device tree are green, the EtherNet/IP communication is successful, as shown in the figure below.



1.2.3 Notes



- The program on the PLC side needs to be transmitted to the controller to complete the controller program update.
- When not switched to remote I/O permission, the robot can transmit data to the PLC, such as position information. The PLC can transmit data to the robot only when the robot has been switched to I/O control permission.

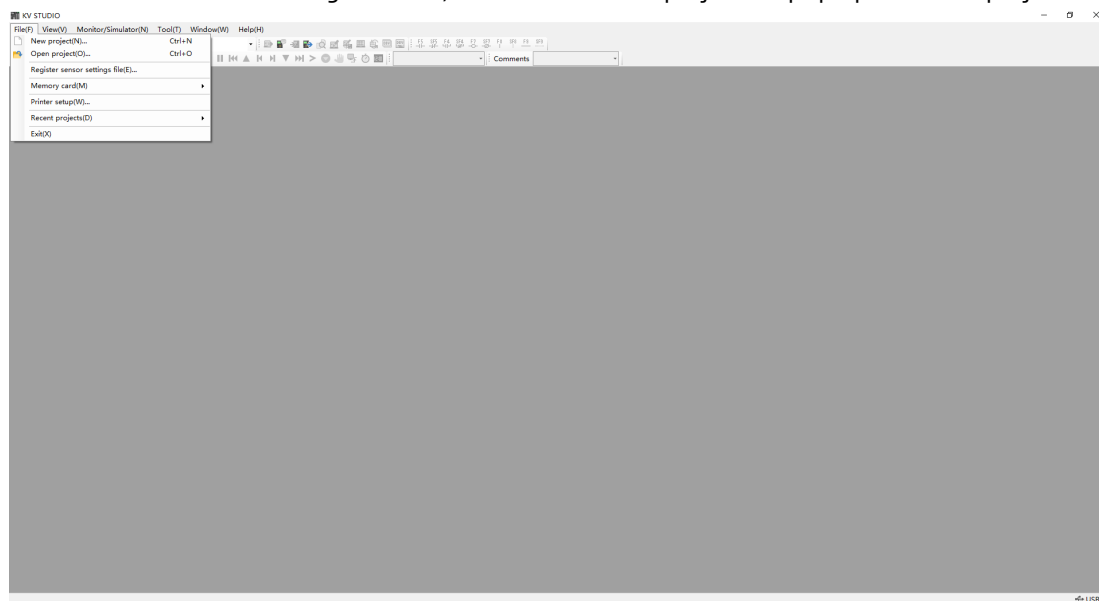
1.3 Communication Between Keyence KV Series and Industrial Robot via EtherNet/IP

1.3.1 Overview

This section provides the configuration method based on KEYENCE KV STUDIO to realize the configuration for KV-8000 and KV-7500. Using KV-7500 as an example, it describes how to realize the EtherCAT communication interaction configuration between the PLCs and Inovance robots via EDS files.

1.3.2 Configuration via KV STUDIO

1. In the KV STUDIO menu navigation bar, click "File" > "New project" to pop up the "New project" dialog box.



2. In the "New Project" dialog box, click the "PLC model" drop-down box to select the PLC model as "KV-7500", and click "OK" to complete the project creation.

New project

Project name(N) PLC model(K)

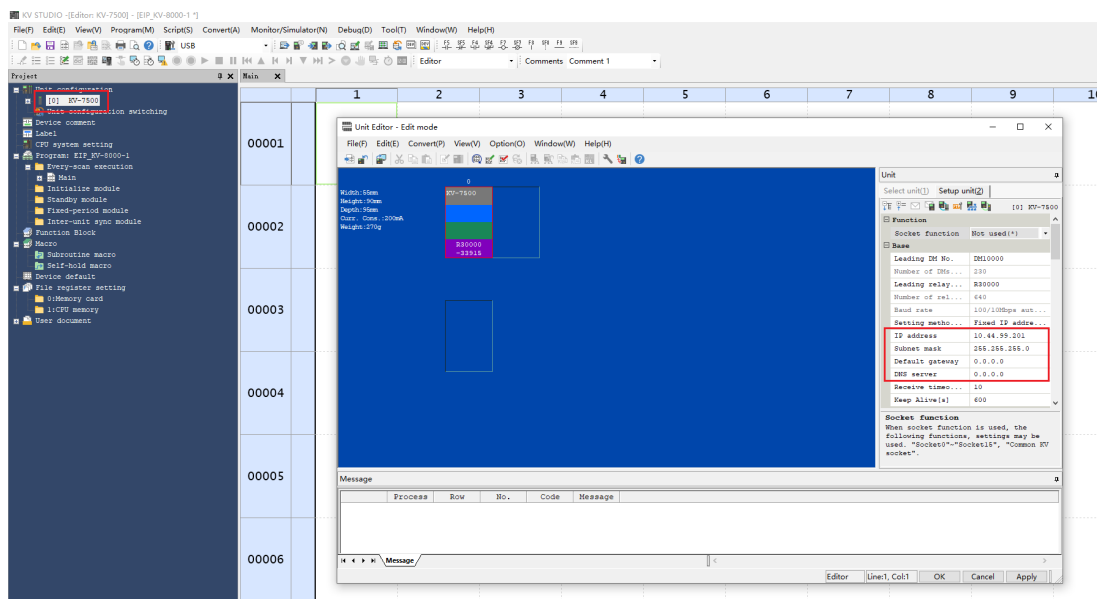
Position(P) Refer(S)...

Comment(C)

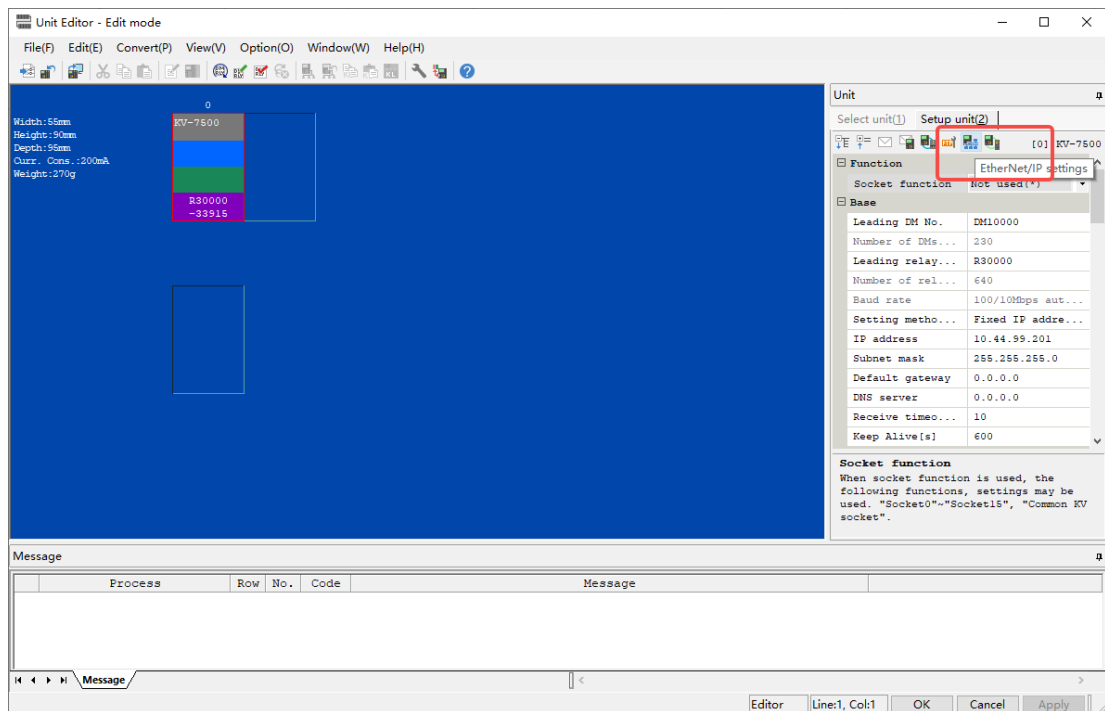
AWV display comments(VV)

☒ Register special device cmnts(M)

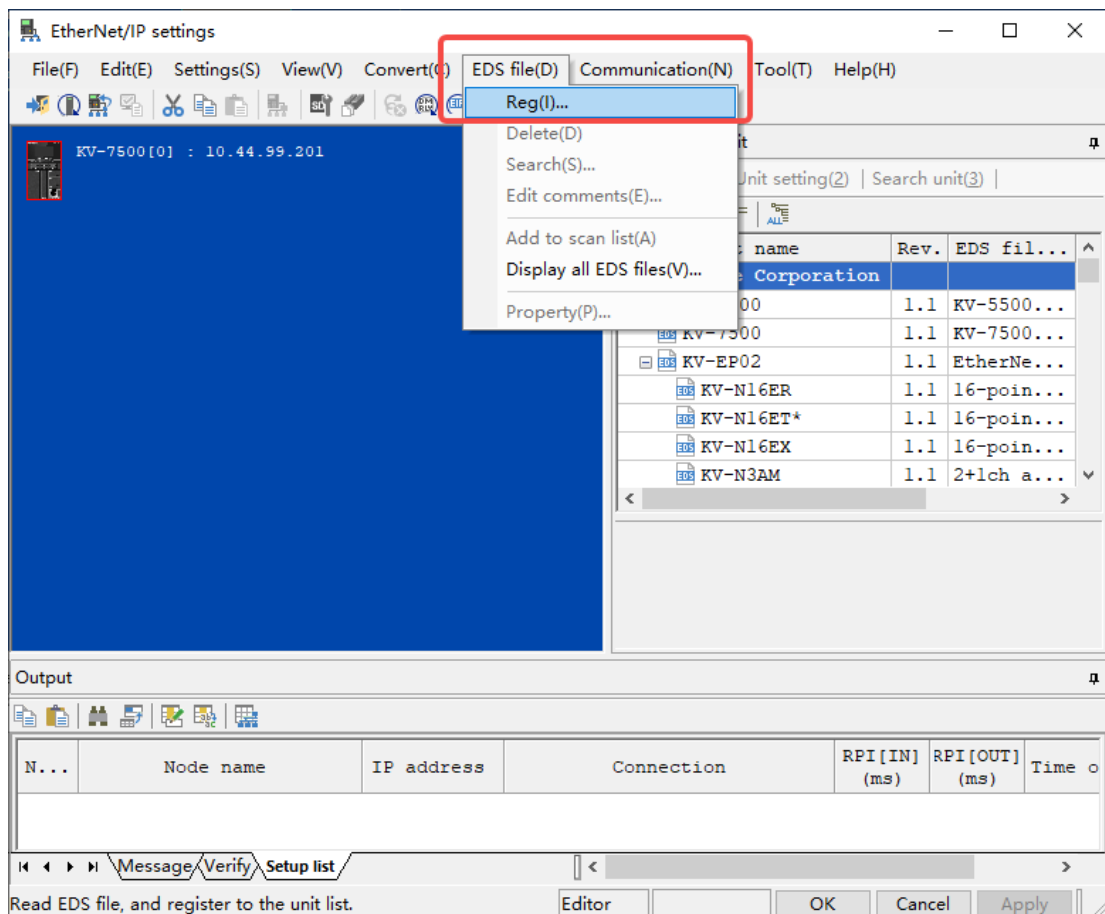
3. On the left navigation tree of the KV STUDIO, double-click "KV-7500" under "Unit configuration". In the pop-up "Unit Editor - Edit mode" dialog box, configure the IP address of the PLC. Ensure the IP addresses of the PLC and the Inovance robot are in the same subnet.



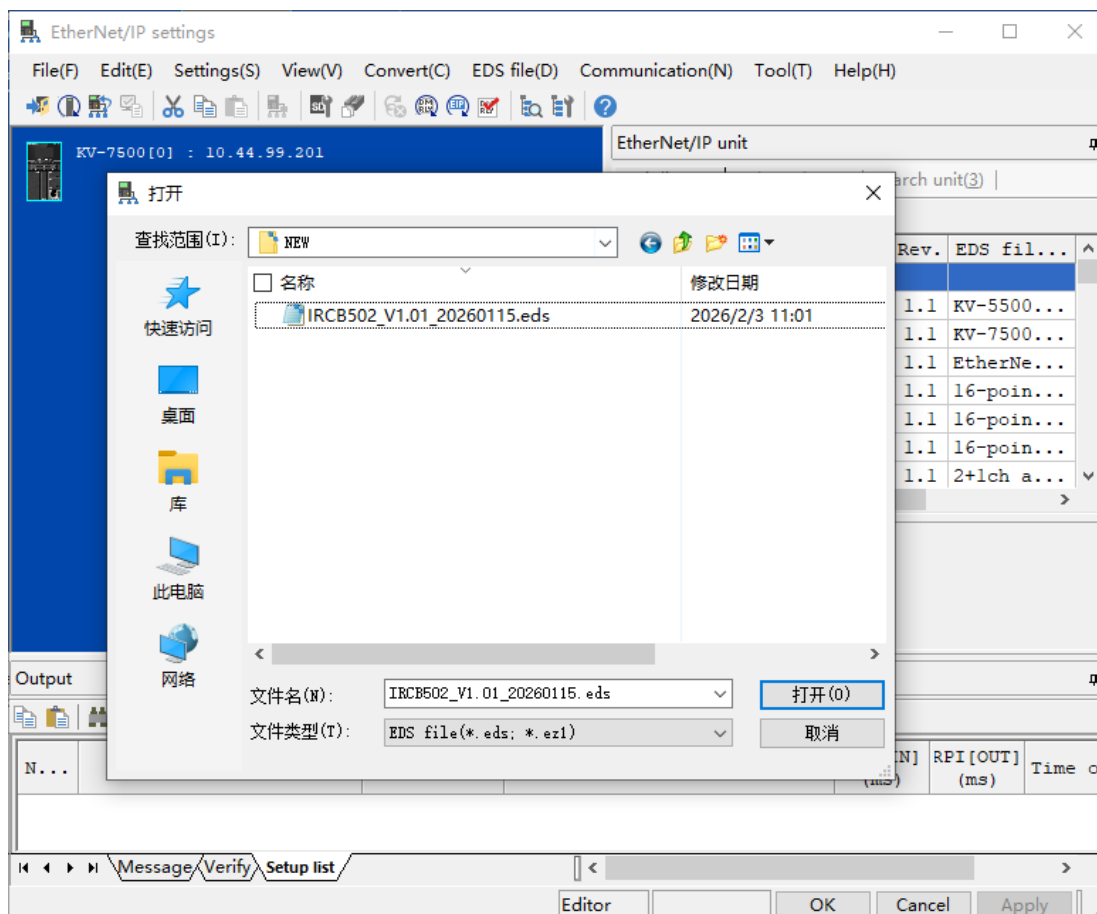
4. In the "Unit Editor - Edit mode" dialog box, click the "EtherNet/IP settings" icon and then click "OK" to complete, as shown in the figure below.



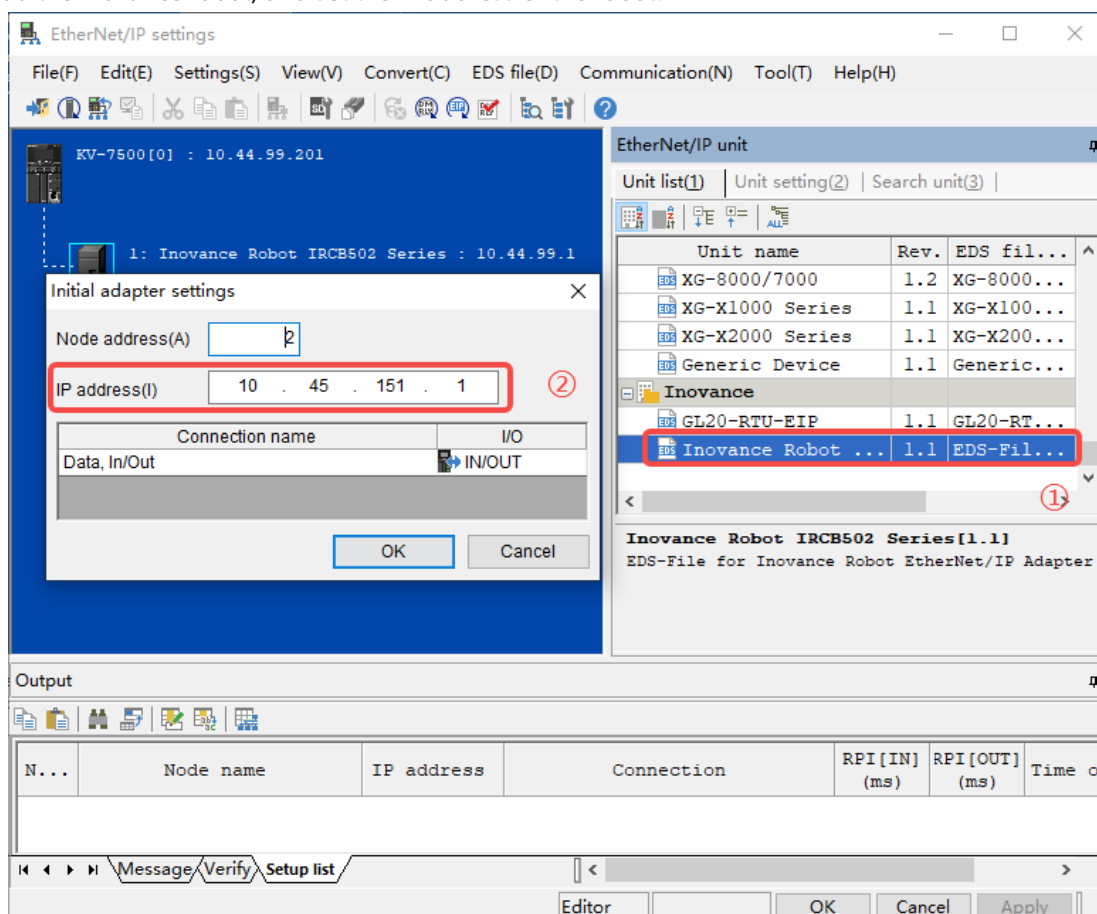
5. Open the "EtherNet/IP settings" dialog box, then select "EDS File" > "Register" to register the EtherNet/IP device.



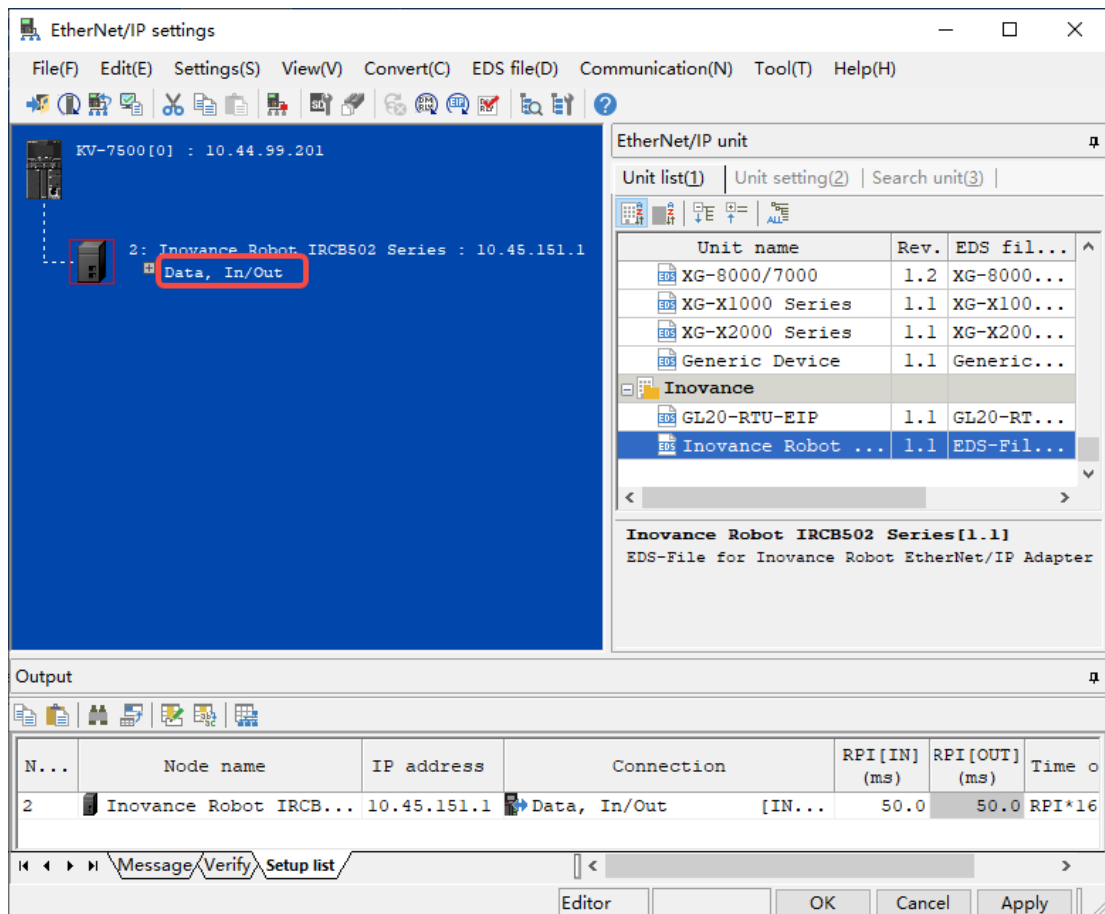
6. In the "EtherNet/IP settings" dialog box, click the registered EtherNet/IP device, select the EDS file to be imported into the Inovance robot, and import it.



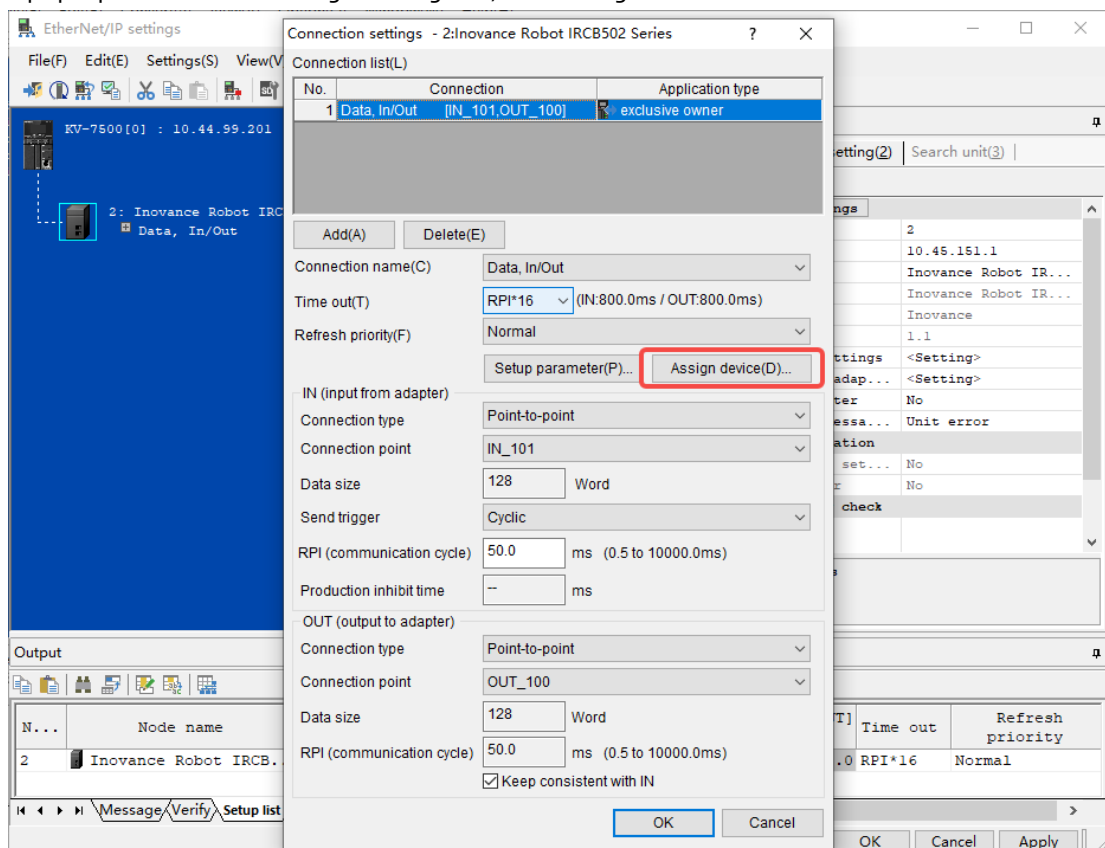
7. In the "EtherNet/IP settings" dialog box, double-click "Inovance Robot IRCB500" in the lower right corner to add the Inovance robot, and set the IP address of the robot.



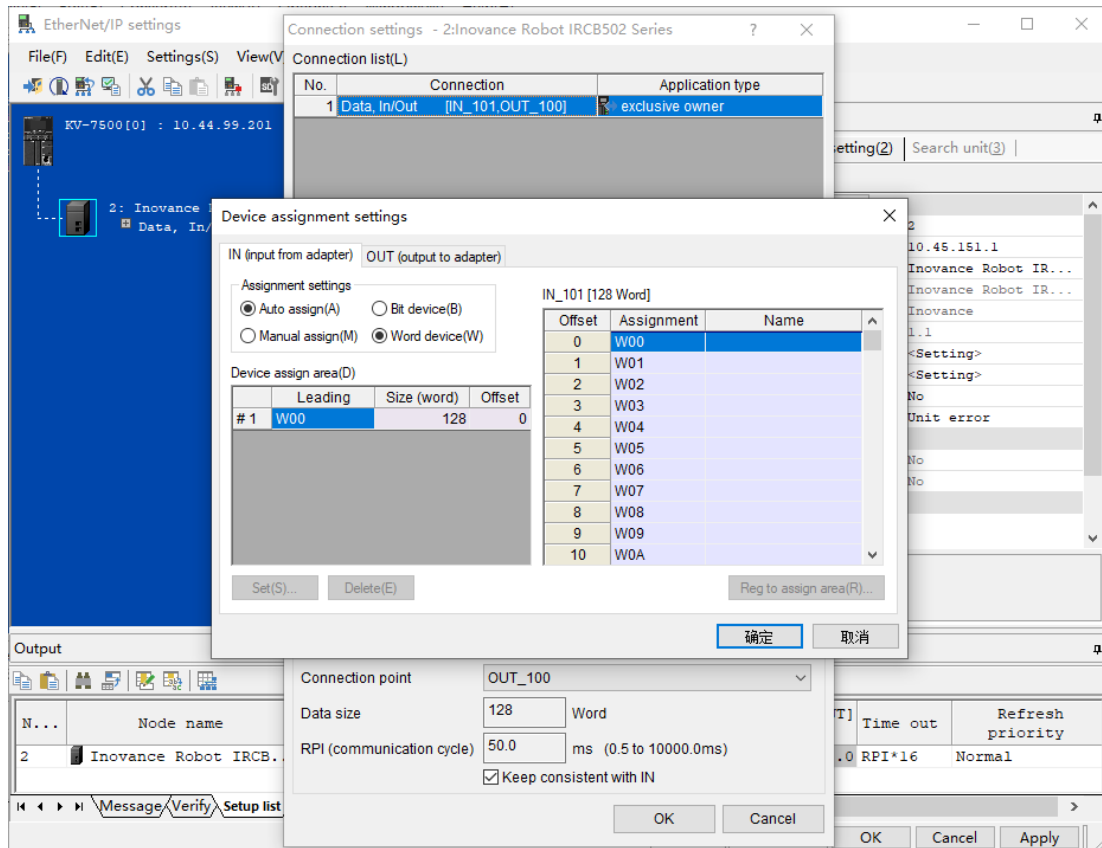
8. In the "EtherNet/IP settings" dialog box, click "Data, In/Out" under "Inovance Robot IRCB502 Series" to open the configuration interface.



9. In the pop-up "Connection settings" dialog box, click "Assign device".



10. In the pop-up "Device assignment settings" dialog box, confirm the assignment of the soft device. You can choose "Manual assign" or "Auto assign", and assign the input and output soft devices accordingly.



1.3.3 Notes



When not switched to remote I/O permission, the robot can transmit data to the PLC, such as position information. The PLC can transmit data to the robot only when the robot has been switched to I/O control permission.

1.4 Communication Between Omron Device and Industrial Robot via EtherNet/IP

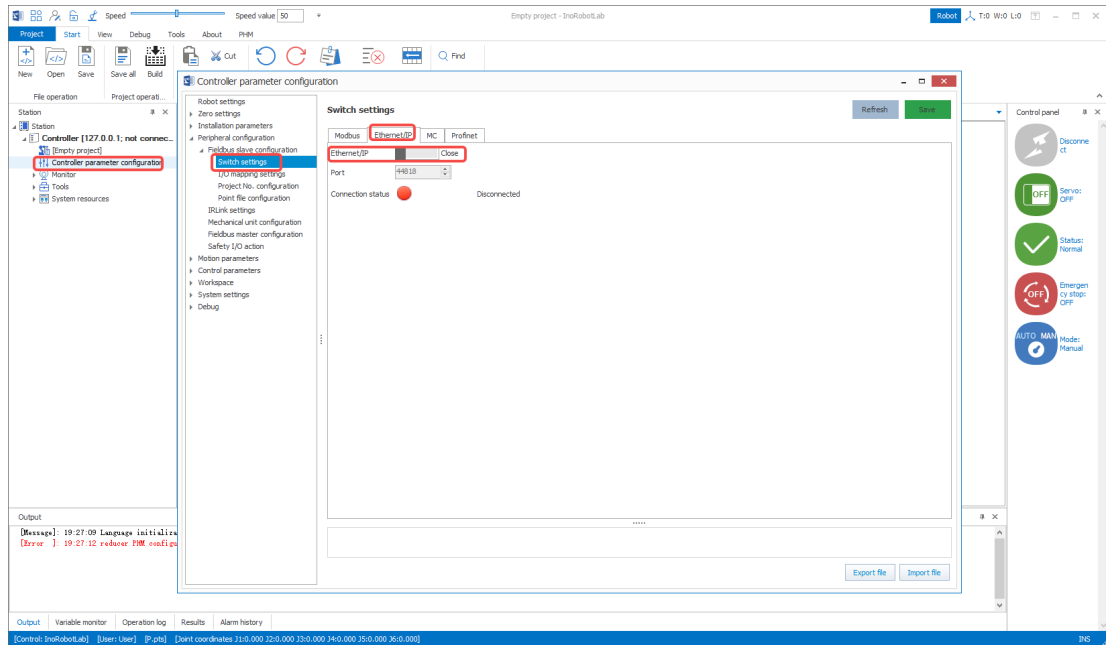
1.4.1 Overview

This section applies to the interaction configuration between Omron PLCs and Inovance robots via EDS. It provides the configuration method based on the Sysmac Studio and CX-One software. You can select one configuration method based on the software type supported by the PLC. It is not necessary to configure both.

1.4.2 Activating the EtherNet/IP Function

1. In the left "Devices" pane of InoRobotLab, double-click "Controller parameter configuration" to open the "Controller parameter configuration" interface.

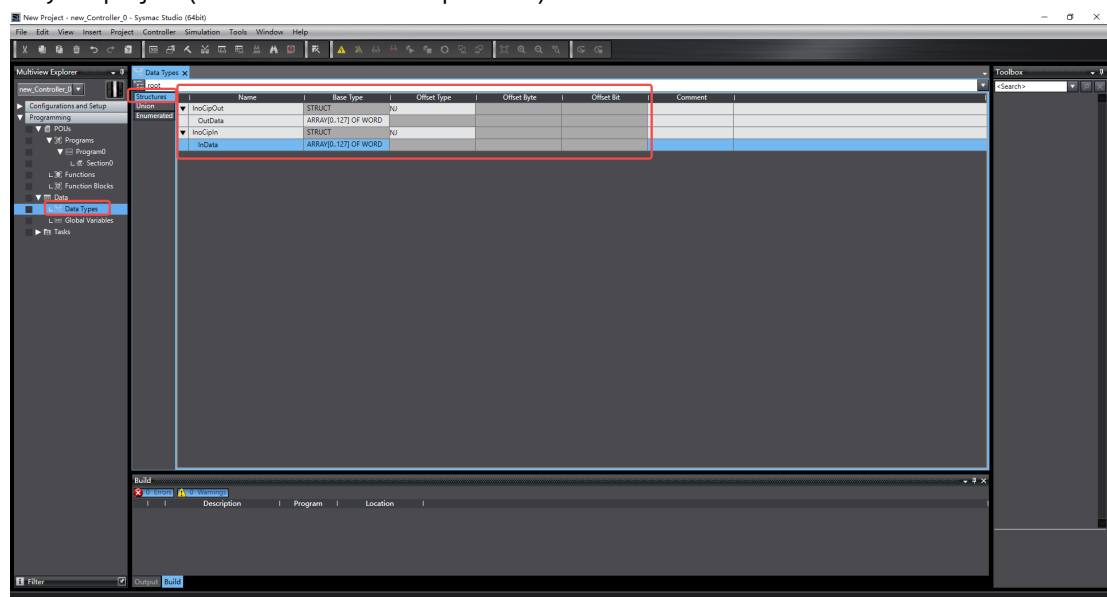
- Click "Peripheral configuration" > "Fieldbus slave configuration" > "Switch settings" > "Ethernet/IP", then activate the EtherNet/IP function by toggling the "EtherNet/IP" to "Open".



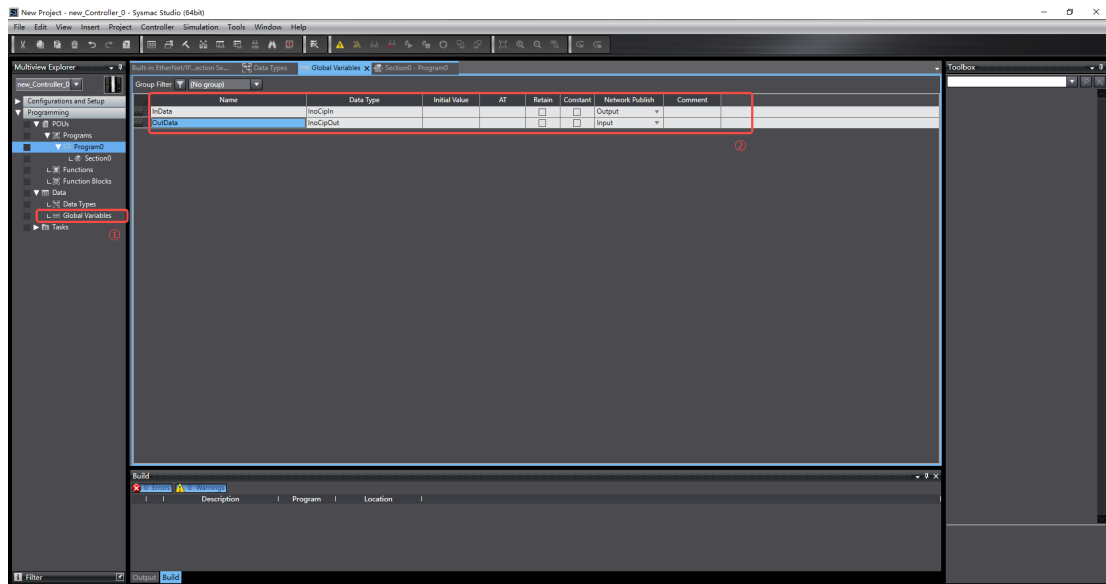
1.4.3 Configuration via Sysmac Studio

1.4.3.1 Copying Variable Type

- In the left device tree, double-click "EipDemo.smc2" to open the project, then click "Data Types" to open the "Data Types" interface.
- On the "Data Types" interface, click "Structures", select all, and copy them to the structure assignment area of your project (customization is also possible).

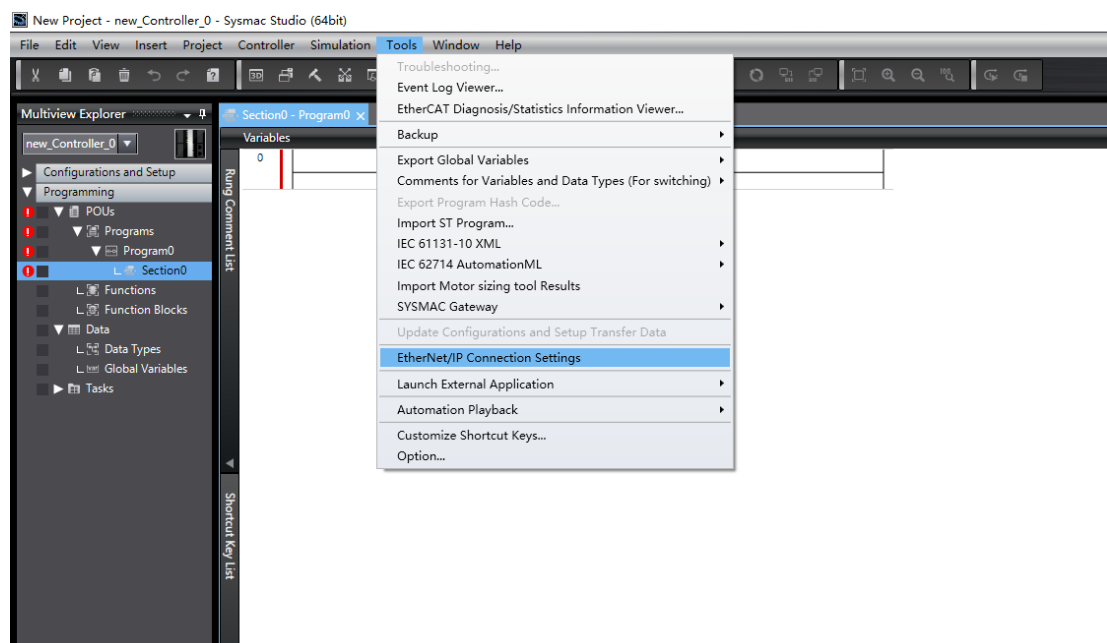


- In the left device tree, click "Global Variables" to open the "Global Variables" interface.
- On the "Global Variables" interface, confirm the structure variables for inputs and outputs (do not change any characters of the copied variables). For controlling multiple robots, you need to copy multiple sets and assign different variable names.

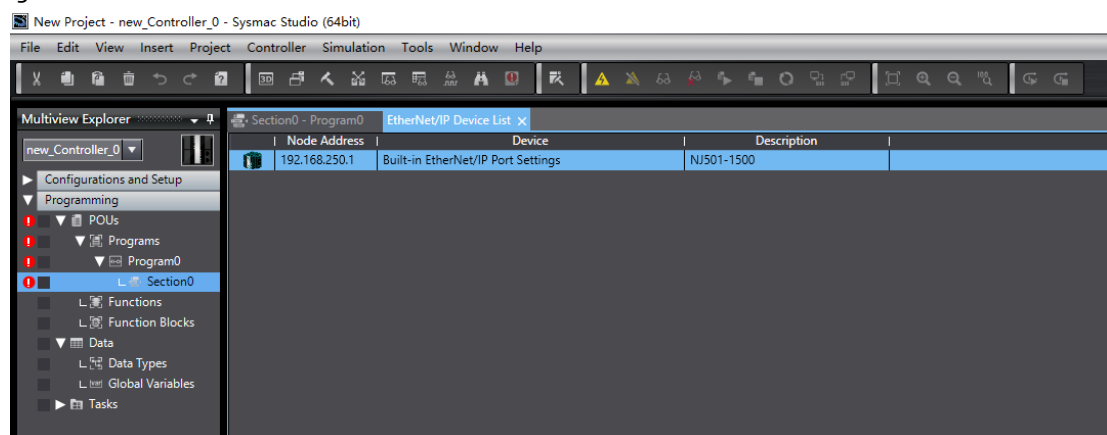


1.4.3.2 Importing the EDS configuration file

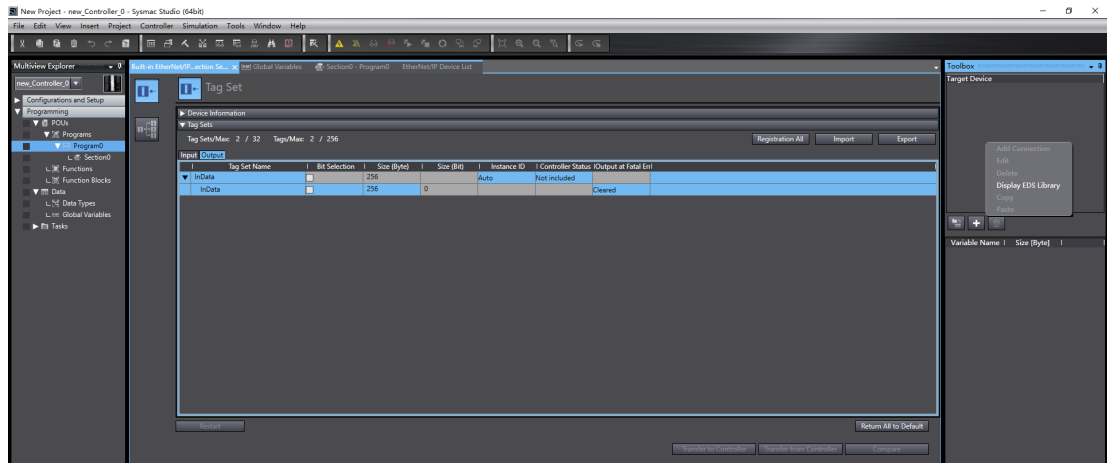
1. In the navigation menu bar, click "Tools" > "EtherNet/IP Connection Settings" to open the "EtherNet/IP Device List" interface.



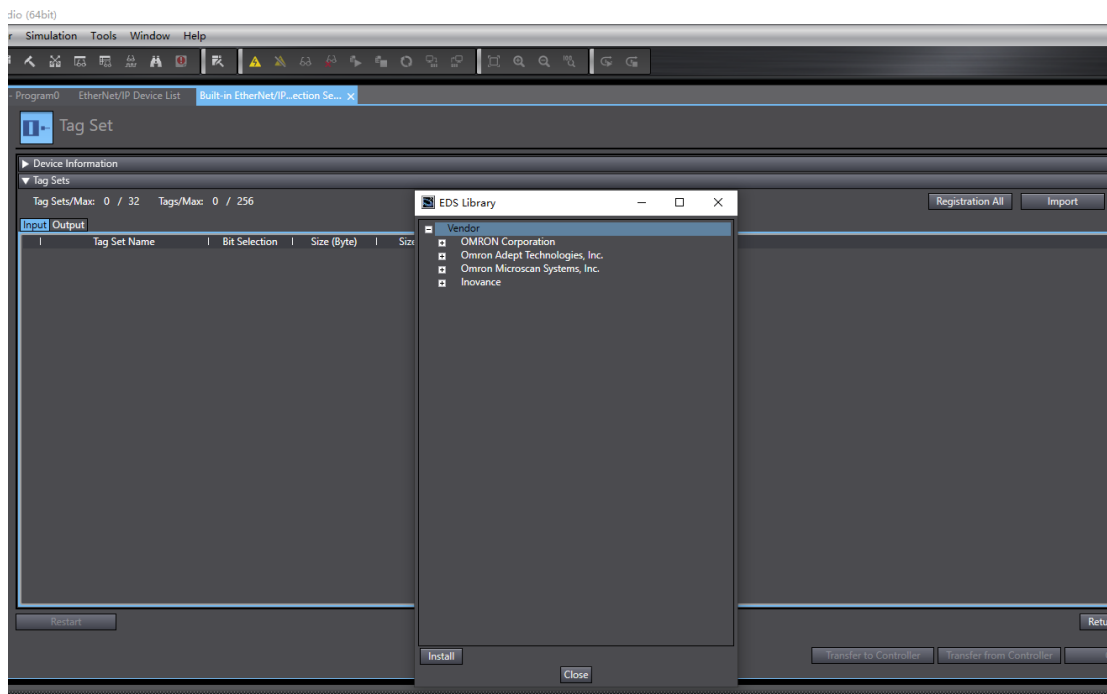
2. On the "EtherNet/IP Device List" interface, double-click "Built-in EtherNet/IP Port Settings" to open the PLC configuration interface.



3. In the upper right corner of the "Built-in EtherNet/IP Port Settings" interface, click "Toolbox" > "Target Device", then right-click and select "Show EDS Library".



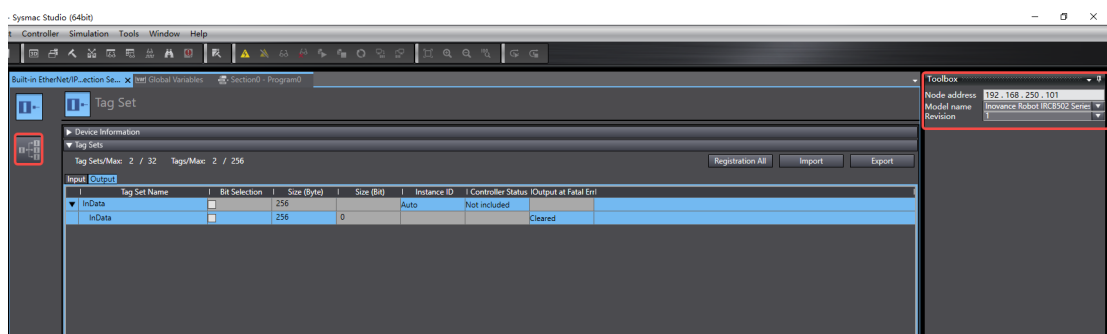
4. In the pop-up "EDS Library" dialog box, click "Install", select the provided EDS file, and complete the import.



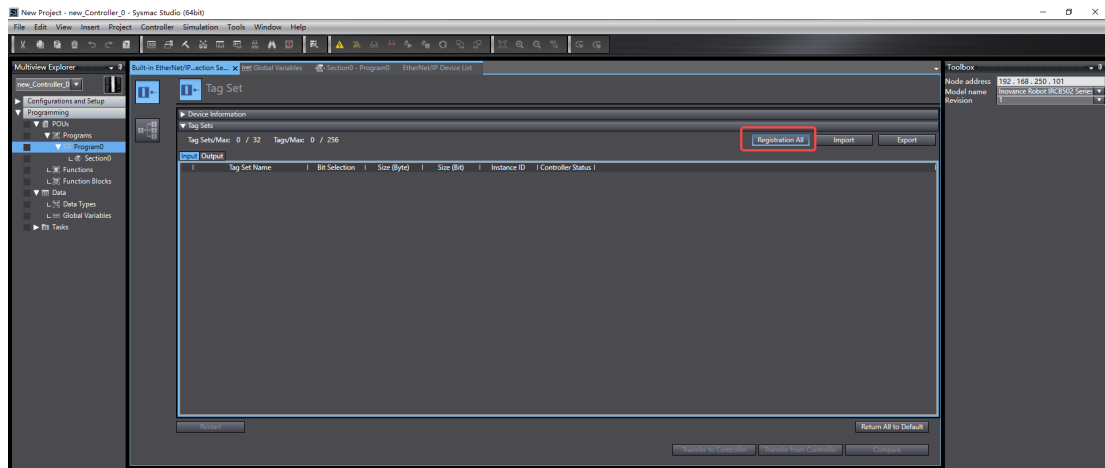
1.4.3.3 Setting PLC Parameters

1. Add a target device.

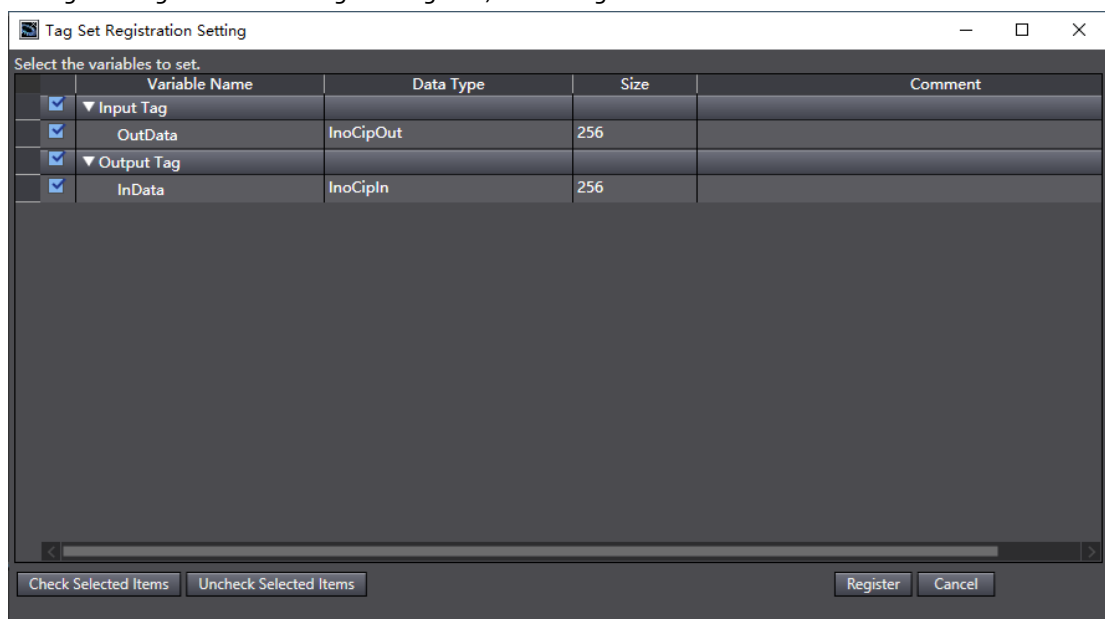
In the "Toolbox" > "Target Device" section at the upper right corner of the "Built-in EtherNet/IP Port Settings" interface, click the "+" symbol to add a target device. The IP address should be the robot's IP address, which must be in the same subnet as the PLC. For the model name, select "Inovance Robot IRCB500". Currently only revision 1 is available.



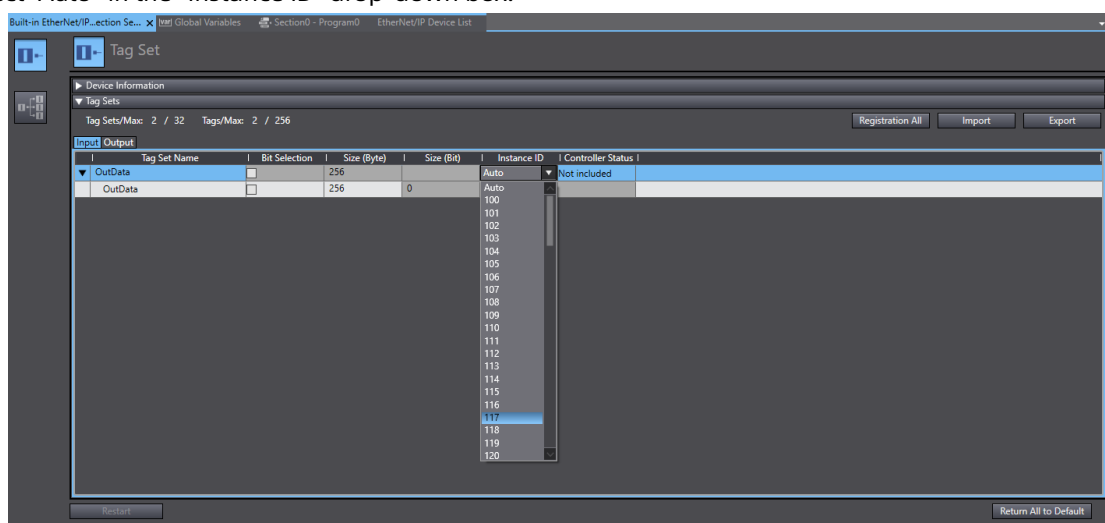
2. On the "Built-in EtherNet/IP Port Settings" interface, click "Tag Sets" > "Registration All" to open the "Tag Set Registration Setting" dialog box.



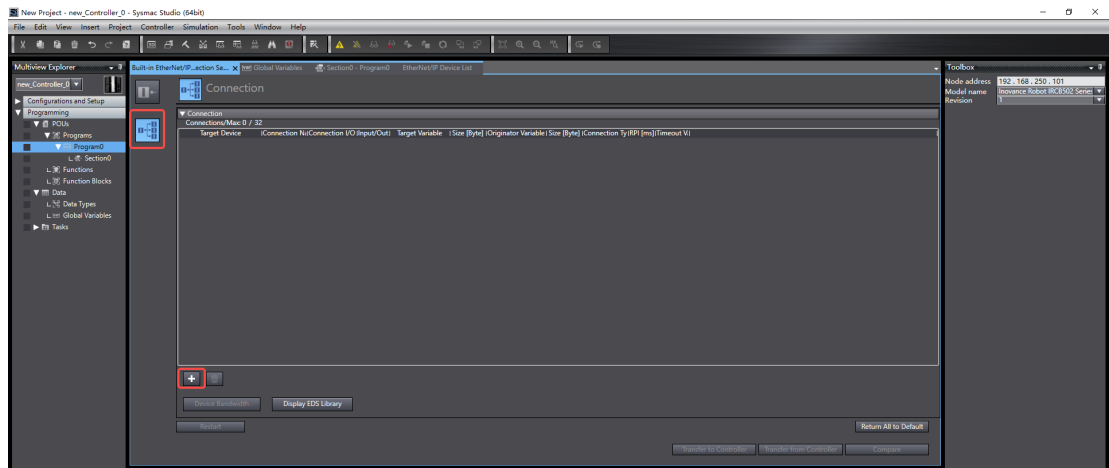
3. In the "Tag Set Registration Setting" dialog box, click "Register".



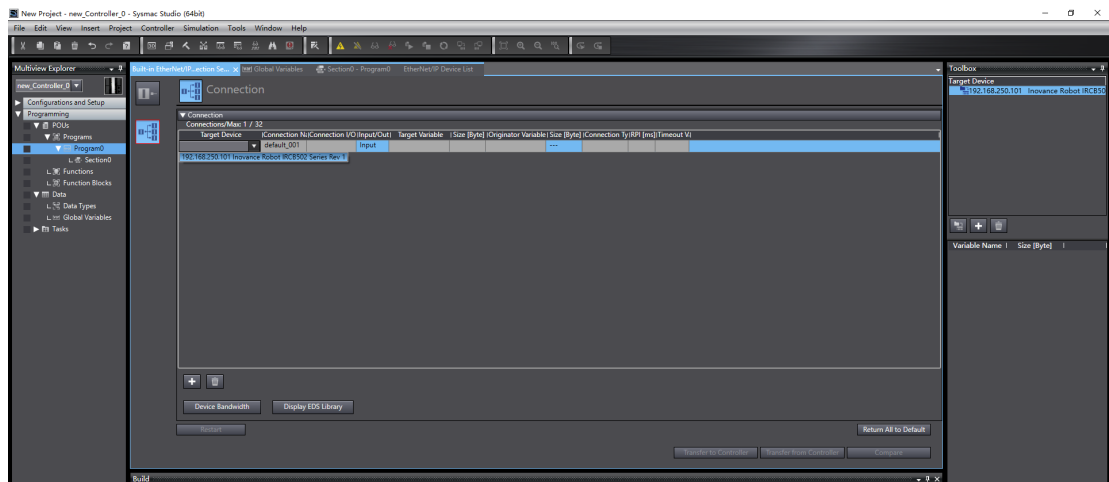
4. Select "Auto" in the "Instance ID" drop-down box.



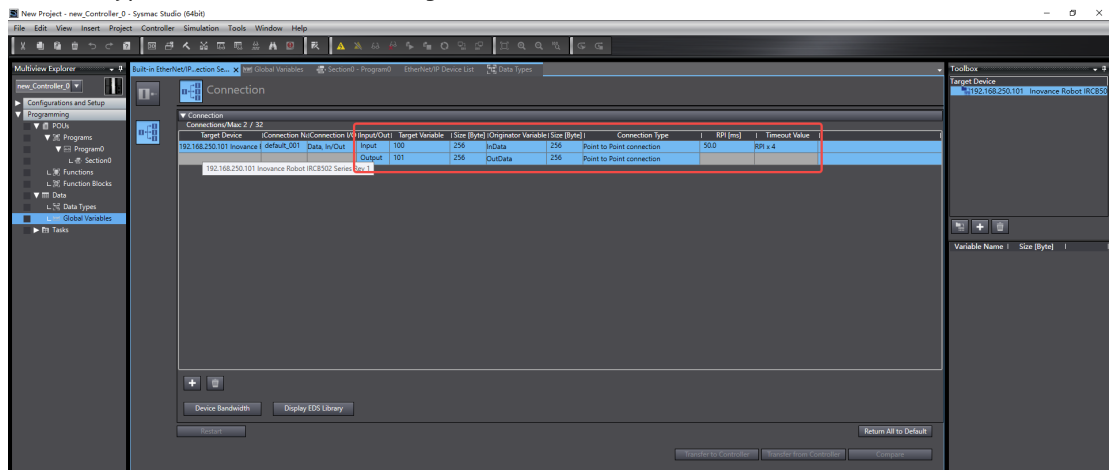
5. Set the connection object.
 - a. On the "Built-in EtherNet/IP Port Settings" interface, click the "+" button to add a target device for connection.



b. On the "Built-in EtherNet/IP Port Settings" interface, select the target device.

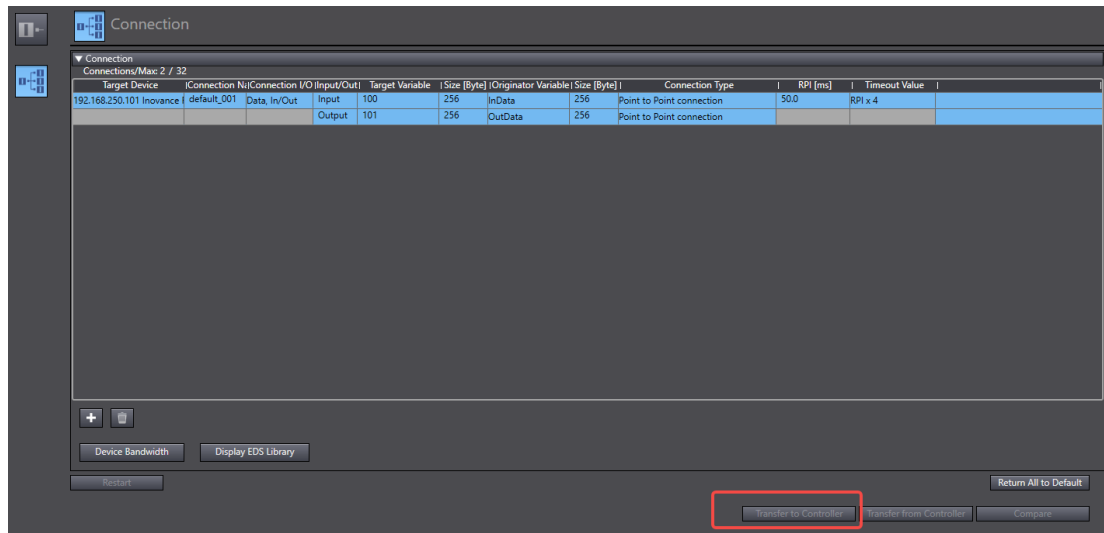


c. On the "Built-in EtherNet/IP Port Settings" interface, set the target variable, byte size, start variable, connection type, etc., as shown in the figure below:



- The number of input targets must be set to 101.
- The number of output targets must be set to 100.
- By default, both the input and output sizes are 256 bytes each.

d. Click "Transfer to Controller" to send the configuration parameters to the PLC.



1.4.4 Configuration via CX-One

1.4.4.1 CX-One: Creating Custom Data Type

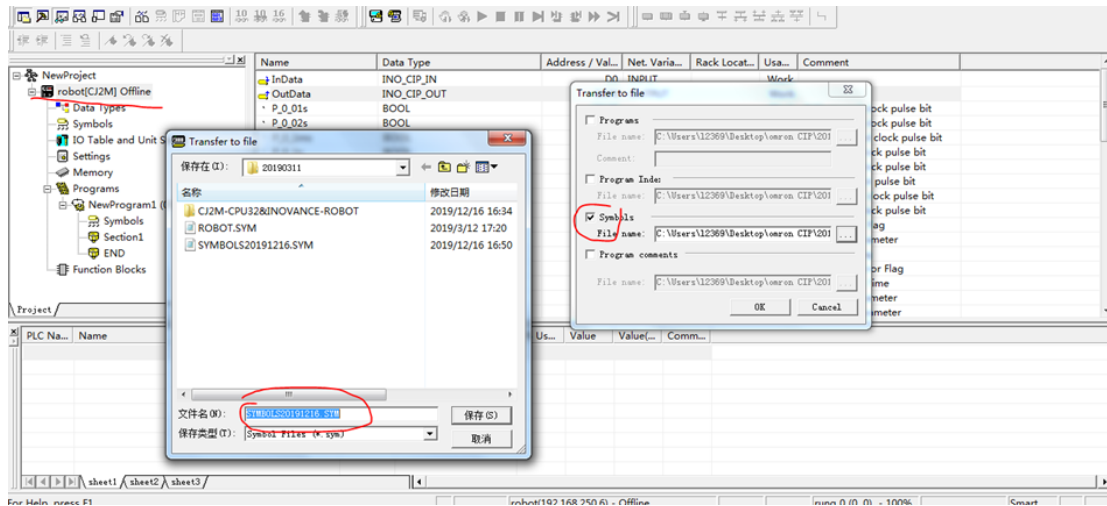
Click "Data Types" under the device tree on the left, and create a robot data type (custom structure) on the right. The input name can be set as InData, and the output name can be set as OutData.

1.4.4.2 CX-One: Creating Input/Output Interface Variables

After clicking "Symbols", you can create variables on the right side.

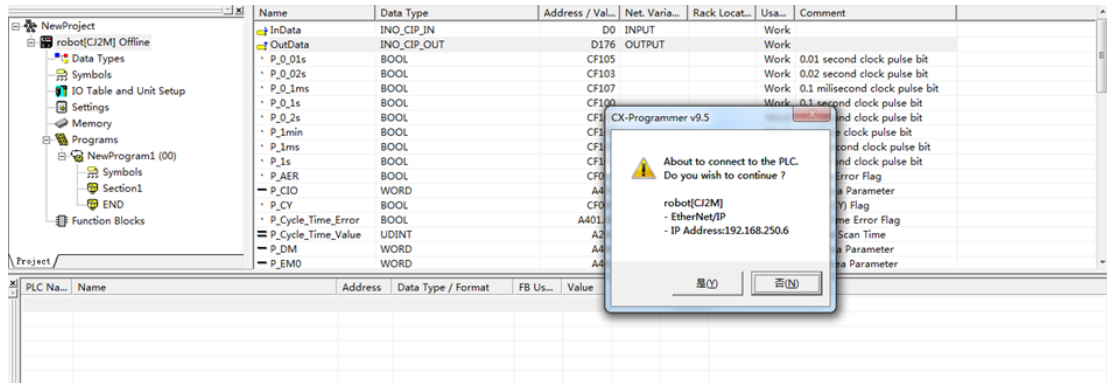
1. InData is the input variable of the PLC, with a size of 128 words, i.e., 256 bytes. This variable serves as the data exchange identification interface between the robot and the PLC. The input start address can be customized. In this case, D0 is defined as the input start address.
2. OutData is the output variable of the PLC, with a size of 128 words, i.e., 256 bytes. This variable serves as the data exchange identification interface between the robot and the PLC. The output start address can be customized. In this case, D176 is defined as the output start address.
3. You can directly call the member variables in InData and OutData to program functions.

Name	Data Type	Address/Variable
InData	INO_CIP_IN	D0
OutData	INO_CIP_OUT	D176
P_0_01s	BOOL	CF105
P_0_02s	BOOL	CF103
P_0_1ms	BOOL	CF107
P_0_1s	BOOL	CF100
P_0_2s	BOOL	CF101
P_1ms	BOOL	CF106
P_1s	BOOL	CF102
P_1	BOOL	CF104
P_AER	BOOL	CF011
P_CIO	WORD	A450
P_CY	BOOL	CF004
P_Cycle_Time_Error	BOOL	A401.08



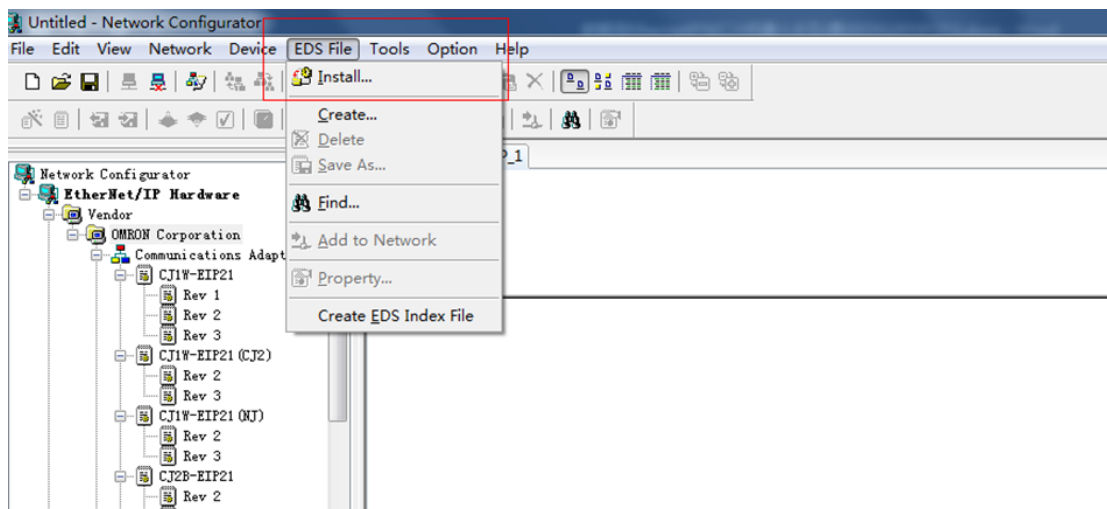
1.4.4.3 CX-One: Downloading Variable Configuration to PLC

Right-click the corresponding project and click "work online" to connect the software to the PLC. Then right-click the project and click "transfer" > "To PLC" to download the project to the PLC. At this point, the input and output configurations set above already exist in the PLC.

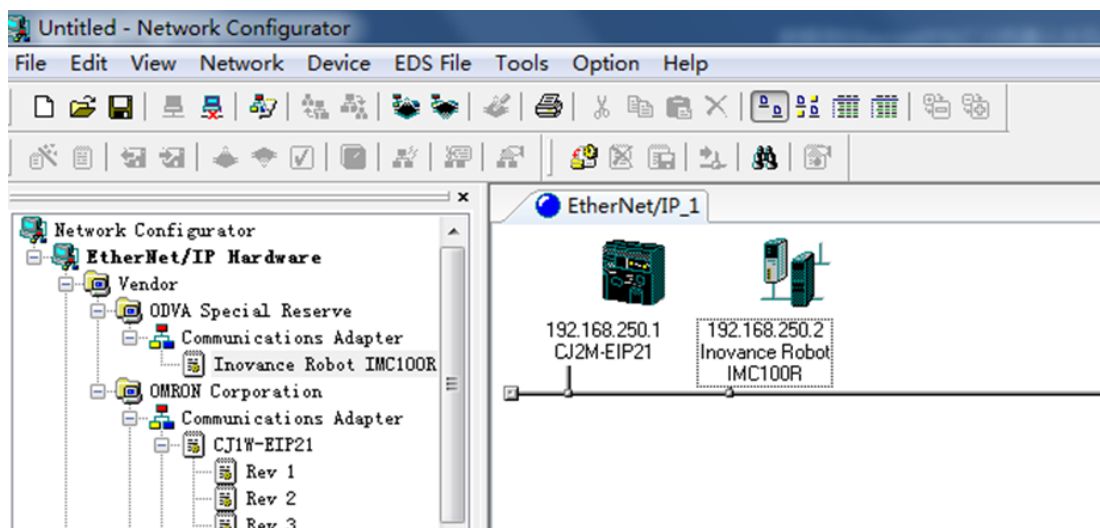


1.4.4.4 Network Configurator: Configuring Network Nodes

1. Open the Network Configurator software, click the "EDS File" function menu to import the EDS file, and enter the "EtherNet/IP" interface.

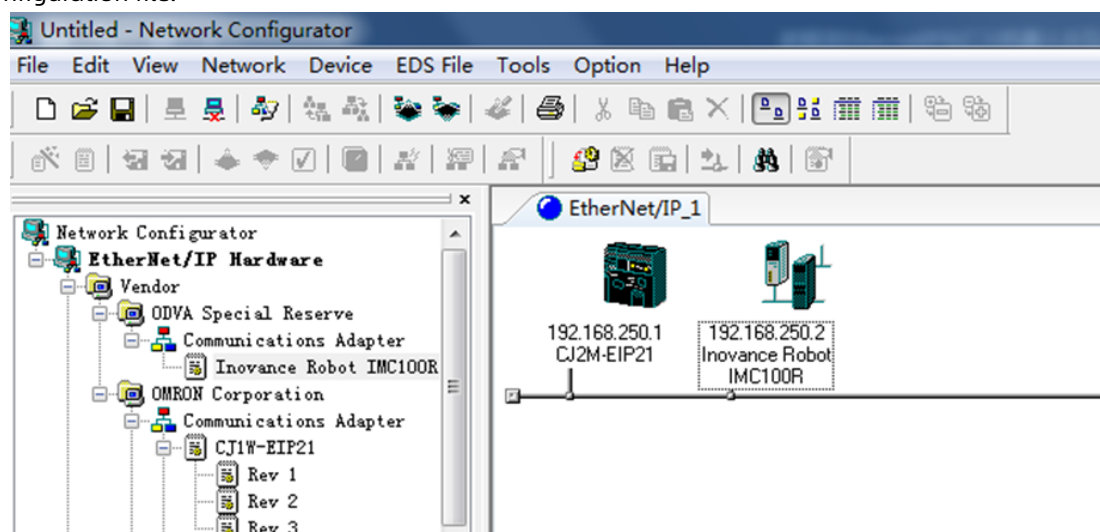


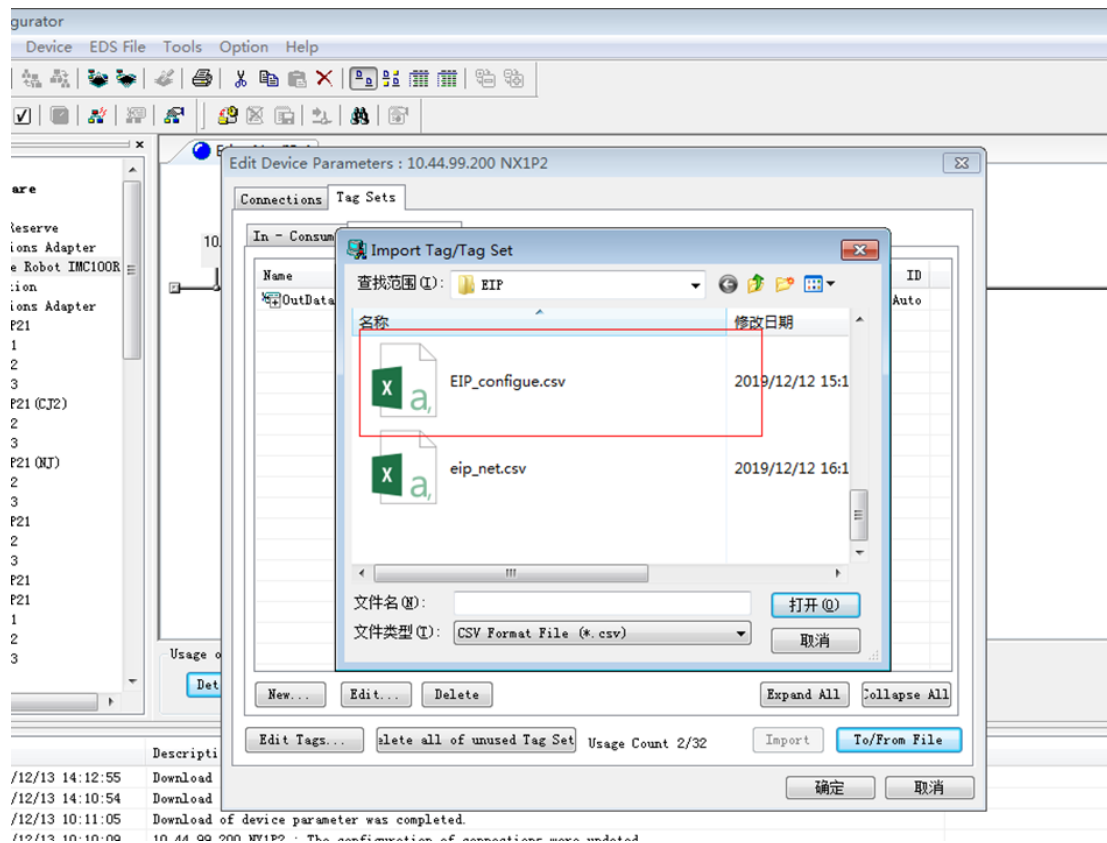
2. On the "EtherNet/IP" interface, drag in the corresponding devices to form network nodes. The CJ2M is an Omron PLC, and the device on the right is the Inovance robot imported via the EDS file.



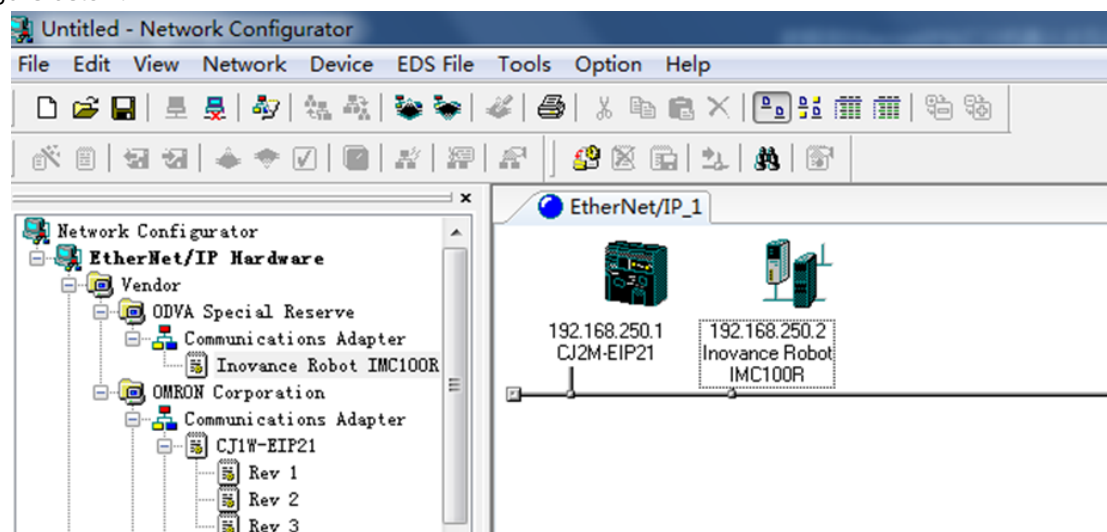
1.4.4.5 Network Configurator: (Importing) Configuring Input/Output Interfaces

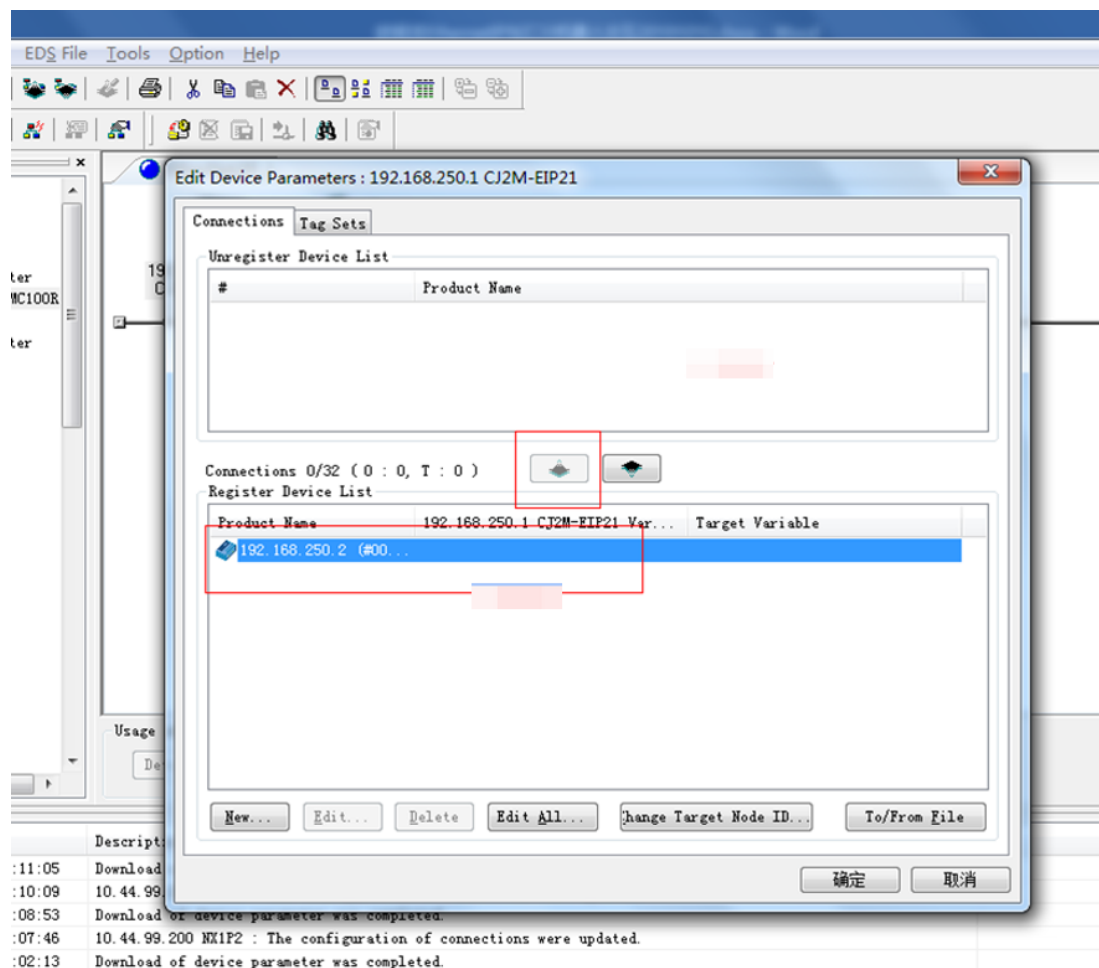
1. Import from the provided .csv file
 - a. Open the Network Configurator software, click "Inovance Robot IMC100R" in the left navigation tree to enter the "EtherNet/IP_1" interface.
 - b. On the "EtherNet/IP_1" interface, double-click the Omron PLC (CJ2M) on the left to enter the "Import Tag/Tag Sets" interface, then click "To/From File" in the lower right corner to import the .csv configuration file.



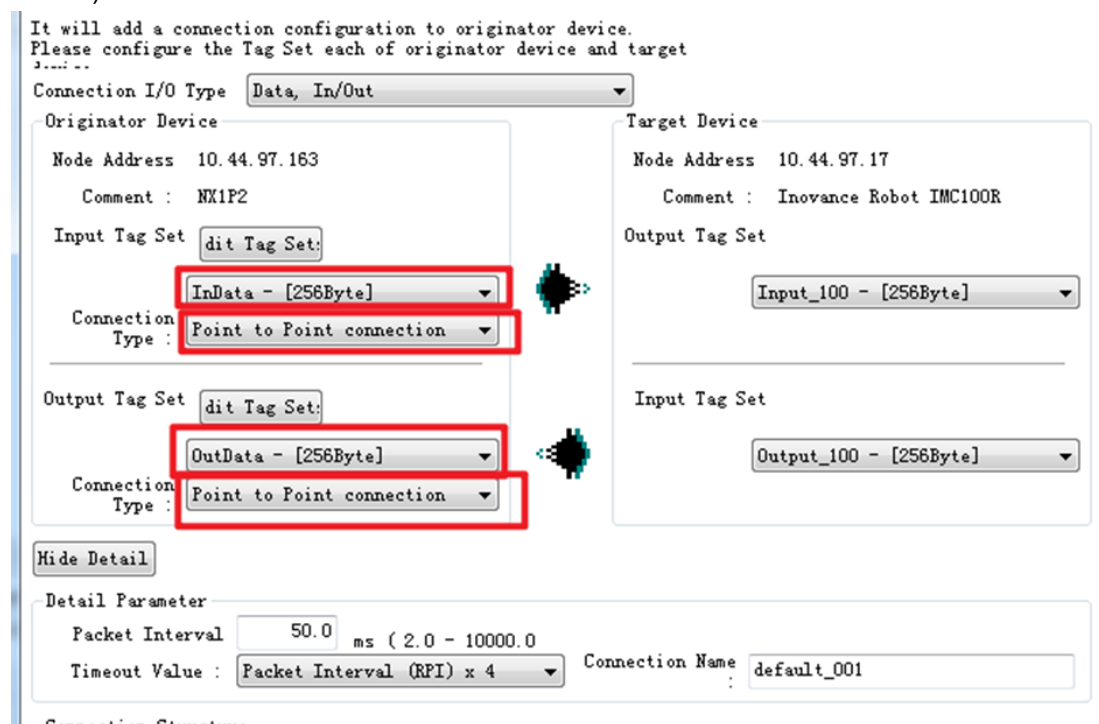


- c. On the "EtherNet/IP_1" interface, double-click the Omron PLC (CJ2M) on the left, and configure the connection properties between the PLC and the Inovance robot in the pop-up window as shown in the figure below.



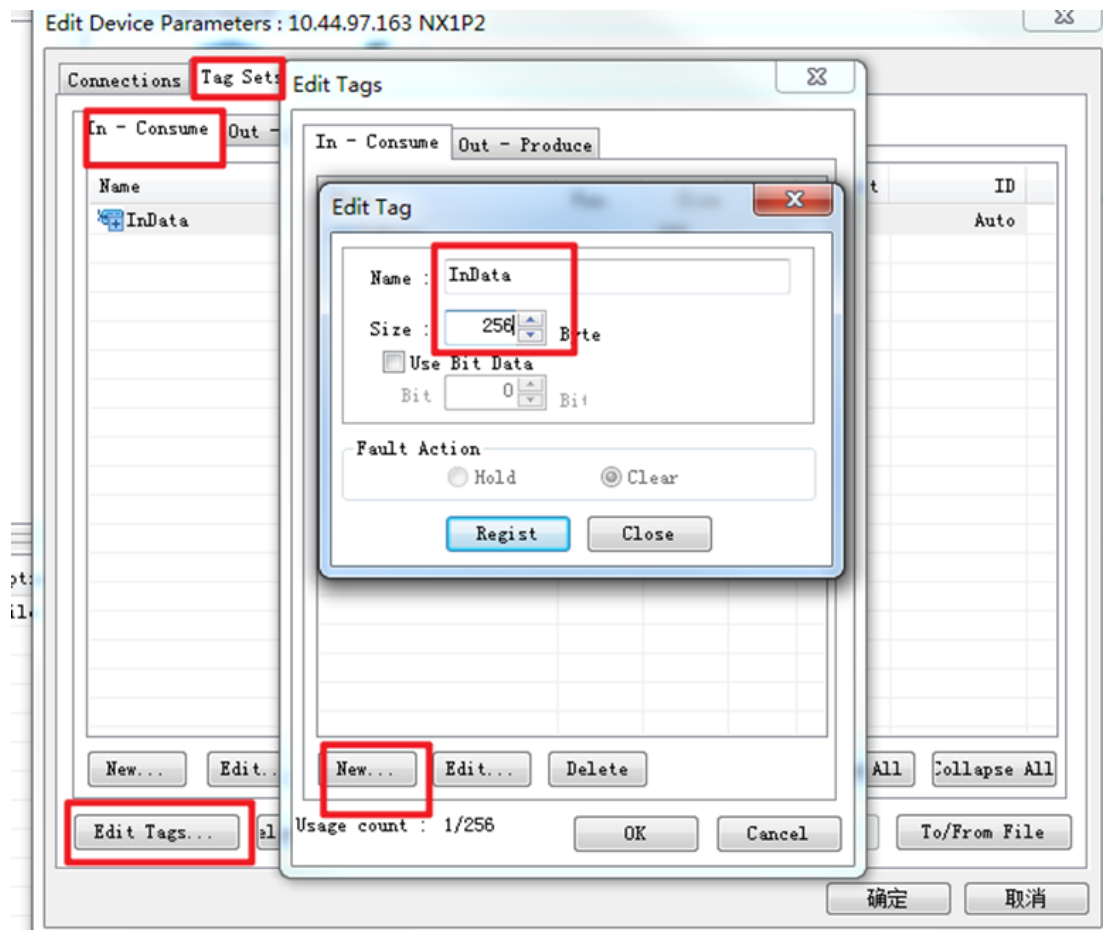


- d. Configure according to the figure below (on S03.20 version, the input and output sizes are 256 bytes by default).



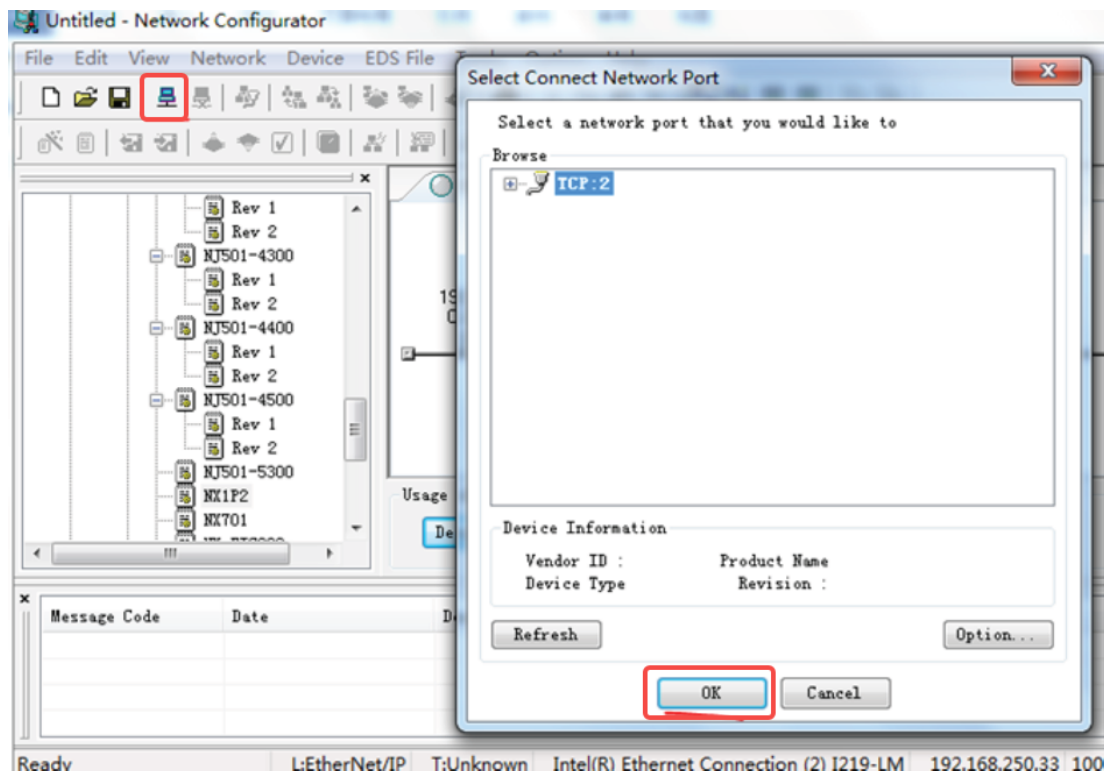
2. Create new tag group

The variable names for the input and output interfaces must be the same as those previously defined in CX-One, namely InData and OutData, where InData is 256 bytes and OutData is 256 bytes.



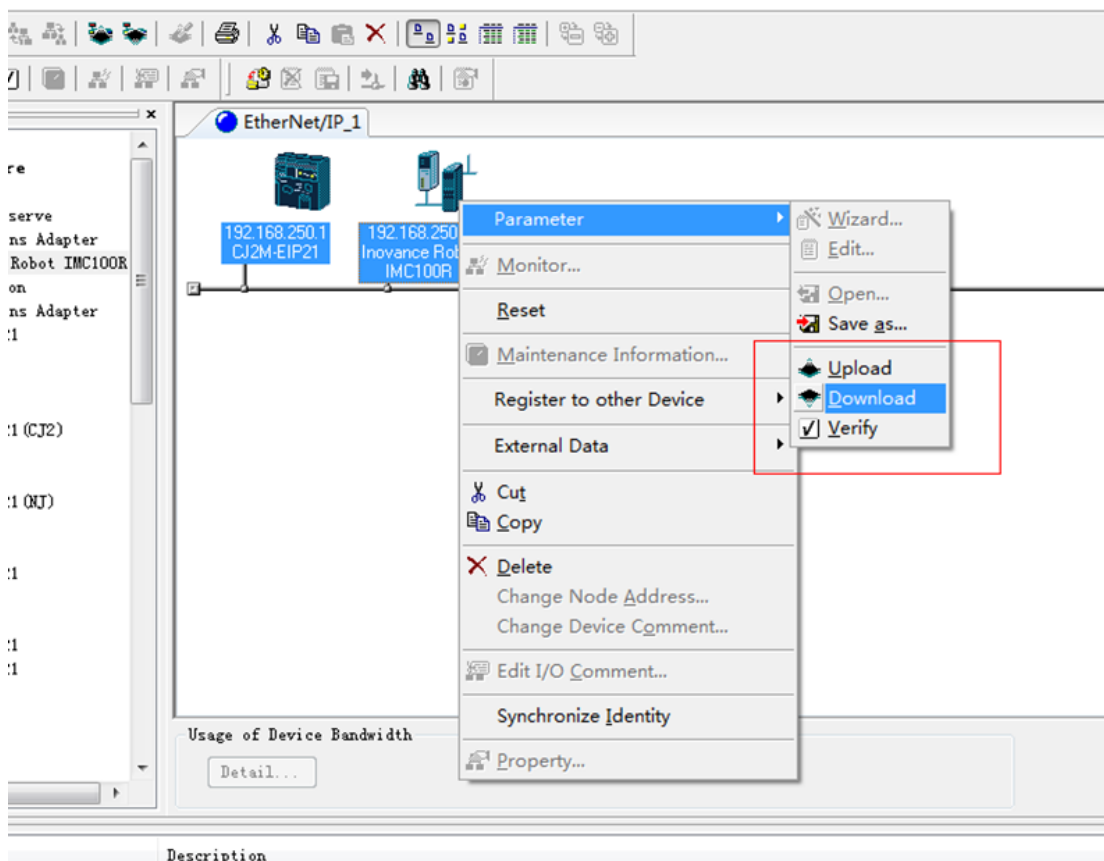
1.4.4.6 Network Configurator: Connecting to PLC

1. On the "Untitled - Network Configurator" interface, click the  icon to open the "Select Connect Network Port" dialog box.
2. Click "OK" in the "Select Connect Network Port" dialog box to successfully connect to the PLC.



1.4.4.7 Network Configurator: Downloading Configuration

1. On the "EtherNet/IP" interface, select the two network device nodes.
2. Right-click and select "Parameter" > "Download" to download to the PLC, then restart the PLC immediately after download is completed.
3. If any error occurs during the download process, ignore them and restart the PLC immediately after downloading to resolve the issue.



1.4.5 Notes



When not switched to remote I/O permission, the robot can transmit data to the PLC, such as position information. The PLC can transmit data to the robot only when the robot has been switched to I/O control permission.

2 Modbus Bus Communication

2.1 Communication Introduction

Modbus is an open serial communication protocol based on RS485/TCP, featuring core advantages such as high standardization of the protocol, clear master-slave architecture, strong cross-platform compatibility, outstanding transmission reliability, and low deployment cost. Its hierarchical data model supports standard data interaction with industrial devices (such as discrete I/O and registers), making it widely adopted in medium- to low-speed control scenarios, like PLC-to-sensor communication.

2.2 Function Overview

2.2.1 Modbus Communication Protocol

- Modbus communication protocol is a standard industrial fieldbus protocol.
- The IMC100 robot system supports the Modbus-RTU protocol over RS485 bus connections, as well as the Modbus TCP protocol over Ethernet connections.

Modbus TCP features:

- Modbus TCP utilizes the standard TCP/IP stack to encapsulate and compress Modbus packets, enabling Modbus TCP devices to connect and communicate over Ethernet and fiber optic networks.
- Compared to the RS485 interface, Modbus TCP supports a larger address space, a multi-master architecture, and transmission rates up to gigabit levels.
- The number of slaves in a Modbus TCP network is limited only by the capability of the physical network layer, typically reaching up to approximately 1024 slaves.

2.2.2 Modbus Control Application Scenarios

- In field Modbus applications, the IMC100 robot typically serves as the controlled object, i.e., as a Modbus slave. This section only addresses this application scenario.
- Devices supporting standard Modbus master functionality, such as touch screens (HMI), PLCs, and industrial PCs, can operate the IMC100 robot over either RS485 or Ethernet connections.
- When the robot is configured as a Modbus slave, the master device can monitor and control the robot's operation based on the corresponding Modbus address table.
- When the robot is not configured as a Modbus slave, the corresponding Modbus functions are invalid.

2.2.3 Modbus Application Process

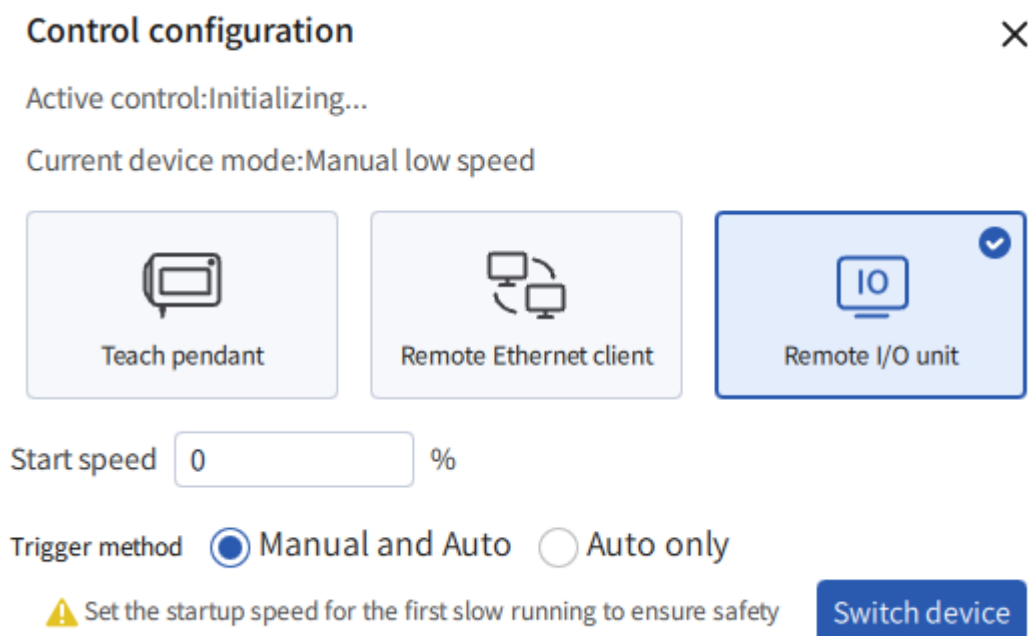
The Modbus application process is divided into two steps: Modbus communication configuration and control application construction.

1. Modbus communication configuration: Configure the Modbus slave related configurations directly in the teach pendant software with version S01.17 or later.

2. Control application construction: Based on the required communication needs, complete the application programming by utilizing the slave address table (for details, see the "Modbus Control Application" section).

Upon completion of the above procedures, Modbus is ready for use.

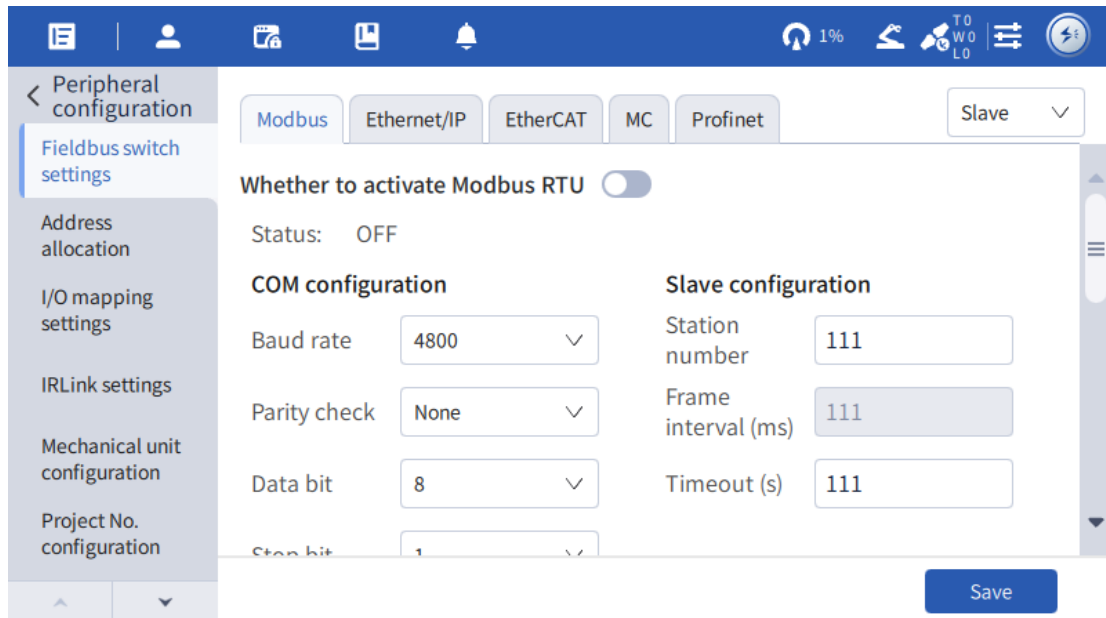
- For the monitoring functions, you can directly query the robot's related statuses and parameters.
- For the control functions, you need to set the robot's "Control configuration" to "Remote I/O unit" via the teach pendant, as shown in the figure below.



2.3 Communication Configuration

You can use the InoRobotTP software to configure the Modbus communication.

1. After logging into the InoRobotTP, click "Settings" > "More" > "Peripheral configuration" > "Fieldbus switch settings" > "Modbus" to enter the Modbus configuration interface.
2. Configure the Modbus connection status. Once the page is activated, it refreshes the active status and connection status in real time:
 - Activate Modbus-RTU or Modbus TCP, then set the parameters.
 - You can activate Modbus-RTU and Modbus TCP simultaneously.



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2.4 Control Application

The Modbus functions of the IMC100 robot are primarily divided into two major categories:

- Monitor functions, including the monitoring on the system status, current parameter values, variable values, etc. The master device can query the robot's related statuses and parameters under any conditions after the robot has started normally.
- Control functions, including controlling system start/stop, modifying system parameters, writing system variables, etc. When the "Control device" of the IMC100 robot is successfully set to "Remote I/O unit" via the teach pendant, the master device can operate and control the robot according to the specific functional procedures.

2.4.1 Modbus Address Description

The allocation of the Modbus slave address table for the IMC100 robot is based on the standard Modbus protocol. The objects are categorized into discrete inputs, coils, input registers, and holding registers. For details, see the following table:

Object	Address range	Object type	Access type	Parameter	Content
Discrete input	0 to 4095	Single bit	Read-only	02	The system provides data (e.g., status, flags)
Coil	4096 to 8191	Single bit	Read/Write	01/05/15	User-modifiable data (e.g., control commands)

Object	Address range	Object type	Access type	Parameter	Content
Input register	0 to 32767	1 word (16-bit)	Read-only	04	The system provides data (e.g., current parameters, variable values)
Holding register	32768 to 65535	1 word (16-bit)	Read/Write	03/06/16	User-modifiable data (e.g., system parameters, variable values)

When it is necessary to control the robot program operation via Modbus communication, set 5 programs in the teach pendant under "Peripheral configuration" > "I/O mapping settings". Then, call the corresponding program based on the above table.

2.4.2 Modbus Control Function Description

In control functions, the types of control commands are divided into level type, reset type, and special type.

- For a level type command, you need to write 1 to the corresponding coil for the command to take effect. To cancel the command, you need to write 0 to the corresponding coil.
- For a reset type command (active on rising edge of the user input command), in order for the command to take effect, you need to write 1 when the corresponding coil is 0, maintain it for a period of time (recommended at least 200 ms), and then clear the value. You can refer to the level sequence relationship of the reset type buttons of the HMI.
- For a special type command, it is actually a special level command, which takes effect only when the value in the address changes.
 - The direct motion command (assigned to read/write coil address 4141), initiates direct motion when its value changes from 0 to 1, and stops the motion when the value changes from 1 to 0.
 - Taking the enable command as an example, a transition from 0 to 1 activates enabling, while a transition from 1 to 0 activates disabling.

Instruction sending interval

Regarding the requirement for instruction sending intervals: It is recommended that the sending interval between two consecutive instructions be greater than 2 ms. Due to the fact that the execution of certain instructions requires a relatively long time, the sending interval for the following instructions must be greater than 200 ms to ensure correct execution:

- Program start/run and program pause
- Program start/run and program stop
- Enable and disable

Description of B, R, and D variables under Modbus

Under Modbus control, the BRD variables support the following range of settings:

- B variable: any integer between 0 and 255

- R variable: same range of a signed 32-bit integer.
- D variable: same range of a 64-bit floating-point number (supports up to 7 significant digits).



- Set strictly according to the variable range.
- When the variable range setting exceeds the limit, due to truncation issues, the B variable may become a value different from the expected setting. Be aware of the potential risks this may cause.

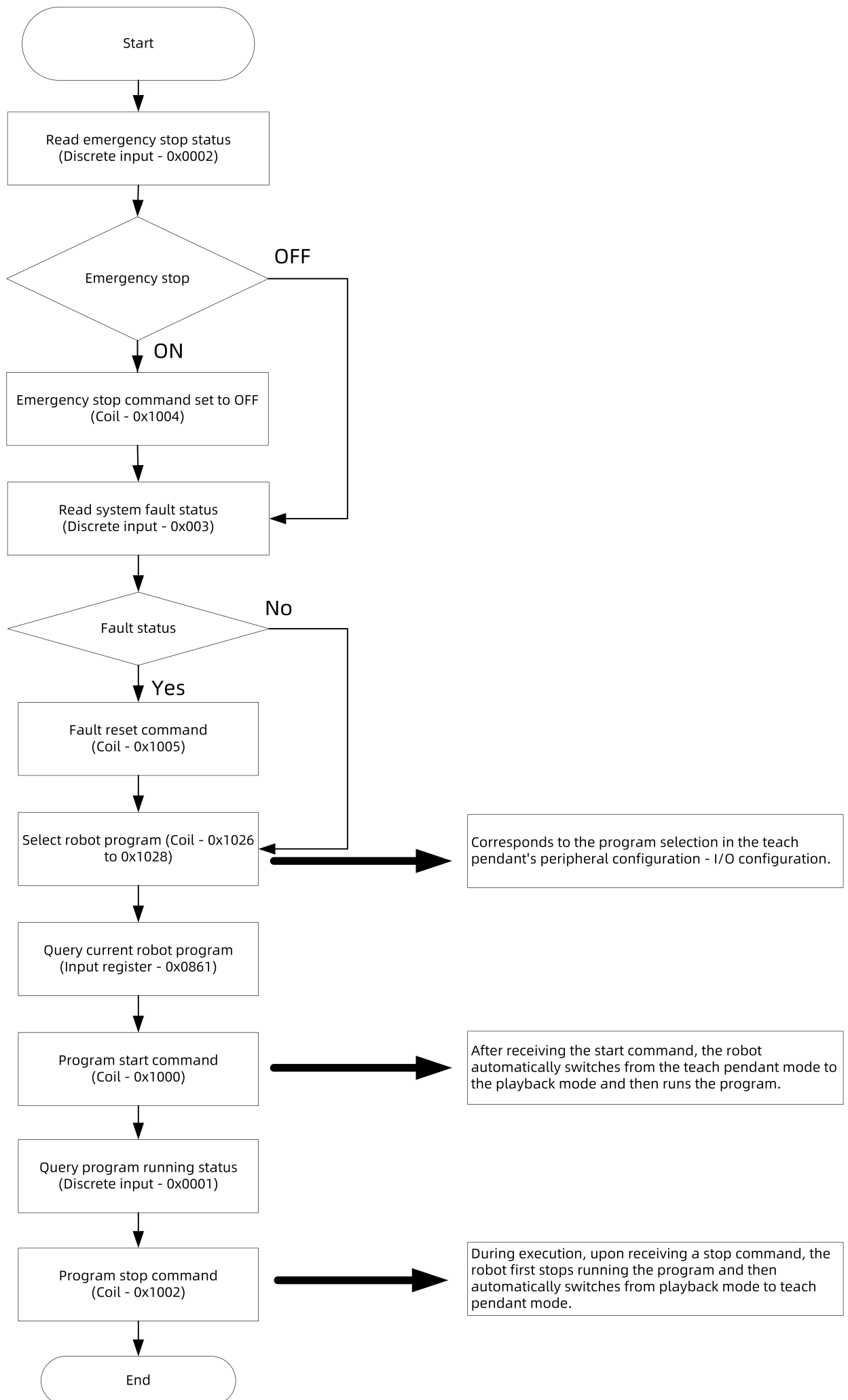
2.4.3 Control Cases

2.4.3.1 Controlling Robot Program Start/Stop via Master

Case description

- Implement remote Modbus control of robot operation.
- Implement functions such as emergency stop control, fault reset, robot program selection, program start, and program stop.

Operation process

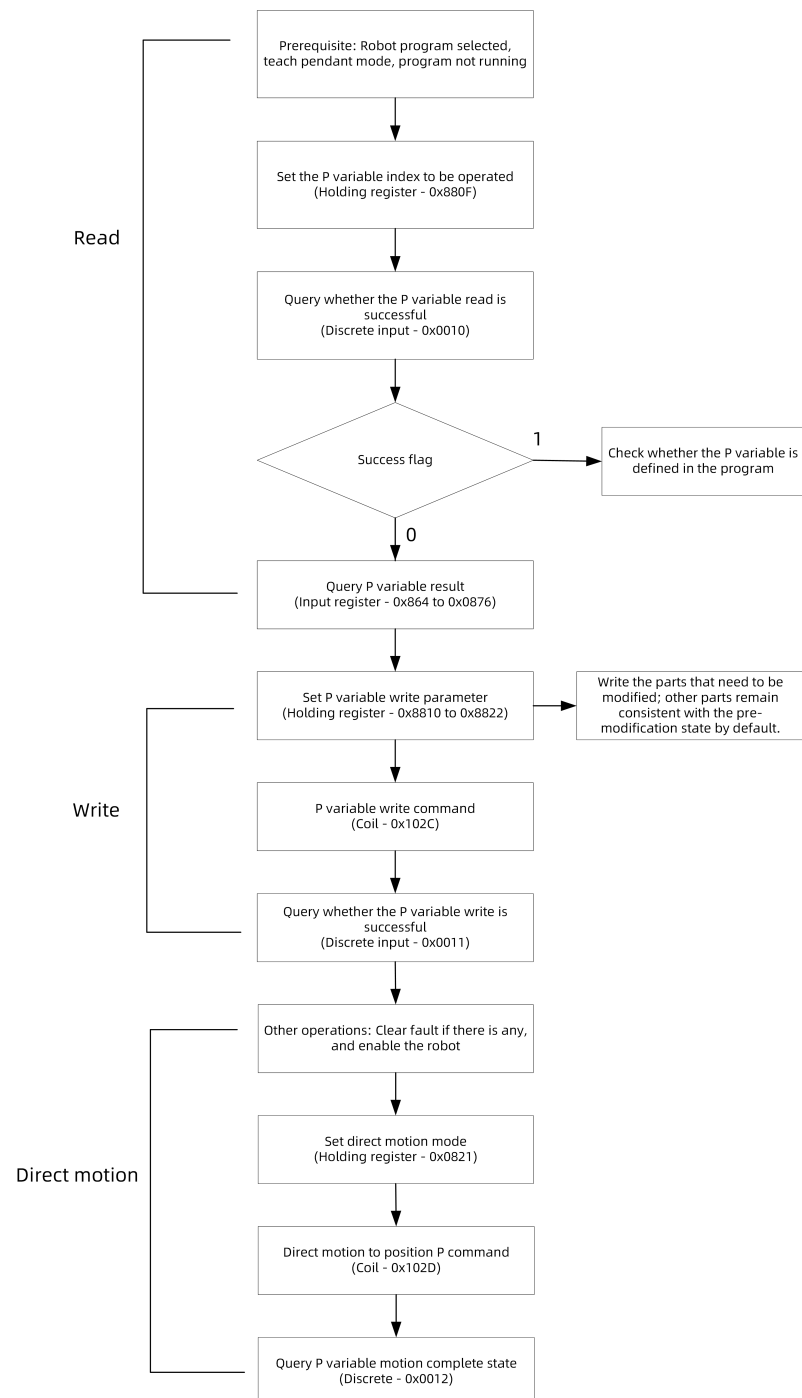


2.4.3.2 Controlling Robot Point Teaching via Master

Case description

- Implement the display and modification of P variables.
- Implement the control of the robot to move to a specified point (the point corresponding to the P variable).

Operation process

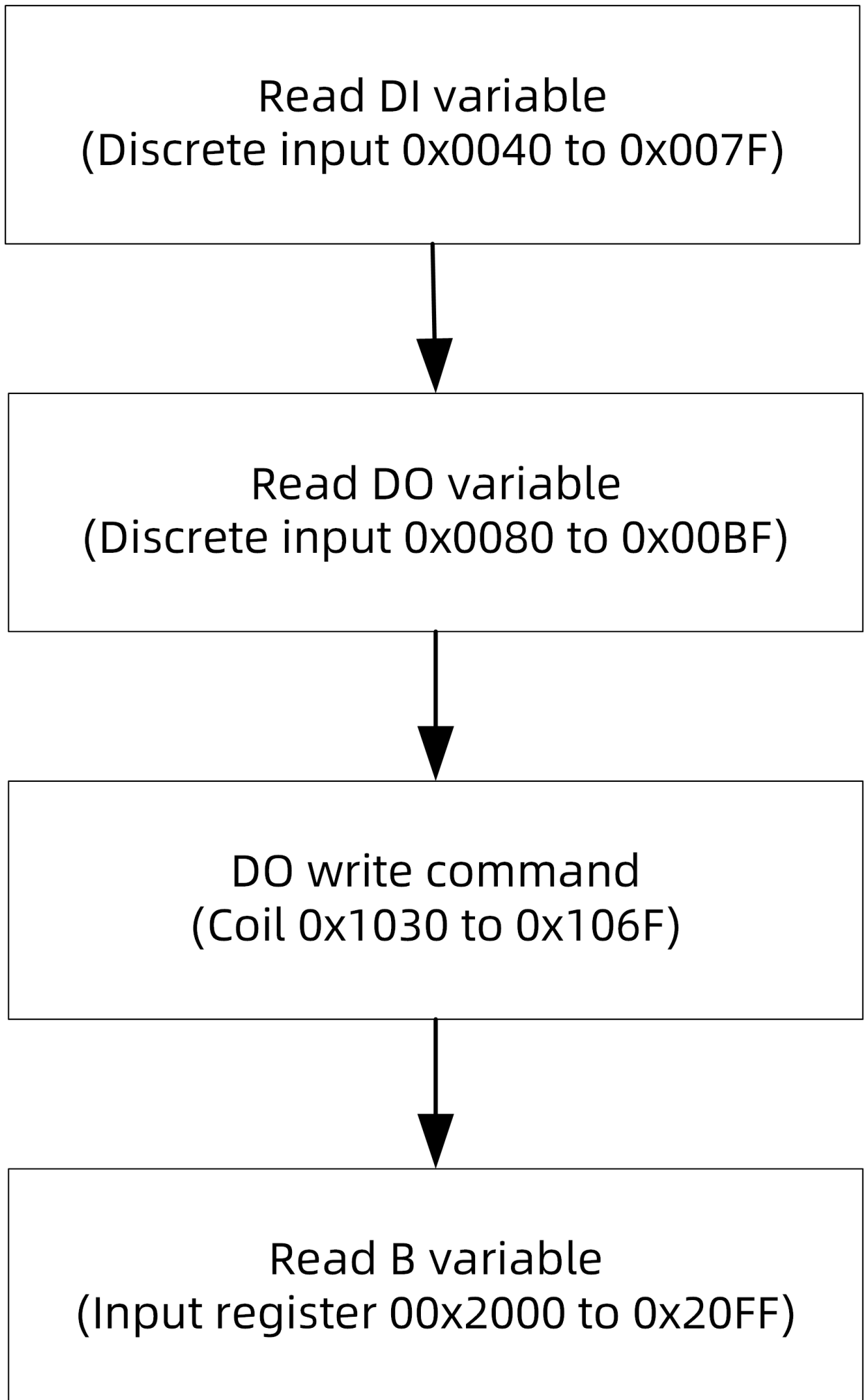


2.4.3.3 Reading/Writing Robot I/O Variables and B Variables via Master

Case description

- Implement reading and writing of robot I/O variables.
- Implement reading and writing of robot B variables.

Operation process

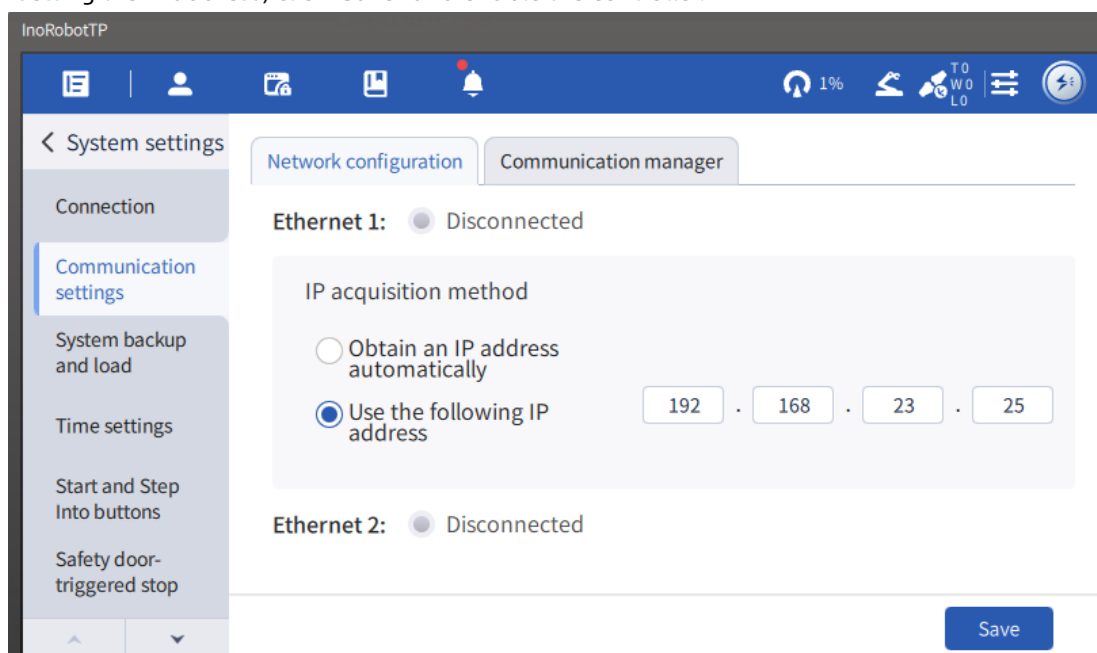


2.5 Comprehensive Case

In this case study, the host controller uses an Inovance HMI to control the robot's motion via Modbus communication.

2.5.1 Controller Network Port Configuration

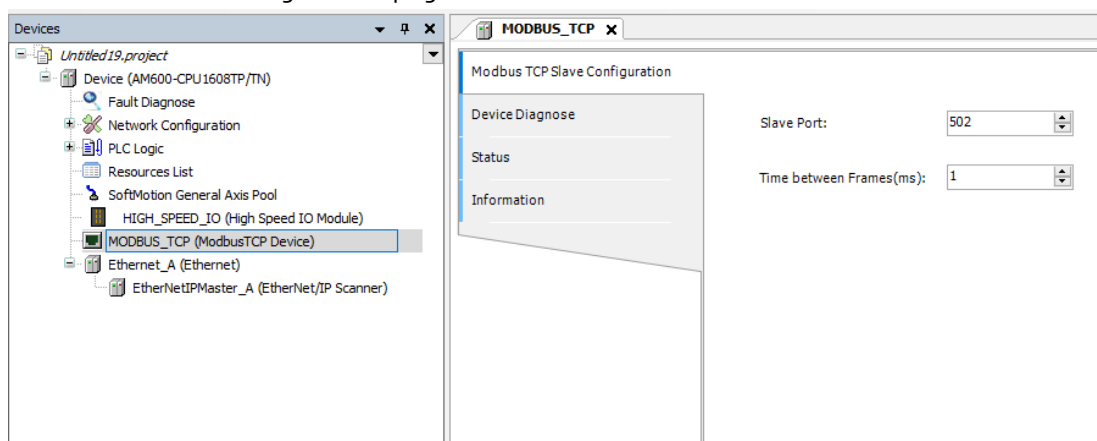
1. Log in to the InoRobotTP software to enter the main interface. Then, click "Settings" > "System settings" > "Communication settings" > "Network configuration".
2. On the "Network configuration" interface, set the IP address to 192.168.23.25.
3. After setting the IP address, click "Save" and enable the controller.



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Slave configuration

1. In the left "Devices" pane of InoProShop, double-click "Modbus_TCP(ModbusTcp_Serial_Device)" to enter the "Modbus TCP Slave Configuration" page.



2. Set the "Slave Port" and "Time between Frames(ms)".

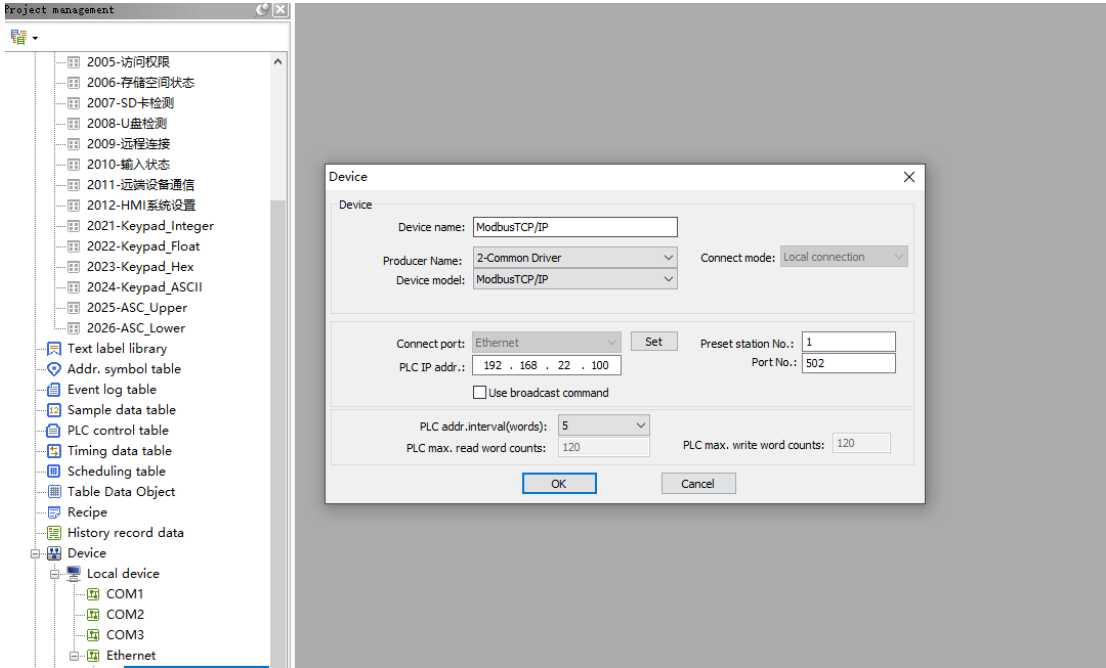
Slave Port:

502

Time between Frames(ms):

1

3. Open the InoTouch Editor (HMI) software. In the left "Project management" pane, expand "Device" > "Local device", right-click "Ethernet" and select "Add Device" to add ModbusTCP/IP, as shown in the figure below.



2.5.2 HMI Function Item Setup

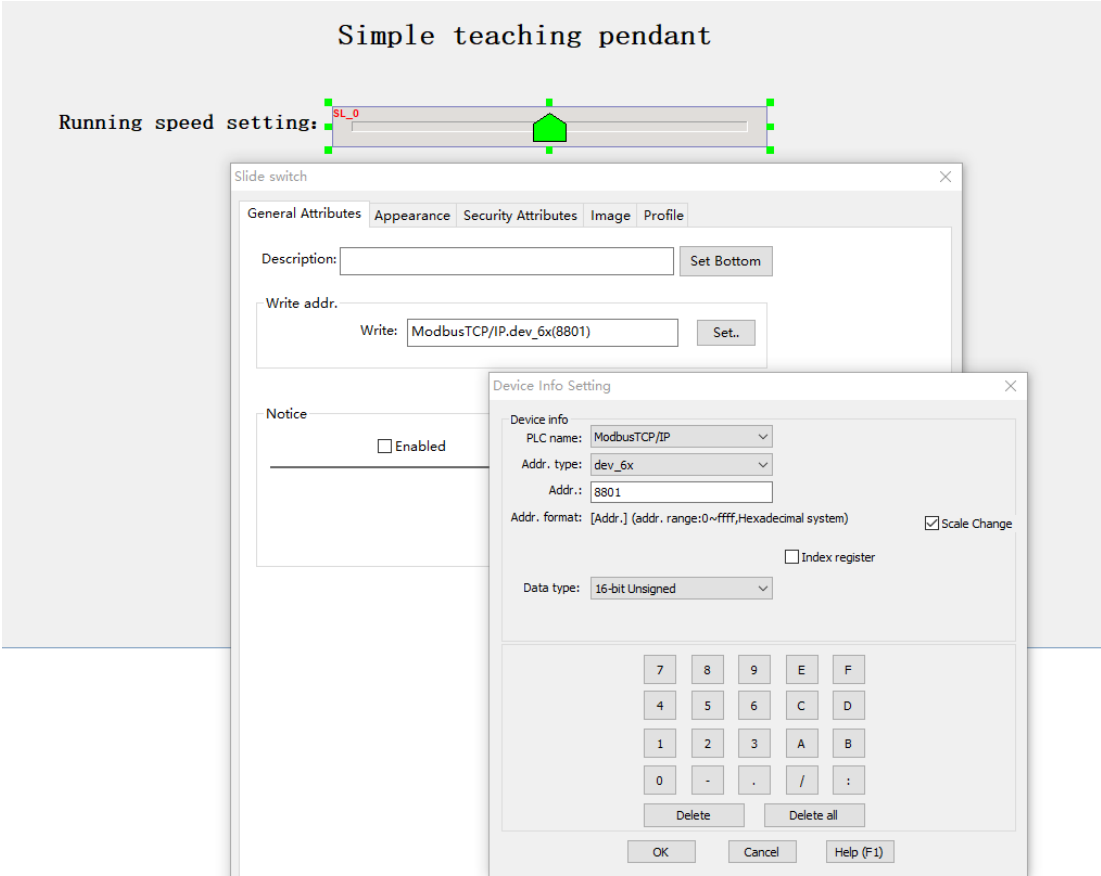
Modbus slave address table:

Address (Decimal)	Address (Hexadecimal)	Variable name	Data type	Content	Note
4112	0x1010	QW65281, bit0	Bit	Teach J1/X+	Reset type
4113	0x1011	QW65281, bit1	Bit	Teach J2/Y+	Reset type
4114	0x1012	QW65281, bit2	Bit	Teach J3/Z+	Reset type
4115	0x1013	QW65281, bit3	Bit	Teach J4/A+	Reset type
4120	0x1018	QW65281, bit8	Bit	Teach J1/X-	Reset type
41210	0x1019	QW65281, bit9	Bit	Teach J2/Y-	Reset type

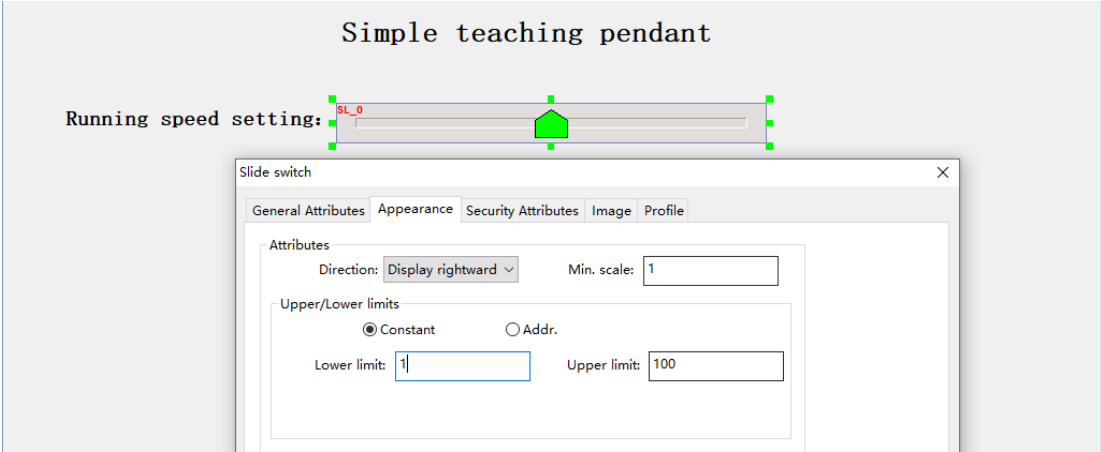
4122	0x101a	QW65281, bit10	Bit	Teach J3/Z-	Reset type
4123	0x101b	QW65281, bit11	Bit	Teach J4/A-	Reset type
34816	0x8800	MW34816	Word	Frame selection: 1: Joint 2: Cartesian 3: Tool 4: User	-
4099	0x1003	QW65280, bit3	Bit	Enable switch: 0: OFF 1: ON	-
34817	0x8801	MW34817	Word	Speed setting (1 to 100)	-
2049	0x0801	MW2049	Word	Current speed	-
2048	0x0800	MW2048	Word	Current frame	-
2052	0x0804	MW2052	Single-precision floating-point	Lower bits of X coordinate of J1	-
2053	0x0805	MW2053	Single-precision floating-point	Higher bits of X coordinate of J1	-
2054	0x0806	MW2054	Single-precision floating-point	Lower bits of Y coordinate of J2	-
2055	0x0807	MW2055	Single-precision floating-point	Higher bits of Y coordinate of J2	-
2056	0x0808	MW2056	Single-precision floating-point	Lower bits of Z coordinate of J3	-
2057	0x0809	MW2057	Single-precision floating-point	Higher bits of Z coordinate of J3	-
2058	0x080a	MW2058	Single-precision floating-point	Lower bits of A coordinate of J4	-
2059	0x080b	MW2059	Single-precision floating-point	Higher bits of A coordinate of J4	-

According to the above address table, set the control functions as follows:

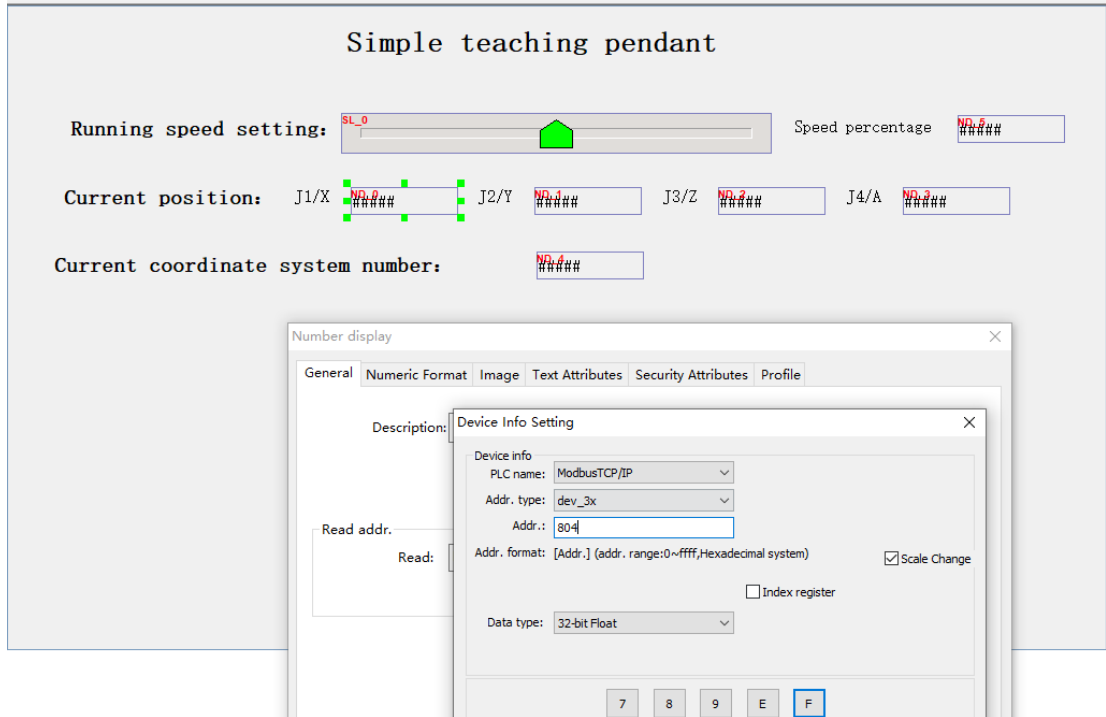
1. Set the control keys of the running speed setting function: Add a slide switch control and write the address 0x8801 (in hexadecimal format).



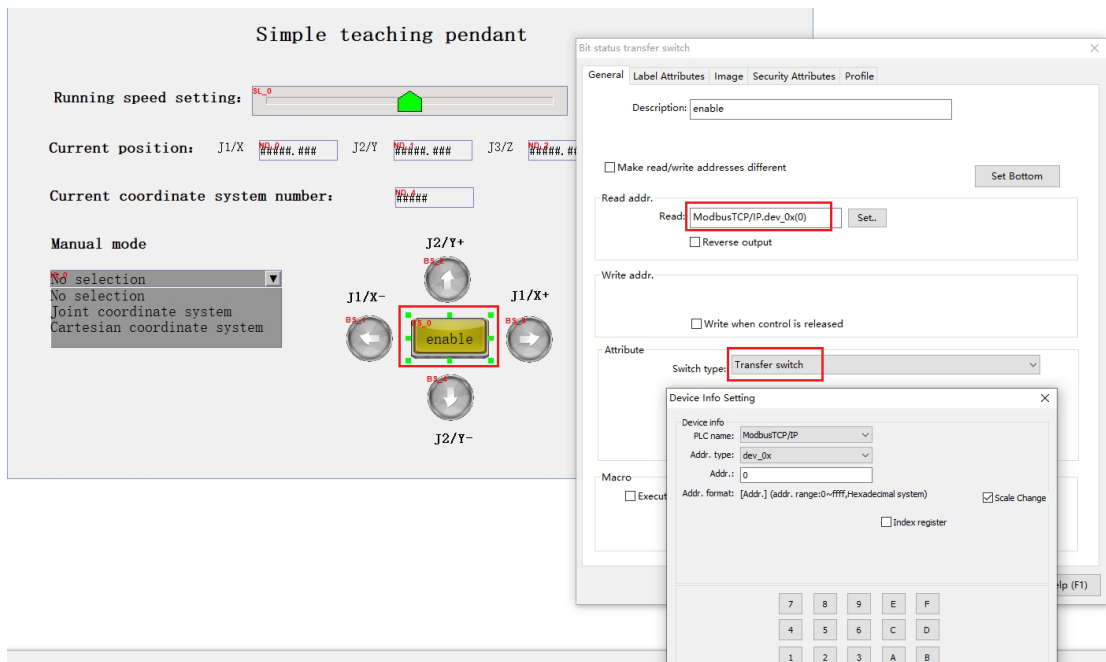
2. Set the lower and upper limits of the control to 1 to 100.



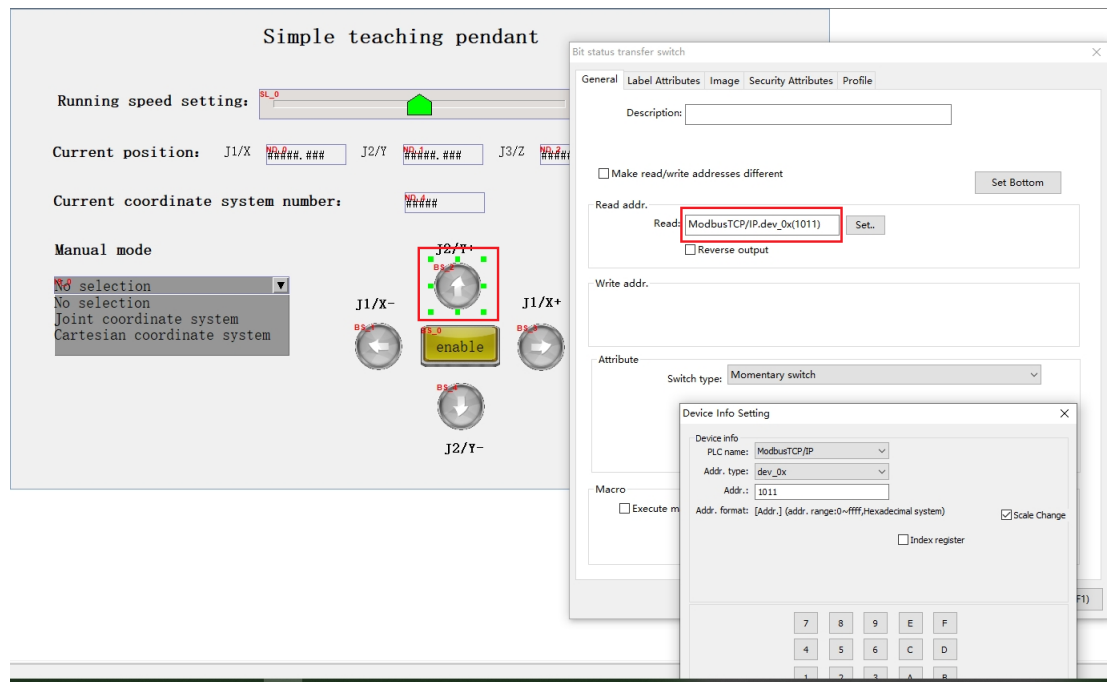
3. Set the monitoring window of current position (each axis orientation): Add a numerical display control, write the address (in hexadecimal format). For example, the address for the J1 axis is 0x0804, and the data type is 32-bit floating point. Set the other axes similarly. For "Speed percentage" and "Current coordinate system number", simply select the number display control. Set the "Data type" and "Value range" according to the specifications in the address table.



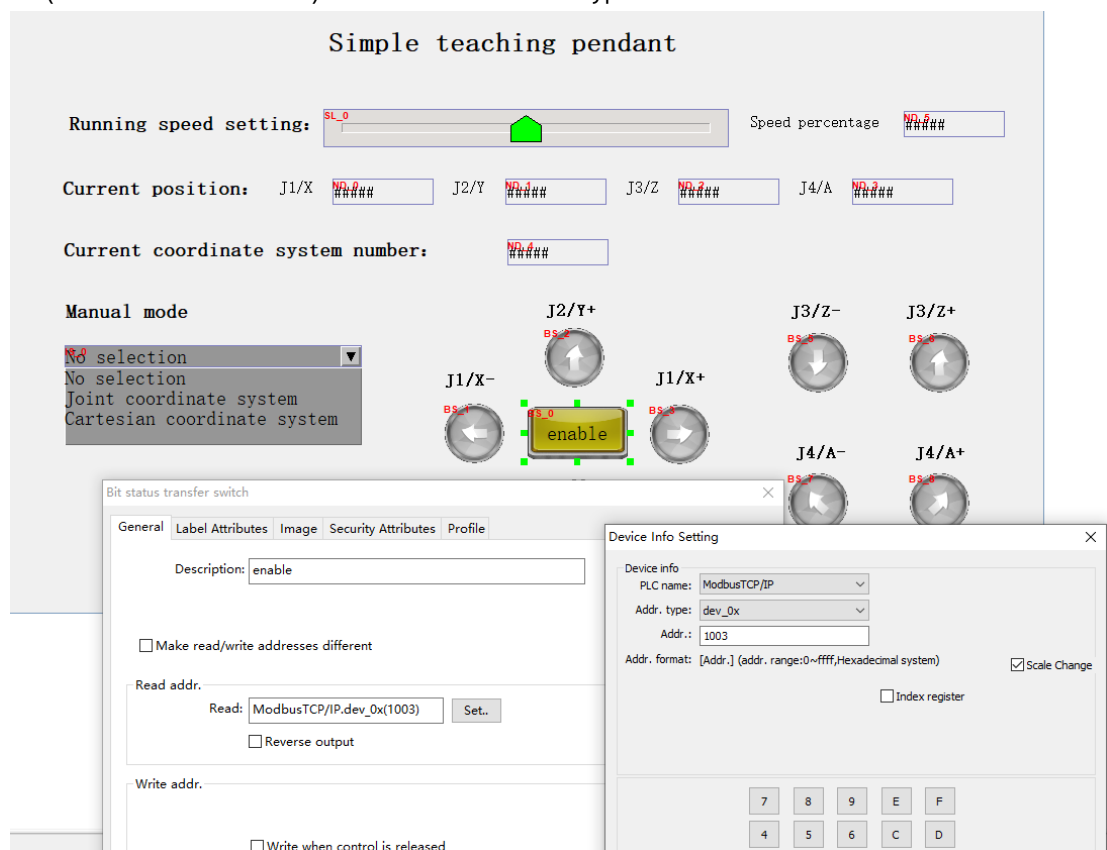
- Set function keys for coordinate system setting: Add a project menu control, input the write address 0x8800 (in hexadecimal format). Set the number of states according to usage requirements. In the example, only "Joint coordinate system" and "Cartesian coordinate system" are used, so the number of states is 3.



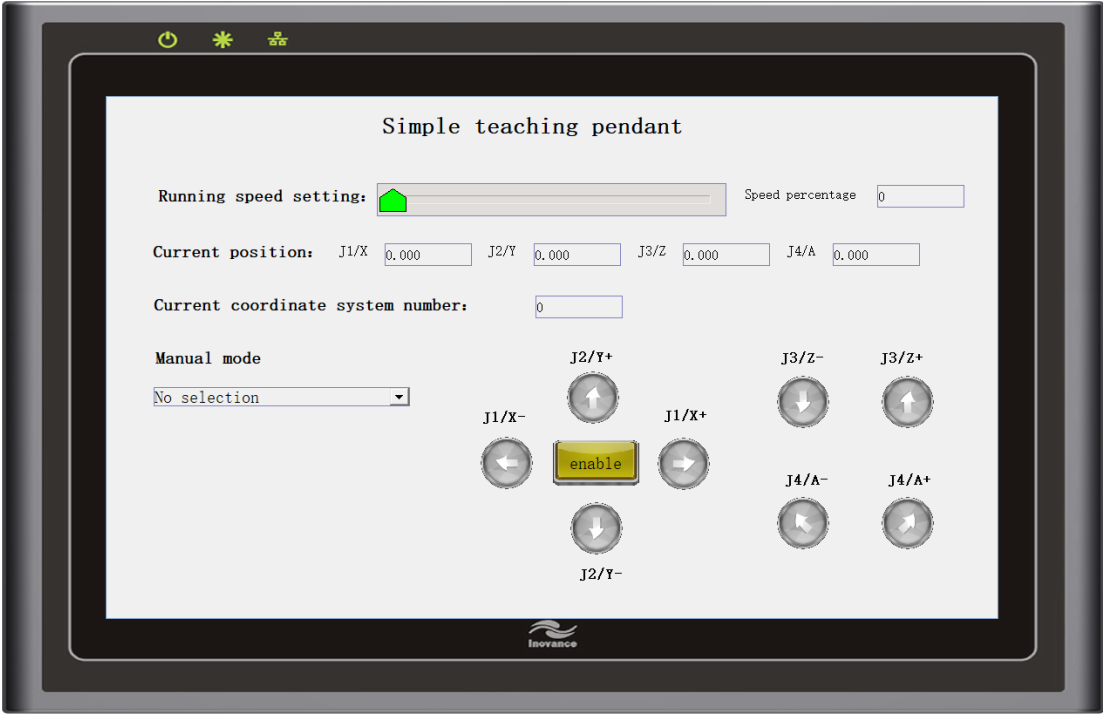
- Set function keys for axis operation: Add a bit status transfer switch control, input the write address (in hexadecimal format). For example, the J2/Y+ operation key address is 0x1011. Set the control switch type to "Momentary type". Set other axes similarly.



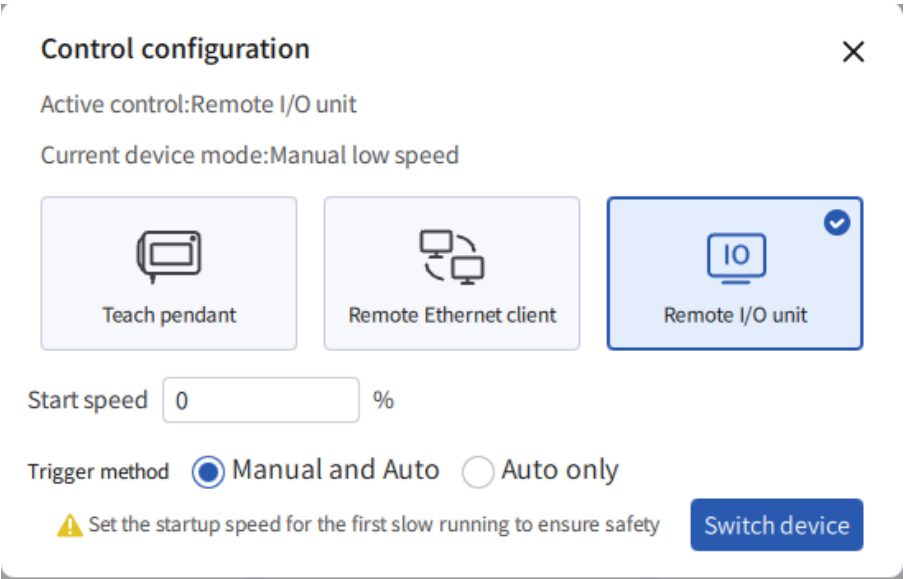
6. Set function key for enable operation: Add a bit status transfer switch control, input the write address 0x1003 (in hexadecimal format). Set the control switch type to "Transfer switch".



7. After completing the operations in the above case, the HMI simple teach pendant operation interface has been created, as shown in the figure below.



8. After the network connection between the robot controller and the HMI is normal, set the robot control permission to remote Modbus unit. On the teach pendant operation interface, click the "Control device" button, and select "Remote I/O unit" as the control permission on the pop-up interface, as shown in the figure below.



3 MC Bus Communication

3.1 Communication Introduction

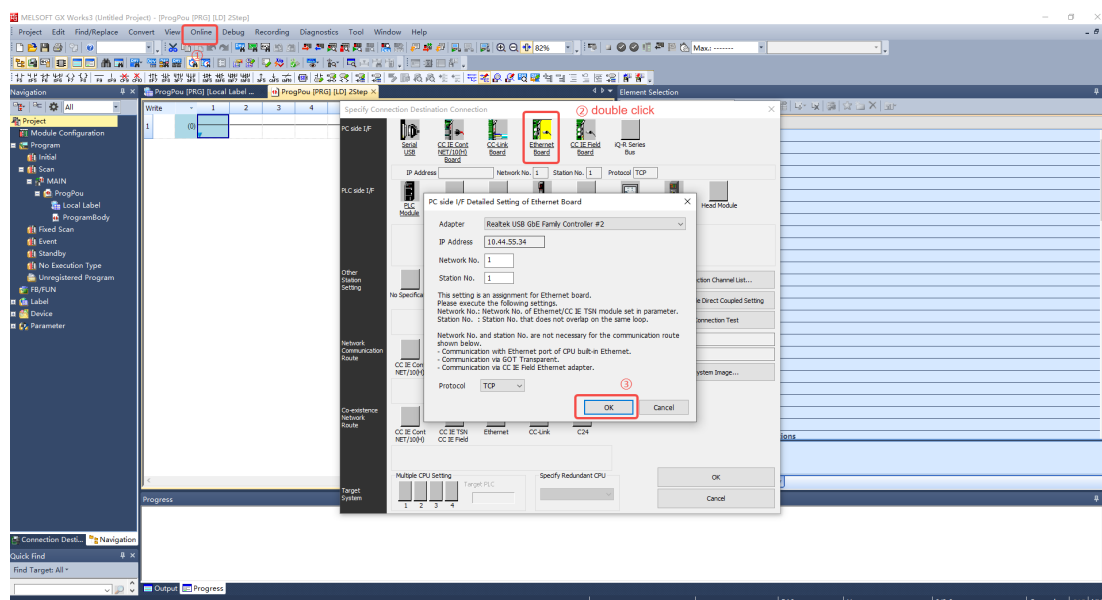
MC communication is a dedicated real-time bus protocol for the motion control field. It adopts a hybrid architecture of high-speed Ethernet physical layer with Time-Sensitive Networking (TSN), featuring microsecond-level synchronization accuracy, optimized multi-axis coordinated control, strong deterministic topology, and comprehensive safety and redundancy mechanisms. Through hardware clock synchronization and deterministic data transmission, it can meet the precise timing requirements of high-dynamic scenarios such as CNC machine tools and robots.

3.2 Communication Between Mitsubishi R Series PLC and Industrial Robot

3.2.1 MELSOFT GX Works3 Software-related Configuration

3.2.1.1 Connection Destination Specification Settings

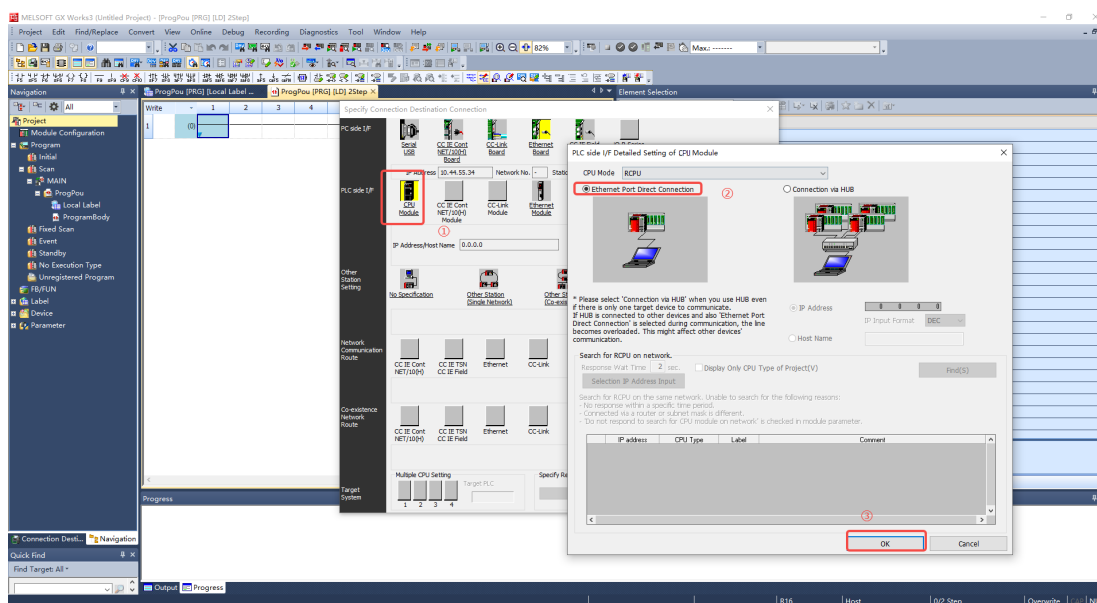
1. Open the MELSOFT GX Works3 software to enter the main software interface, and click "Online" on the menu bar to enter the "Specify Connection Destination Connection" interface.
2. On the "Specify Connection Destination Connection" interface, double-click to activate the "Ethernet Board".
3. Click "OK" to complete the PC side I/F detailed settings of Ethernet board.



3.2.1.2 CPU Module Activation

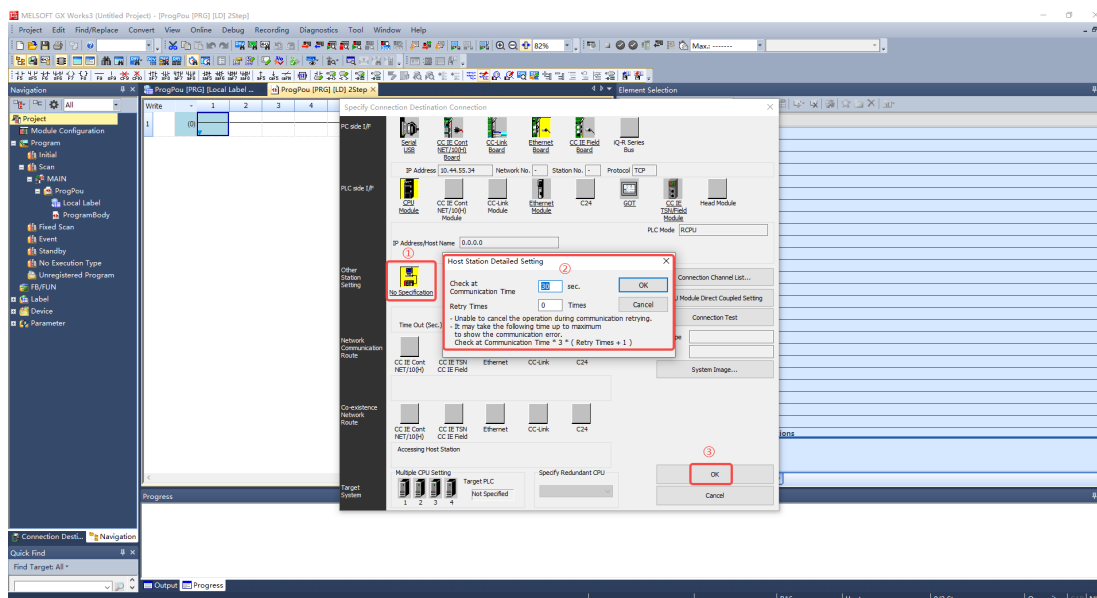
1. Open the MELSOFT GX Works3 software to enter the software main interface, and click "Online" on the menu bar to enter the "Specify Connection Destination Connection" interface.
2. On the "Specify Connection Destination Connection" interface, double-click "CPU Module" to enter the "PLC side I/F Detailed Settings of CPU Module" interface.

3. On the "PLC side I/F Detailed Settings of CPU Module" interface, click to select "Ethernet Port Direct Connection".
4. Click "OK" to complete the Ethernet port direct connection.



3.2.1.3 Other Station Specification

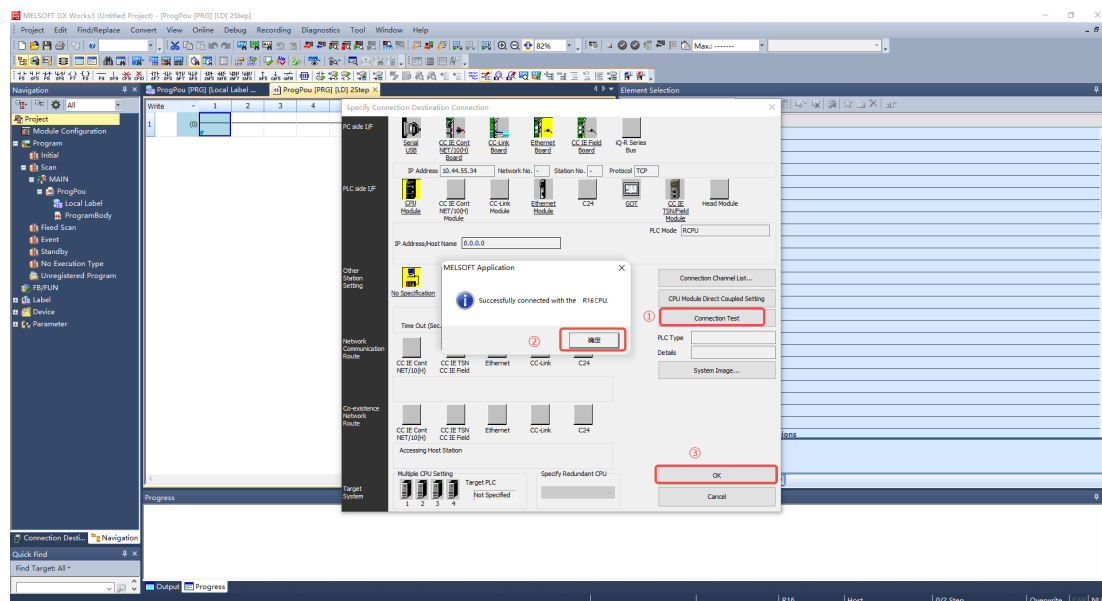
1. Open the MELSOFT GX Works3 software to enter the main software interface, click "Online" on the menu bar to access the "Specify Connection Destination Connection" interface, and double-click "No Specification" to enter the "Host Station Detailed Setting" interface.
2. On the "Host Station Detailed Setting" interface, set the "Check at Communication Time" to 3s.
3. Click "OK" to complete the local station settings.



3.2.1.4 Communication Test

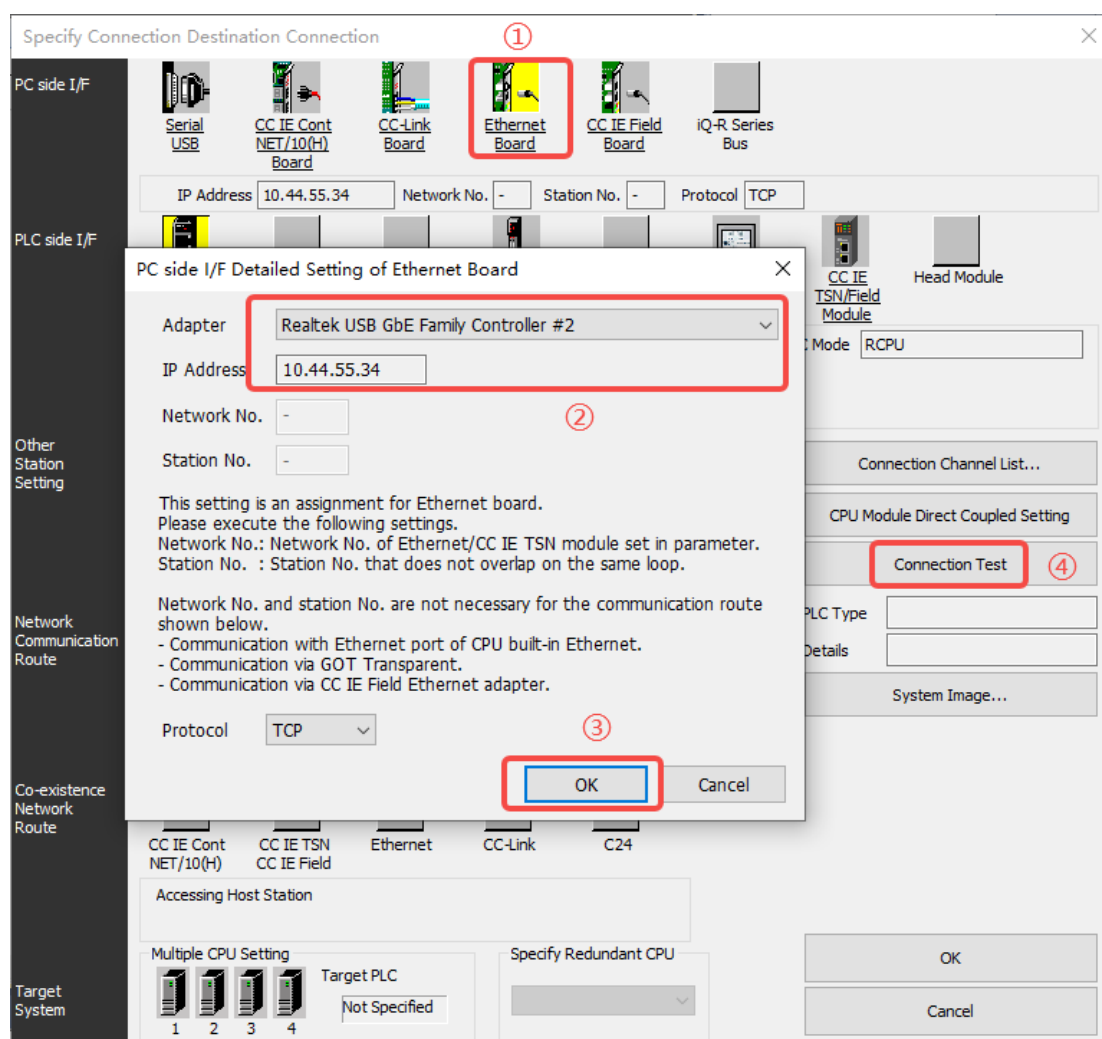
1. In the left device tree on the main interface of the MELSOFT GX Works3 software, click "Parameter" > "R16CPU" > "Module Parameter" to enter the "Specify Connection Destination Connection" interface.
2. On the "Specify Connection Destination Connection" interface, click "Communication Test". A prompt box will appear, displaying "Successfully connected with the R16CPU".

3. Click "OK" to complete the communication test.



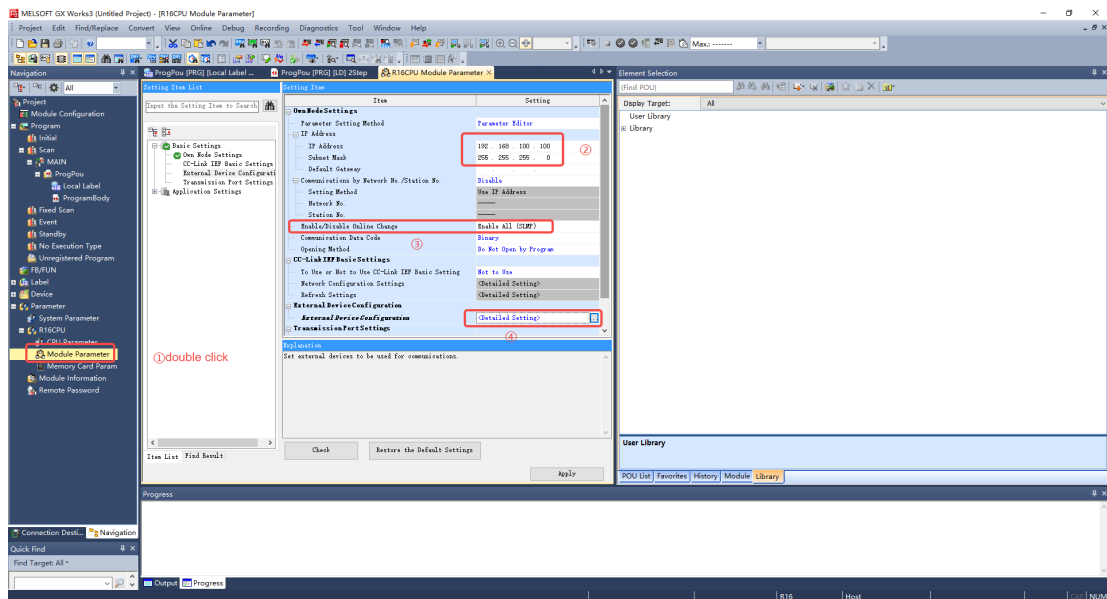
If the communication test fails to connect, configure according to the figure below.

1. On the "Specify Connection Destination Connection" interface, click "Ethernet Board" to enter the "PC side I/F Detailed Setting of Ethernet Board" interface.
2. On the "PC side I/F Detailed Setting of Ethernet Board" interface, select "PLC IP" from the adapter drop-down list, and then click "OK".
3. Click "Communication Test" to complete the communication test.



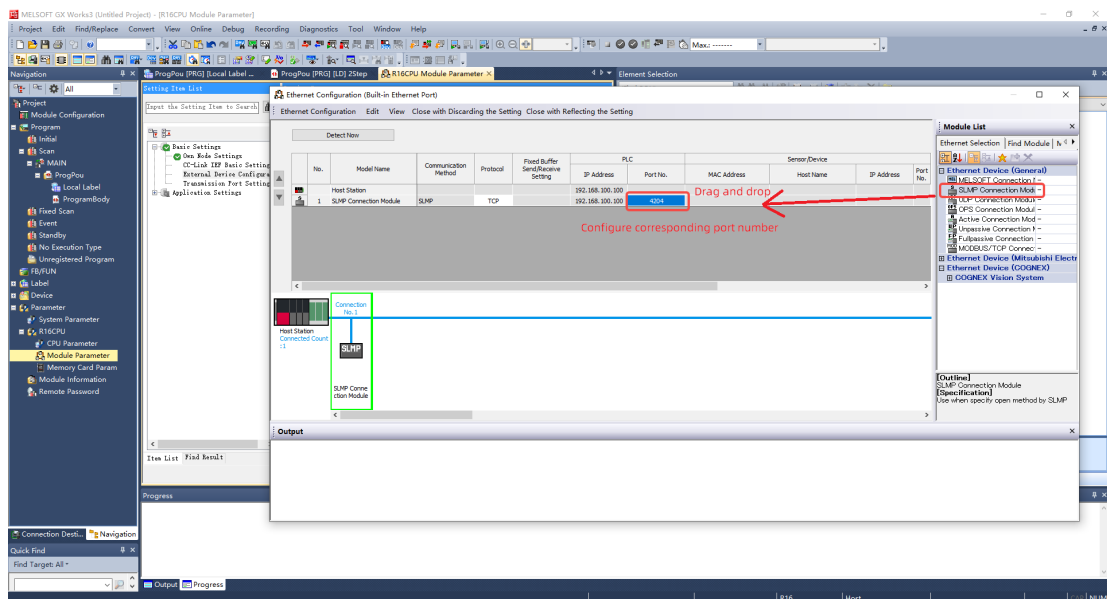
3.2.1.5 PLC Module Parameter Settings

1. In the left device tree on the main interface of the MELSOFT GX Works3 software, double-click "Module Parameter" to enter the "R16CPU Module Parameter" interface.
2. On the "Setting Item" interface, set the "IP Address" and "Subnet Mask".
3. Select "Enable All (SLMP)" for "Enable/Disable Online Change".
4. In the "External Device Configuration" item, double-click to view "Detailed Settings".
5. Click "Apply" to complete the PLC module parameter settings.



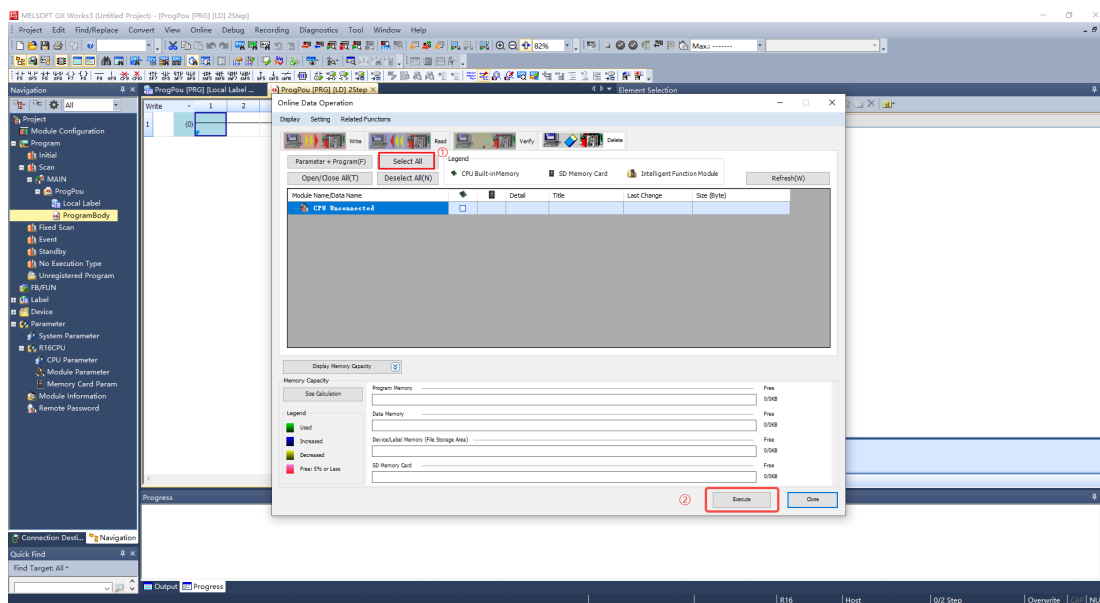
3.2.1.6 Ethernet Settings

1. In the "Module List" > "Ethernet Device (General)" selection box on the left side, drag and drop the "SLMP Connection Module" to the "Ethernet Configuration (Built-in Ethernet Port)" interface.
2. On the "Ethernet Configuration (Built-in Ethernet Port)" interface, select "TCP" for the protocol, set the corresponding port number and press Enter, then save and exit.



3.2.1.7 Configuration Parameter Download

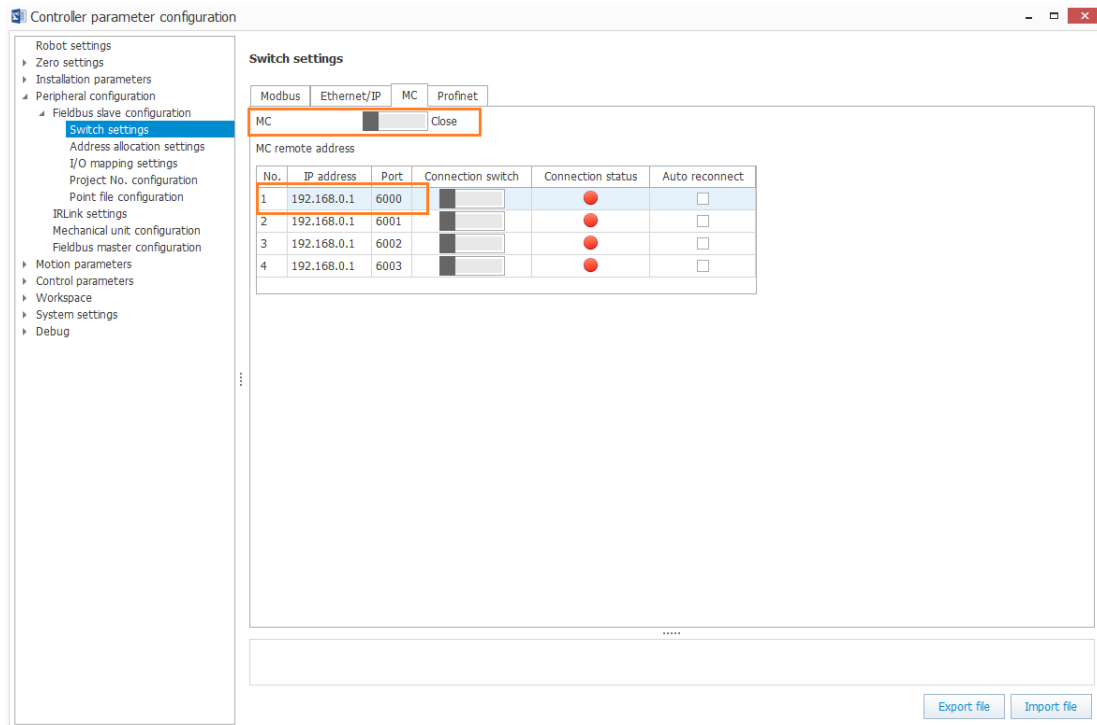
1. Open the MELSOFT GX Works3 software to enter the main software interface, and click "Online" on the menu bar to enter the "Online Data Operation" interface.
2. On the "Online Data Operation" interface, click "Select All".
3. Click "Execute" to complete the parameter configuration.
4. After the download is complete, power off and restart the PLC. Once the restart is finished, communication with the robot can be established.



3.2.2 Robot Interface-related Settings

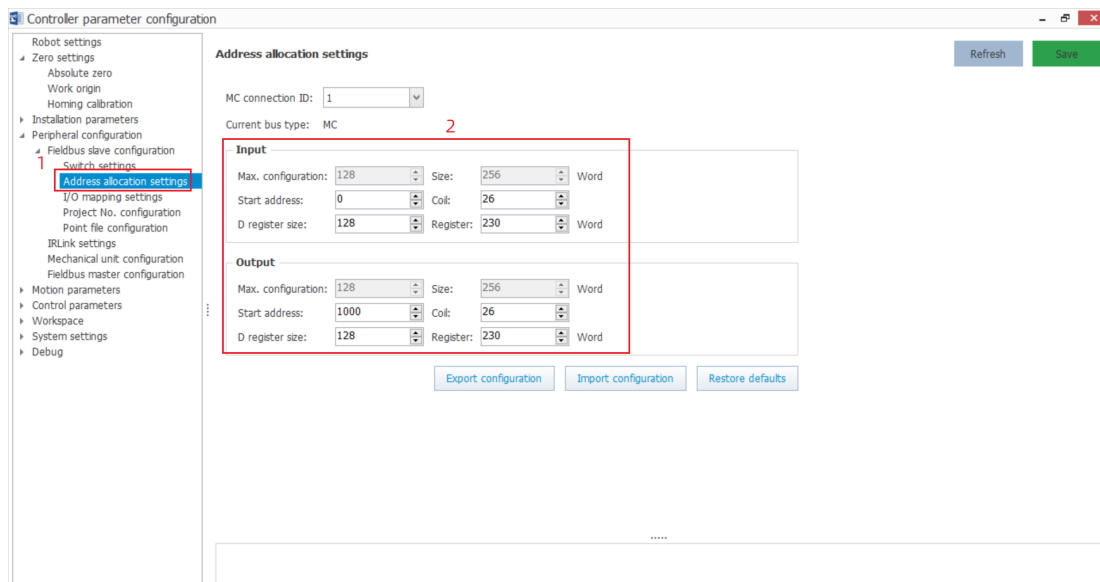
3.2.2.1 Robot MC Protocol Settings

1. On the main interface of the InoRobotLab software, double-click "Controller parameter configuration" in the left directory tree to enter the "Controller parameter configuration" interface.
2. On the "Controller parameter configuration" interface, click "Peripheral configuration" > "Fieldbus slave configuration" > "Switch settings" > "MC".
3. Toggle the switch to activate MC, and fill in the PLC's "IP address" and "Port" number.



3.2.2.2 Address Allocation Settings

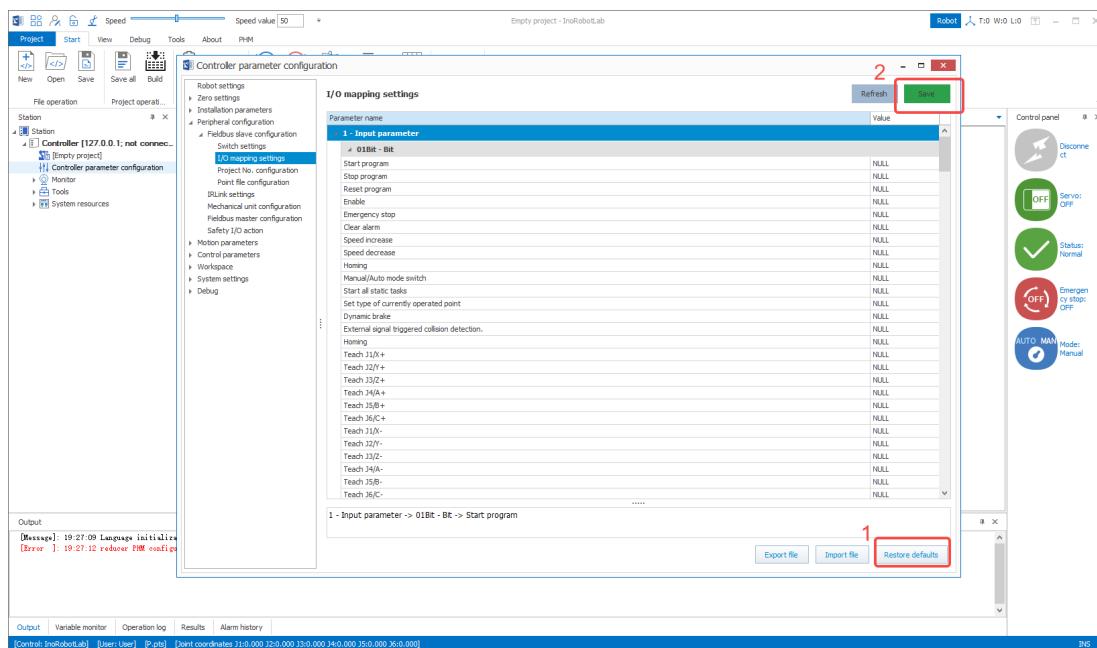
1. On the main interface of the InoRobotLab software, double-click "Controller parameter configuration" in the left directory tree to enter the "Controller parameter configuration" interface.
2. Click "Peripheral configuration" > "Fieldbus slave configuration" > "Address allocation settings".
3. Fill in the address for interaction with the PLC on the "Address allocation settings" interface, then click "Save" in the upper right corner to complete the address allocation settings.



- Set the MC connection identifier according to the configuration index.
- Set the input and output start addresses according to the PLC registers.

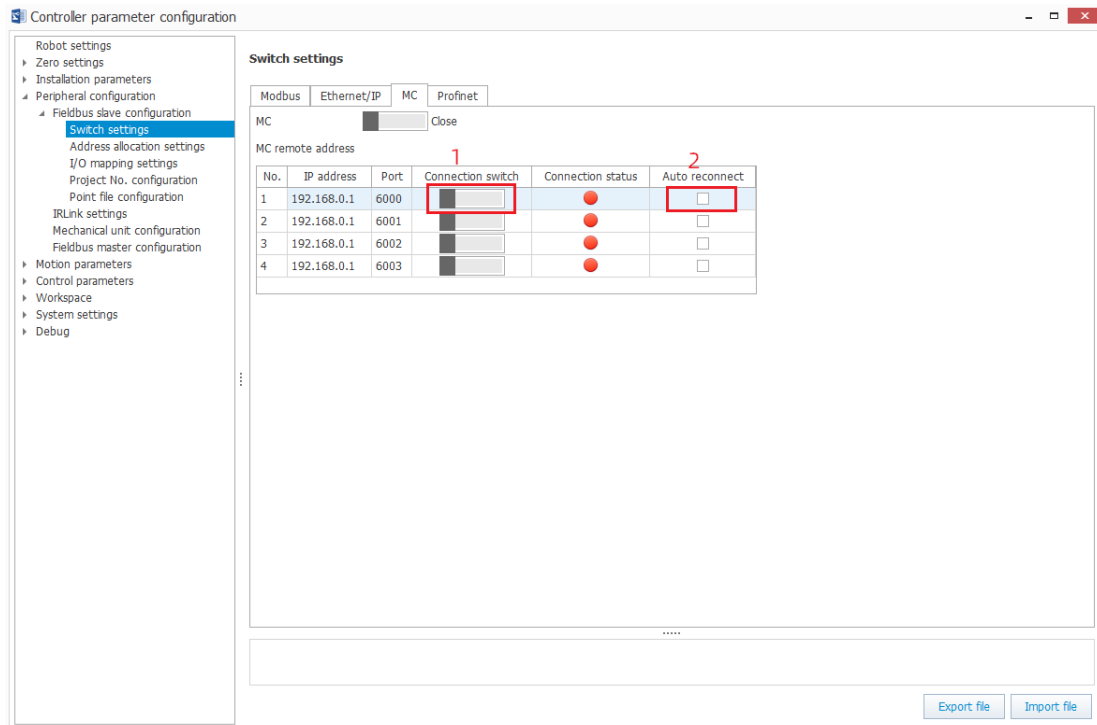
3.2.2.3 I/O Mapping Settings

1. On the main interface of the InoRobotLab software, double-click "Controller parameter configuration" in the left directory tree to enter the "Controller parameter configuration" interface.
2. On the "Controller parameter configuration" interface, click "Peripheral configuration" > "Fieldbus slave configuration" > "I/O mapping settings".
3. On the "I/O mapping settings" interface, click "Restore defaults" to use the default allocation, then click "Save" in the upper right corner to complete the I/O mapping setting.



3.2.2.4 Establishing Connection with PLC

1. After the PLC configuration is complete and restarted, on the main interface of the InoRobotLab software, double-click "Controller parameter configuration" in the left directory tree to enter the "Controller parameter configuration" interface.
2. On the "Controller parameter configuration" interface, click "Peripheral configuration" > "Fieldbus slave configuration" > "Switch settings" > "MC".
3. On the "MC" interface, toggle the MC switch to activate the "Connection switch", and check the "Auto reconnect" option. The connection status indicator will then turn green.

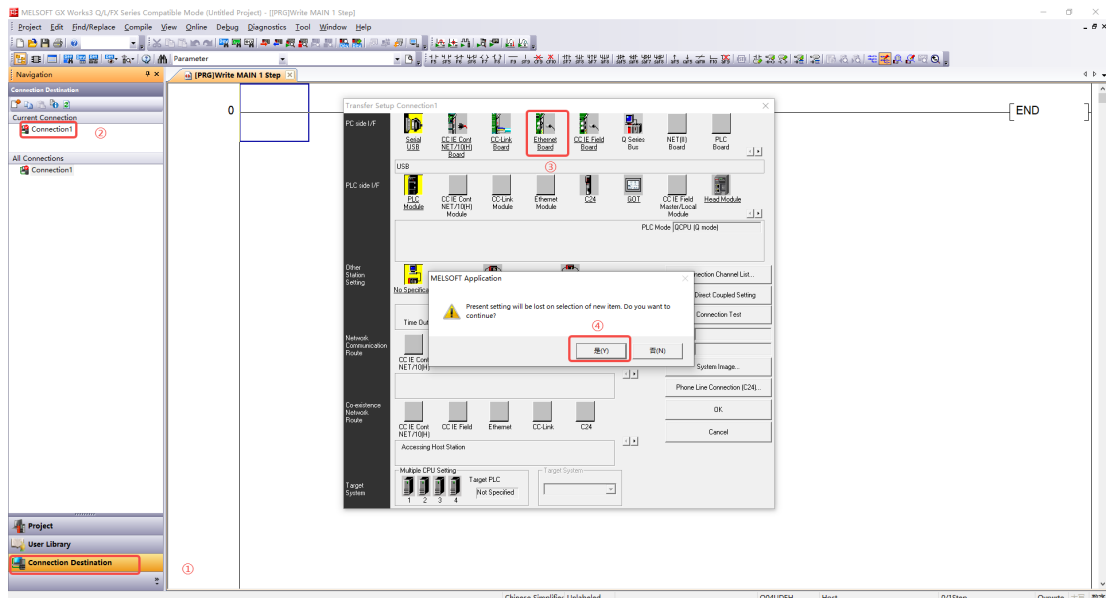


3.3 Communication Between Mitsubishi Q Series PLC and Industrial Robot

3.3.1 MELSOFT GX Works3 Software-related Configuration

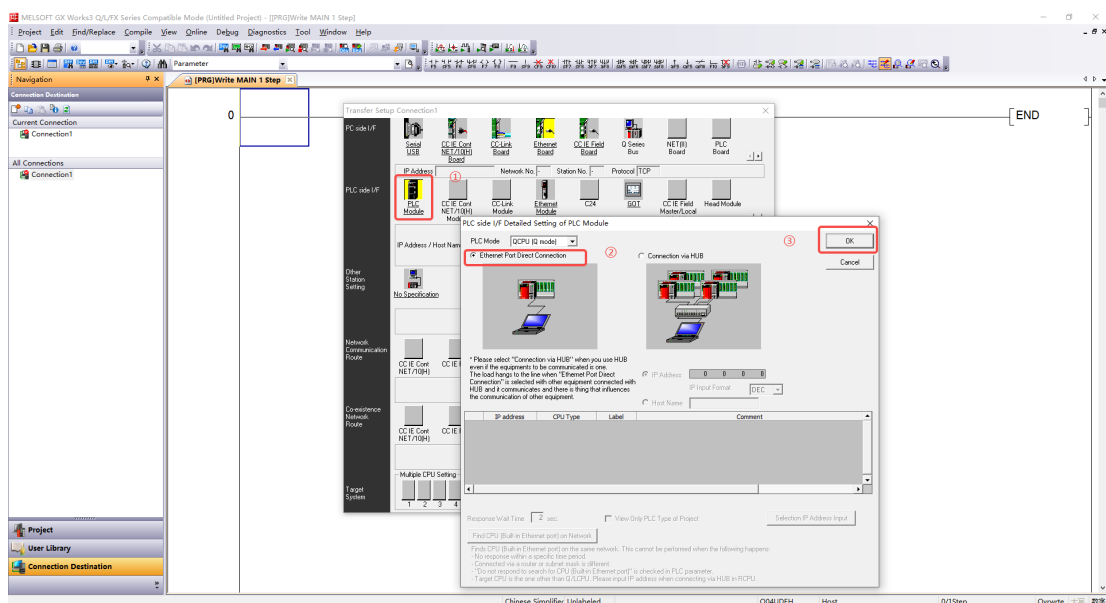
3.3.1.1 Connection Destination Specification Settings

1. In the left device tree on the main interface of the MELSOFT GX Works3 software, click "Connection Destination" to enter the "Connection Destination" interface.
2. On the "Connection Destination" interface, double-click "Connection1" under "Current Connection" to enter the "Transfer Setup Connection1" interface.
3. On the "Transfer Setup Connection1" interface, double-click to activate "Ethernet Board", which will pop up the "MELSOFT Application" prompt box.
4. In the "MELSOFT Application" prompt box, click "Yes" to complete the connection destination specification settings.



3.3.1.2 PLC Module Activation

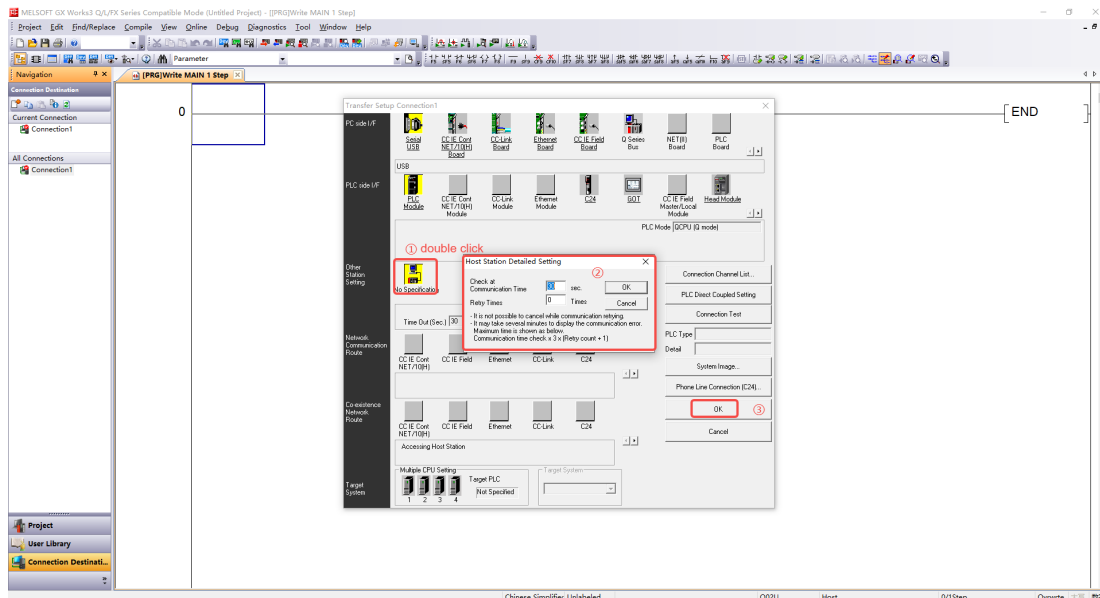
1. In the left device tree on the main interface of the MELSOFT GX Works3 software, click "Connection Destination" to enter the "Connection Destination" interface.
2. On the "Connection Destination" interface, double-click "Connection1" under "Current Connection" to enter the "Transfer Setup Connection1" interface.
3. On the "Transfer Setup Connection1" interface, double-click to activate "PLC Module" to enter the "PLC side I/F Detailed Setting of PLC Module" interface.
4. On the "PLC side I/F Detailed Setting of PLC Module" interface, click to select "Ethernet Port Direct Connection".
5. Click "OK" to complete the PLC Modbus activation.



3.3.1.3 Other Station Specification

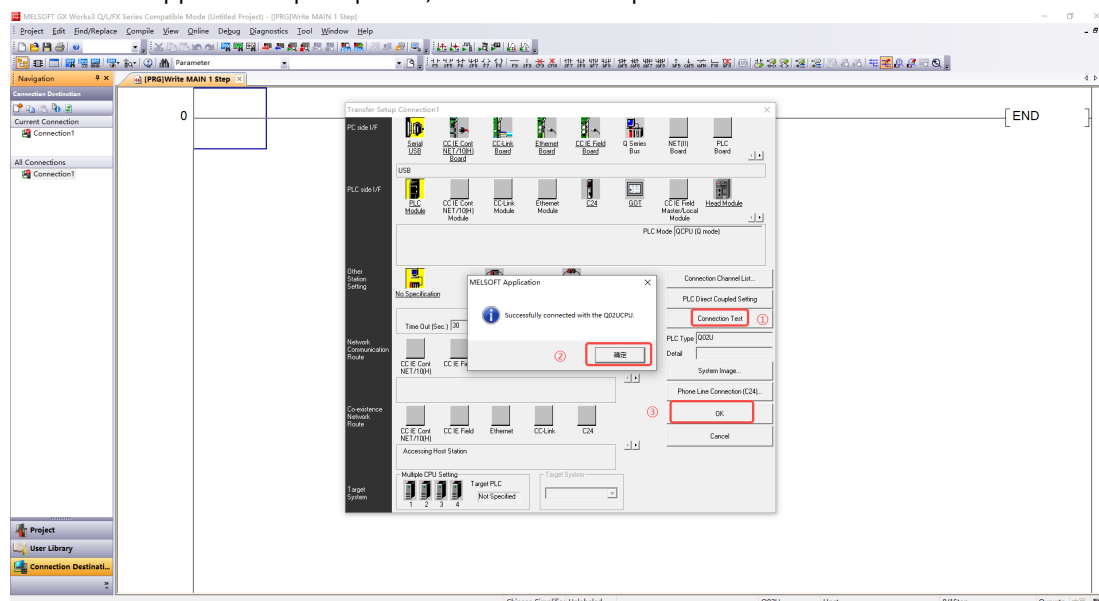
1. In the left device tree on the main interface of the MELSOFT GX Works3 software, click "Connection Destination" to enter the "Connection Destination" interface.

2. On the "Connection Destination" interface, double-click "Connection1" under "Current Connection" to enter the "Transfer Setup Connection1" interface.
3. On the "Transfer Setup Connection1" interface, double-click to activate "No Specification" and then enter the "Host Station Detailed Setting" interface.
4. On the "Host Station Detailed Setting" interface, set the "Check at Communication Time" to 3s.
5. Click "OK" to complete the station specification settings.



3.3.1.4 Communication Test

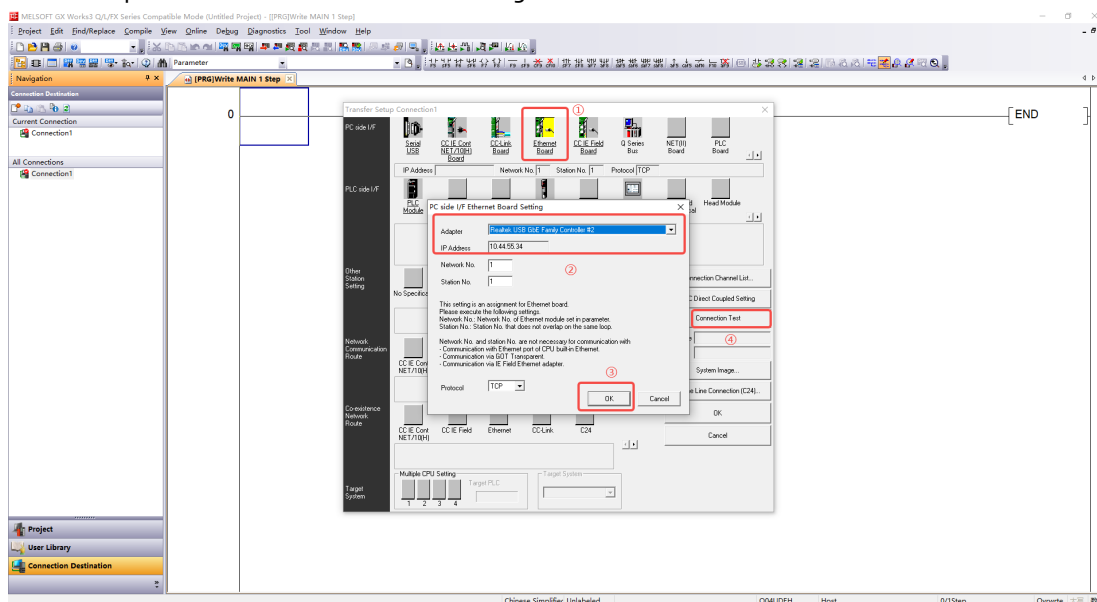
1. In the left device tree on the main interface of the MELSOFT GX Works3 software, click "Connection Destination" to enter the "Connection Destination" interface.
2. On the "Connection Destination" interface, double-click "Connection1" under "Current Connection" to enter the "Transfer Setup Connection1" interface.
3. On the "Transfer Setup Connection1" interface, click "Communication Test", which will pop up the "MELSOFT Application" prompt box.
4. In the "MELSOFT Application" prompt box, click "OK" to complete the communication test.





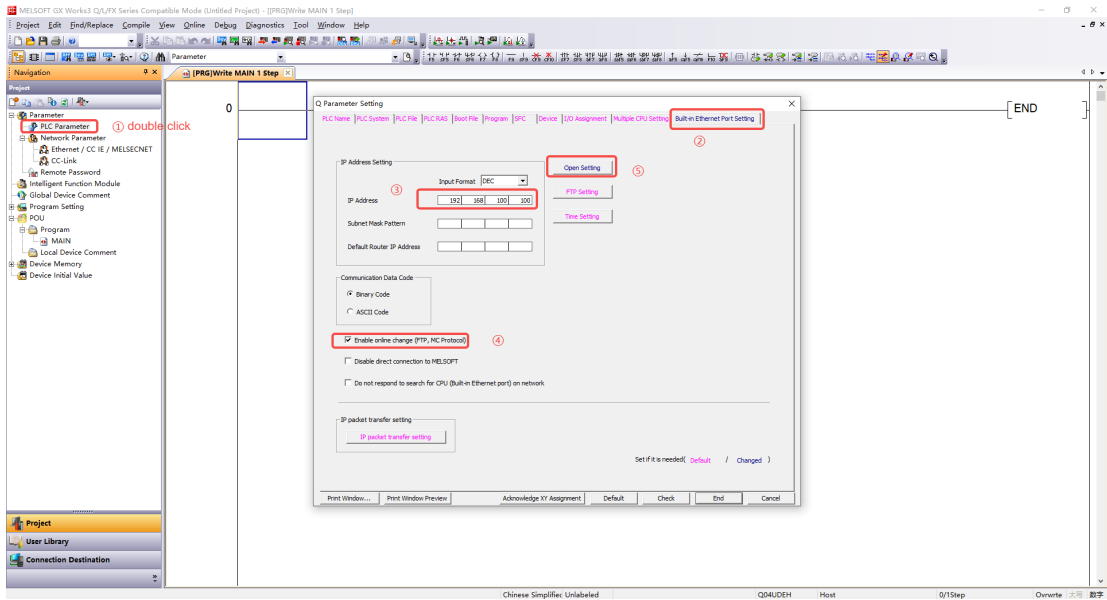
If the communication test is unsuccessful, select the PLC IP address corresponding to the adapter and perform the communication test connection again.

5. On the "Transfer Setup Connection1" interface, click "Ethernet Board" to enter the "PC side I/F Ethernet Board Setting" interface.
6. On the "PC side I/F Ethernet Board Setting" interface, select "PLC IP" from the "Adapter" drop-down list.
7. Click "OK" and perform the communication test again.



3.3.1.5 Mitsubishi Q Series Communication Settings

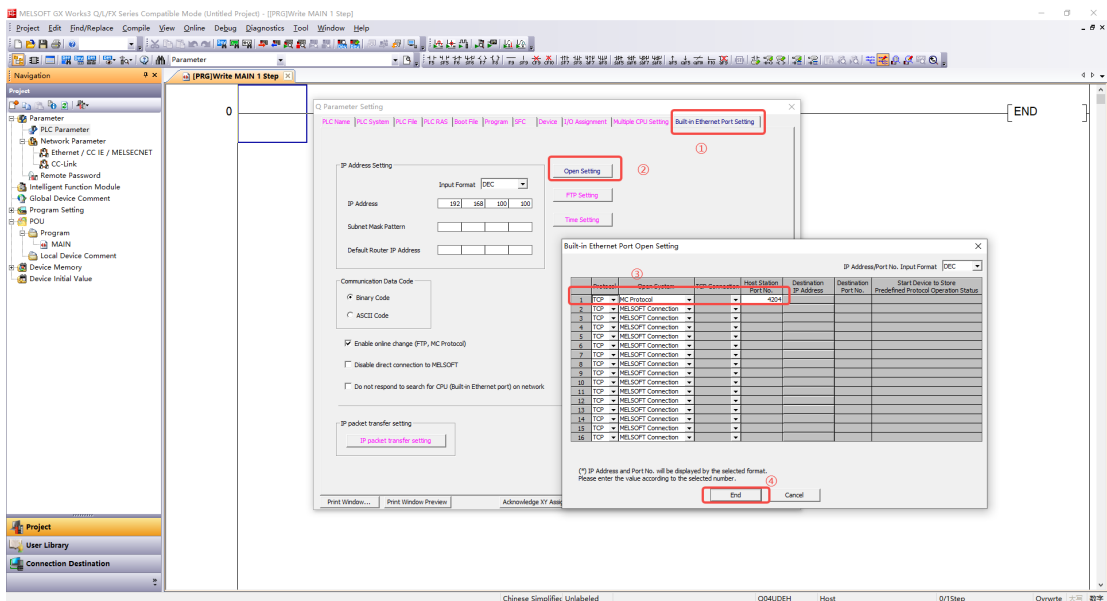
1. In the left device tree on the main interface of the MELSOFT GX Works3 software, double-click "PLC Parameter" under "Parameter" to enter the "Q Parameter Setting" interface.
2. On the "Q Parameter Setting" interface, click to select "Built-in Ethernet Port Setting".
3. On the "Q Parameter Setting" interface, set the PLC IP address.
4. Check "Enable online change (FTP, MC Protocol)".
5. Click "Open Setting" to complete the Mitsubishi Q series communication settings.



The subnet mask type is not required.

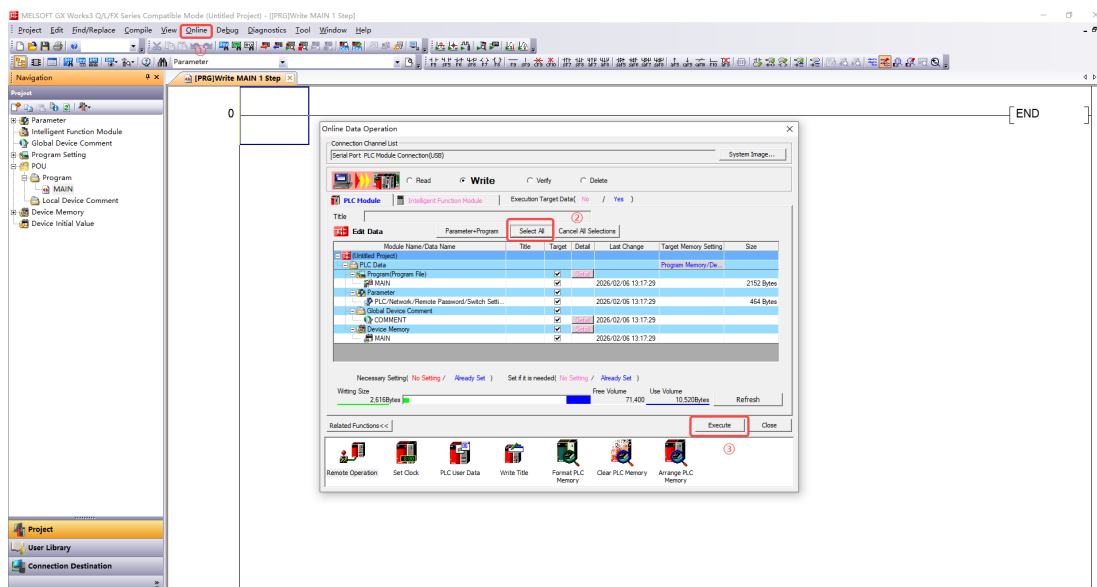
3.3.1.6 Ethernet Settings

1. In the left device tree on the main interface of the MELSOFT GX Works3 software, double-click "PLC Parameter" under "Parameter" to enter the "Parameter Setting" interface.
2. On the "Q Parameter Setting" interface, click "Built-in Ethernet Port Setting".
3. Click "Open Setting" to enter the "Built-in Ethernet Port Open Setting" interface.
4. On the "Built-in Ethernet Port Open Setting" interface, select "TCP" for the protocol, "MC protocol" for the "Open System", and configure the "Host Station Port No".
5. After completing the above operations, click "End" to complete the Ethernet configuration.



3.3.1.7 Configuration Parameter Download

1. Open the MELSOFT GX Works3 software to enter the main software interface, and click "Online" on the menu bar to enter the "Online Data Operation" interface.
2. On the "Online Data Operation" interface, click "Select All".
3. Click "Execute" to complete the parameter configuration.
4. After the download is complete, power off and restart the PLC. Once the restart is finished, communication with the robot can be established.



3.3.2 Robot Interface-related Settings

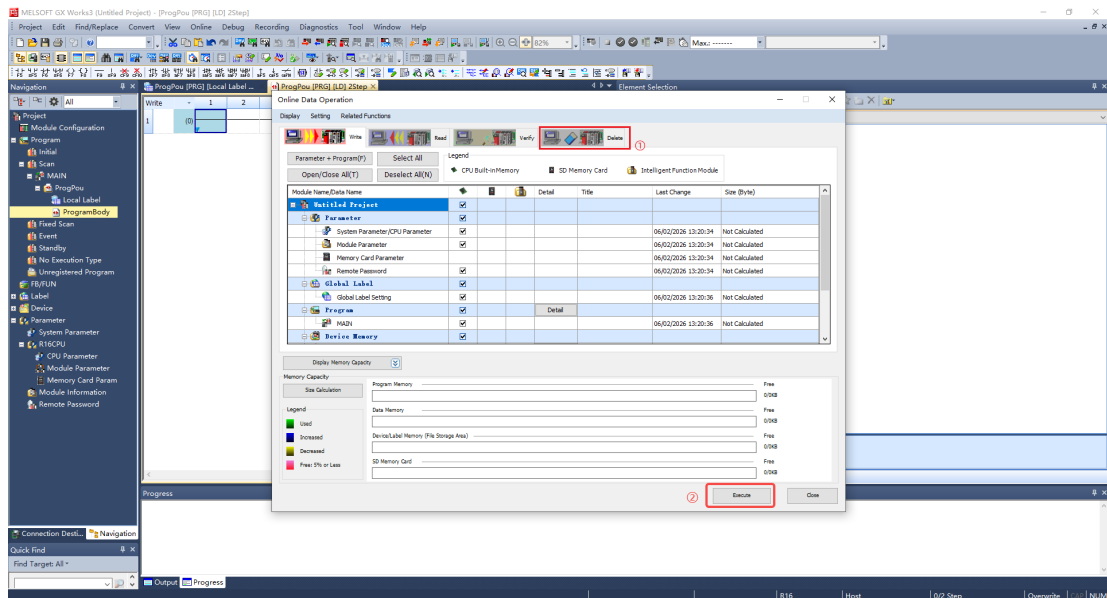
For detailed operations, see [3.2.2 Robot Interface-related Settings](#) section.

3.4 FAQ

- **Issue 1: The InoRobotLab software persistently displays alarm messages**

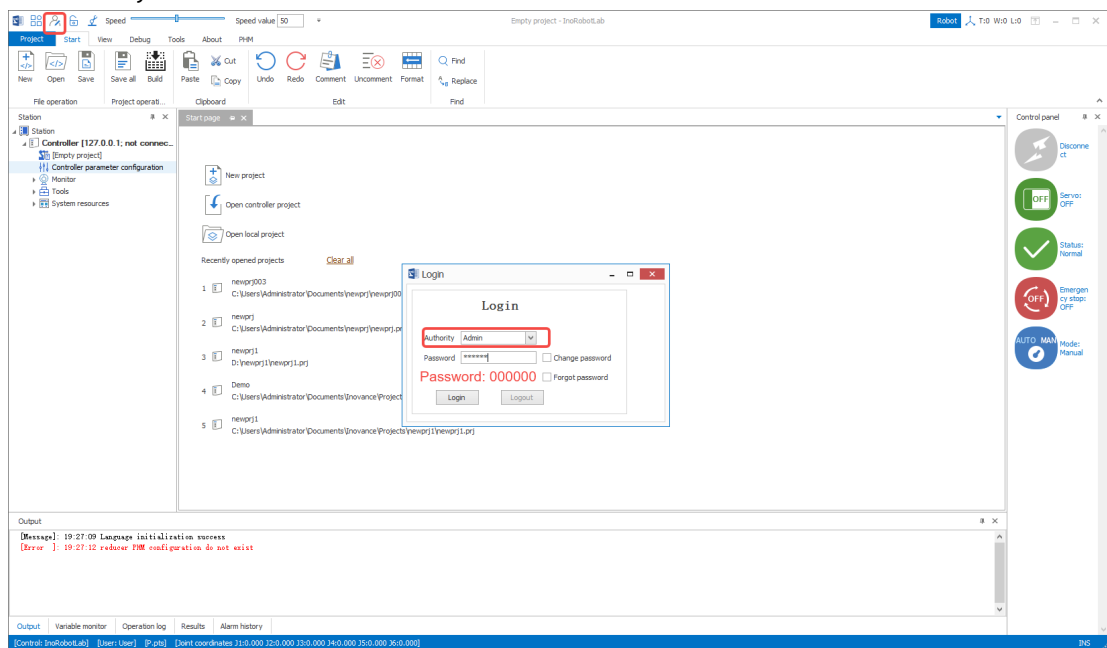
If the InoRobotLab software keeps prompting alarm messages, resolve the issue according to the following method:

1. For a new PLC, first delete the PLC configuration parameters by clicking "Delete", and then "Execute".
2. Power off and restart after completion.



• Issue 2: Process for robot startup via PLC

Click the "Login" button in the upper left corner of the InoRobotLab software interface, log in with "Admin" > complete the programming, switch to remote I/O unit mode. PLC continuously provides the emergency stop signal (level type) > clear alarm signal once (reset type) > program start signal (reset type), then robot starts and runs normally.



4 PROFINET Bus Communication

4.1 Communication Introduction

PROFINET communication is a real-time industrial fieldbus communication protocol based on Ethernet, offering advantages such as high performance, flexible topology, low cost, high precision, and ease of application.

4.2 PROFINET Communication with a Siemens PLC

4.2.1 Expansion Card Panel Introduction

The PROFINET expansion card panel includes a nameplate, network ports, status indicators, and locking bolts. The network ports use standard Ethernet RJ45 sockets with their pin assignments compliant with the standard Ethernet pinout, supporting both straight-through and crossover cables.



Table 2-1 Terminal functions of expansion card

Terminal mark	Terminal name	Description
PROFINET1	Network port 1	- If the PLC has not been configured with device topology, the expansion card ports are direction-insensitive. In this case, you can connect either port to the PLC. - If the PLC has been configured with device topology, you must select corresponding expansion card port when connecting to the PLC to avoid communication failure. - It is recommended to use shielded Cat 5e twisted pair cables to ensure stable operation.
PROFINET2	Network port 2	

Table 2-2 Indicators of expansion card

Indicator	Status display	Status description	Handling method
NS	Steady OFF	Communication timeout between expansion card and controller	- Re-plug the extension card. - Power cycle the controller. - Replace the expansion card.
	Slow flashing (approximately once per second, equal duration of on and off time)	Communication error between expansion card and controller	

Indicator	Status display	Status description	Handling method
	Fast flashing (approximately once every 0.5s, with on time longer than off time)	Normal communication between expansion card and controller	N/A
	Steady ON	Expansion card not activated	Activate via InoRobotLab software or teach pendant.
	Alternatively flashing with MS indicator (green light)	Expansion card in BOOT state	N/A
	Synchronously flashing with MS indicator (green light)	Extension card in station search state	N/A
MS	Steady OFF	Expansion card not activated	Activate via InoRobotLab software or teach pendant.
	Steady ON in green	Expansion card working normally	N/A
	Flashing in red	Extension card fault	Check the fault code via the controller LED panel or InoRobotLab software to troubleshoot the corresponding fault.
	Synchronously flashing in green with NS indicator	Extension card in station search state	N/A
	Alternately flashing in green with NS indicator	Expansion card in BOOT state	N/A

4.2.2 Expansion Card Installation Method

1. The PROFINET expansion card supports hot-plugging. During installation, as shown in [Figure 3-1](#), orient the expansion card with its front side facing upward and insert it along the controller's expansion card slot until it is fully seated. After approximately 5s, as shown in [Figure 3-2](#), the NS indicator on the expansion card panel displays a steady green light, indicating that the expansion card startup is complete.
2. If the NS indicator does not light up for an extended period after the expansion card is inserted, or if the NS indicator flashes slowly (for the definition of slow flashing, see Table 2-2), check whether the controller is powered on. If it is already powered on, power it off and restart it.



After the expansion card is fully seated, tighten the bolts on both sides of the panel to prevent the expansion card from loosening due to cable pulling.

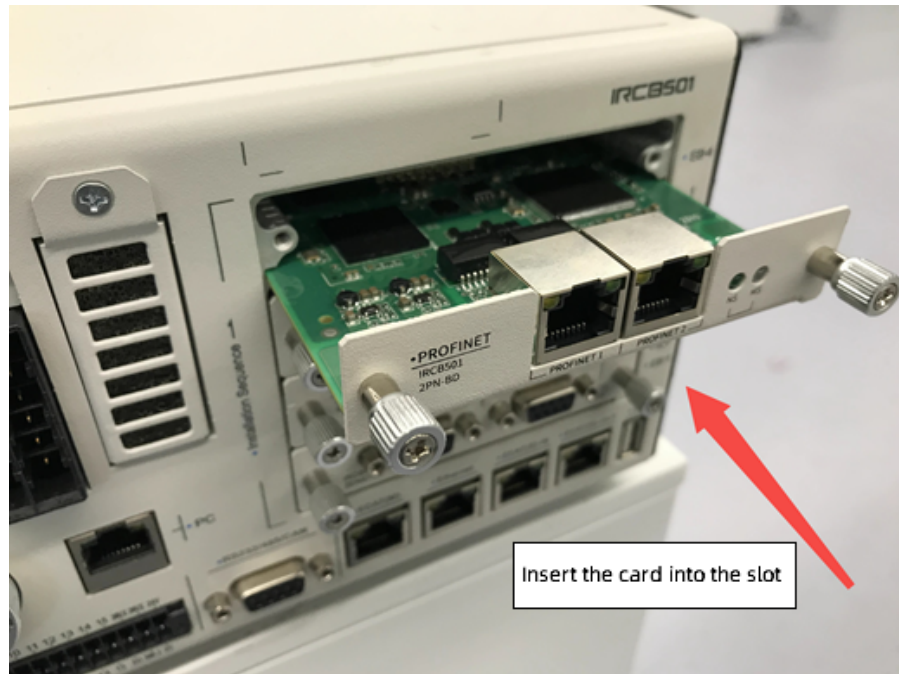


Figure 3-1 Expansion card installation

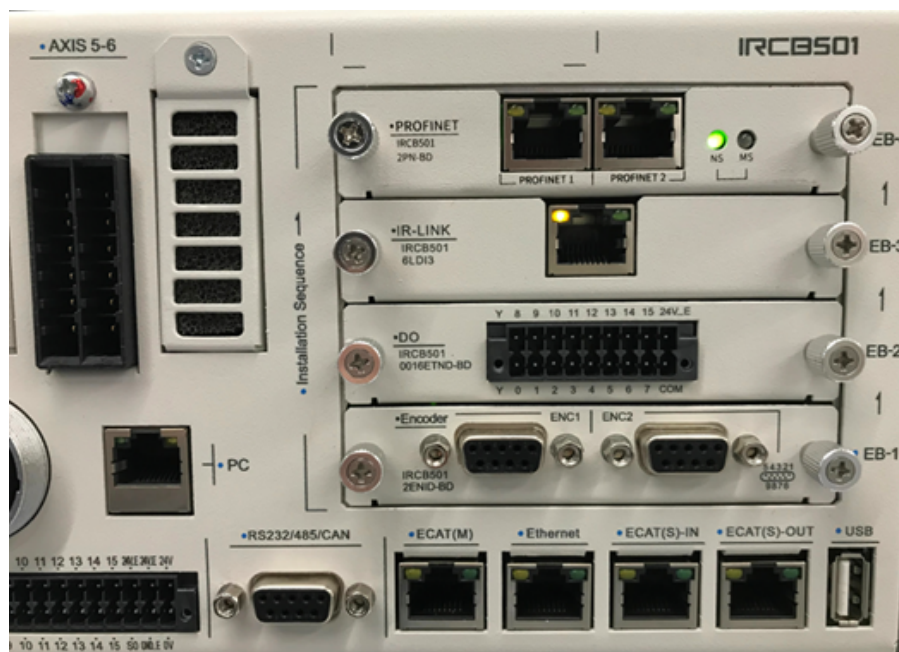


Figure 3-2 Completed startup of expansion card

4.2.3 Network Topology Methods

After the PROFINET expansion card is successfully installed into the robot controller, the next step is to configure the network with other devices on the PROFINET bus.

The PROFINET bus supports line (bus), star, and tree topologies, as illustrated in the figure below. The appropriate use of switches enables the implementation of all three topology types.

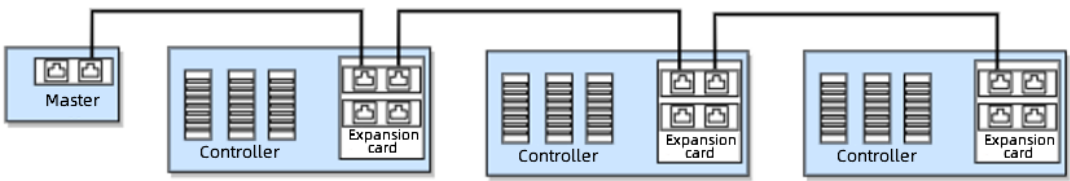


Figure 4-1 Bus topology

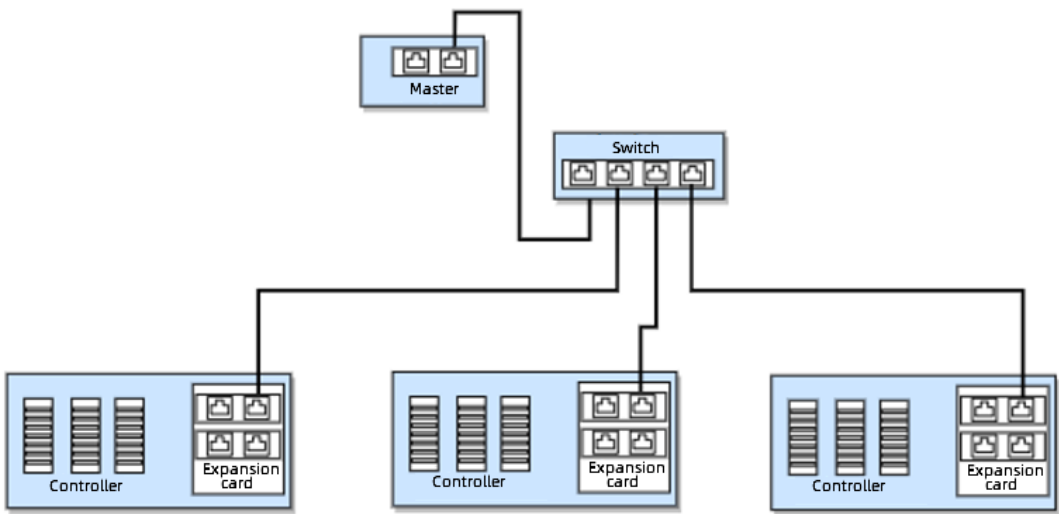


Figure 4-2 Star topology

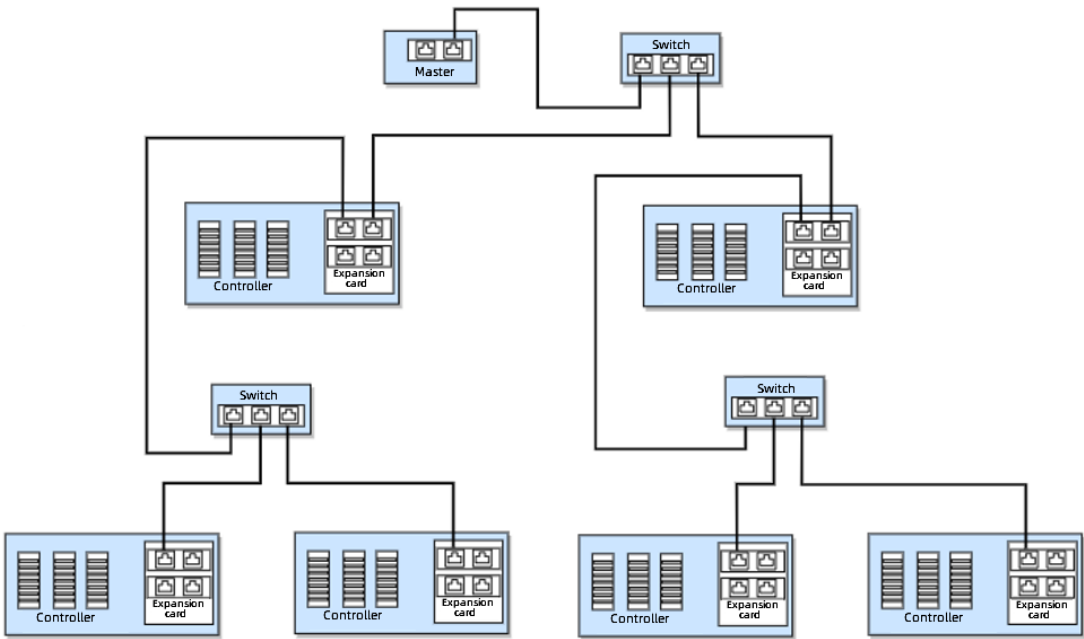


Figure 4-3 Tree topology

4.2.4 Communication Configuration

4.2.4.1 Siemens PLC Configuration

4.2.4.1.1 Configuration Settings

1. Create a project.

- a. Start the TIA Portal software to open the "PORTAL" interface. Click "Create new project", set the "Project name" and storage path, then click "Create" to complete the new project creation.

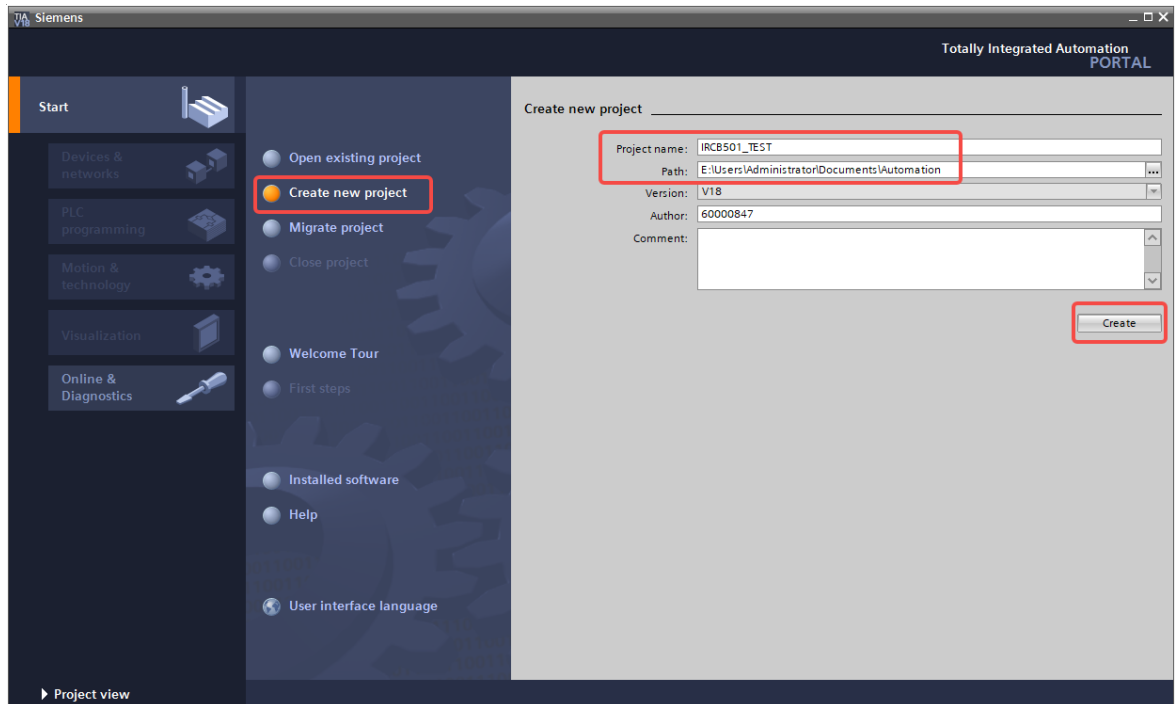


Figure 5-1 Creating a project

- b. When the newly created project is successfully open, click "Configuring a device" on the "Beginner Interface" to enter the "Devices & networks" interface.

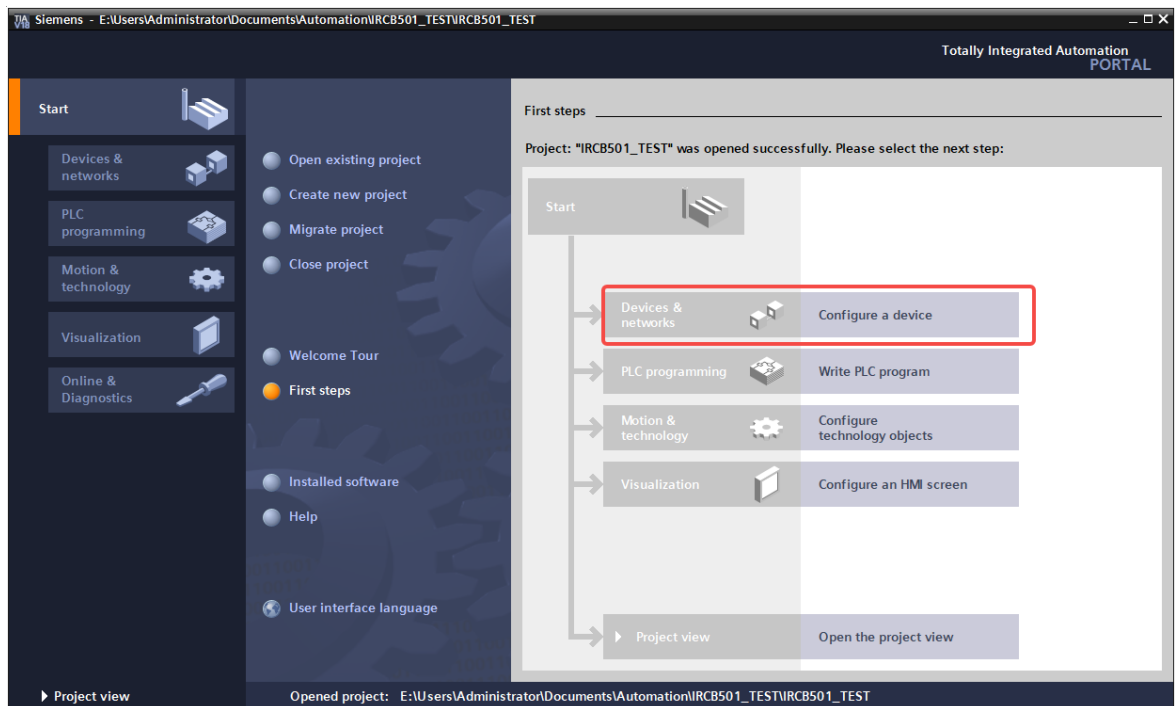


Figure 5-2 Configuring a device

- c. On the "Devices & networks" interface, click "Add new device" to open the setting interface.

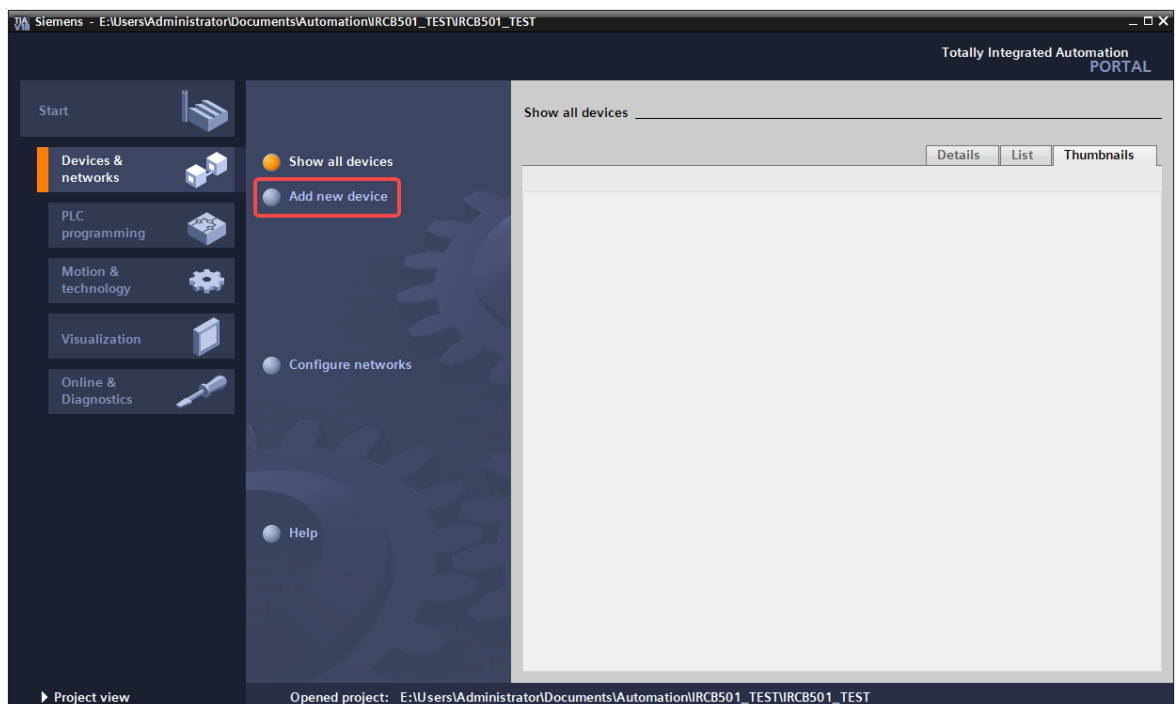


Figure 5-3 Adding a device to the project

- d. On the "Add new device" interface, select the firmware version of the PLC master according to the article number. Click "Add" or double-click the PLC master to complete the setting.



Note: The selected PLC must match the article No. Otherwise, the project cannot be downloaded.

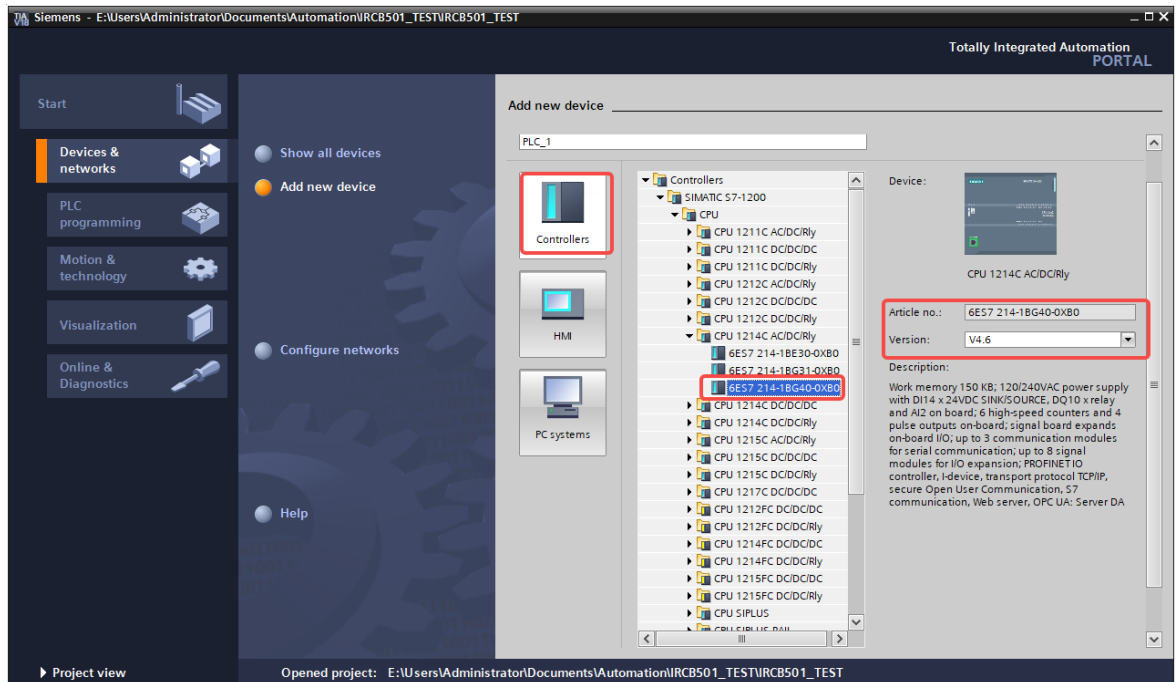


Figure 5-4 Adding a controller to the project

2. Install the GSD File (If the GSDML file for the Inovance robot controller has not been installed previously, perform this installation procedure.)
 - a. Open the TIA Portal software, click on the top toolbar "Options" > "Manage general station description files (GSD)".

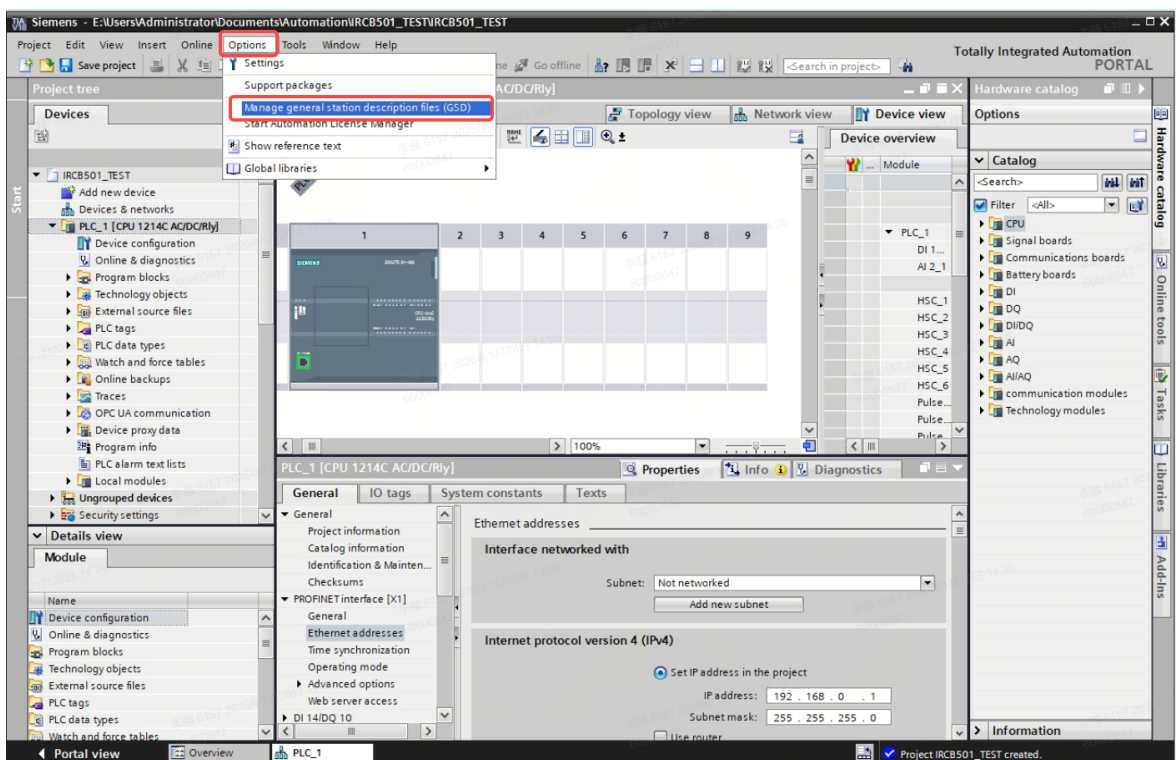


Figure 5-5 Manage general station description files (GSD)

- b. In the "Manage general station description files" dialog box, select the path where the GSDML file is stored (Note: Do not store the GSDML file in a path containing Chinese characters, as this may cause errors). Check the GSDML files that need to be installed, and click "Install". After installation is done, click "Close".

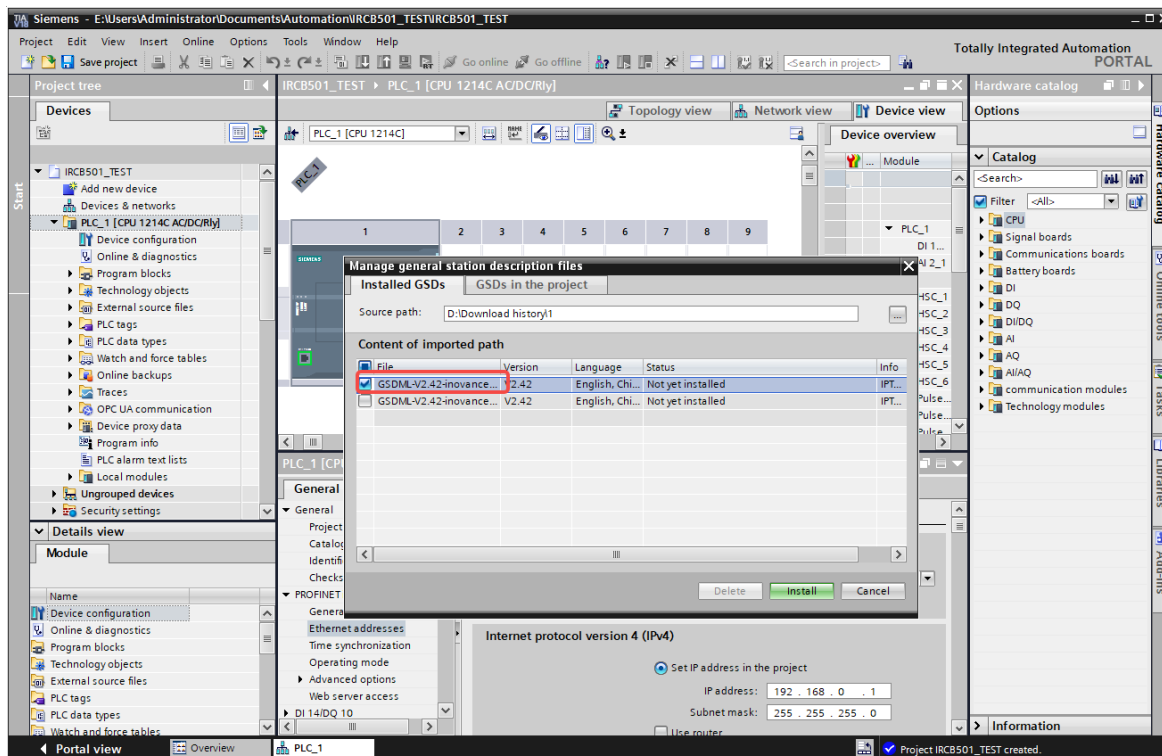


Figure 5-6 Installing device GSD files in TIA Portal

3. Create a network.

- a. Open the TIA Portal software, double-click "Device configuration" on the left interface, and switch to the "Network view" on the opened interface.

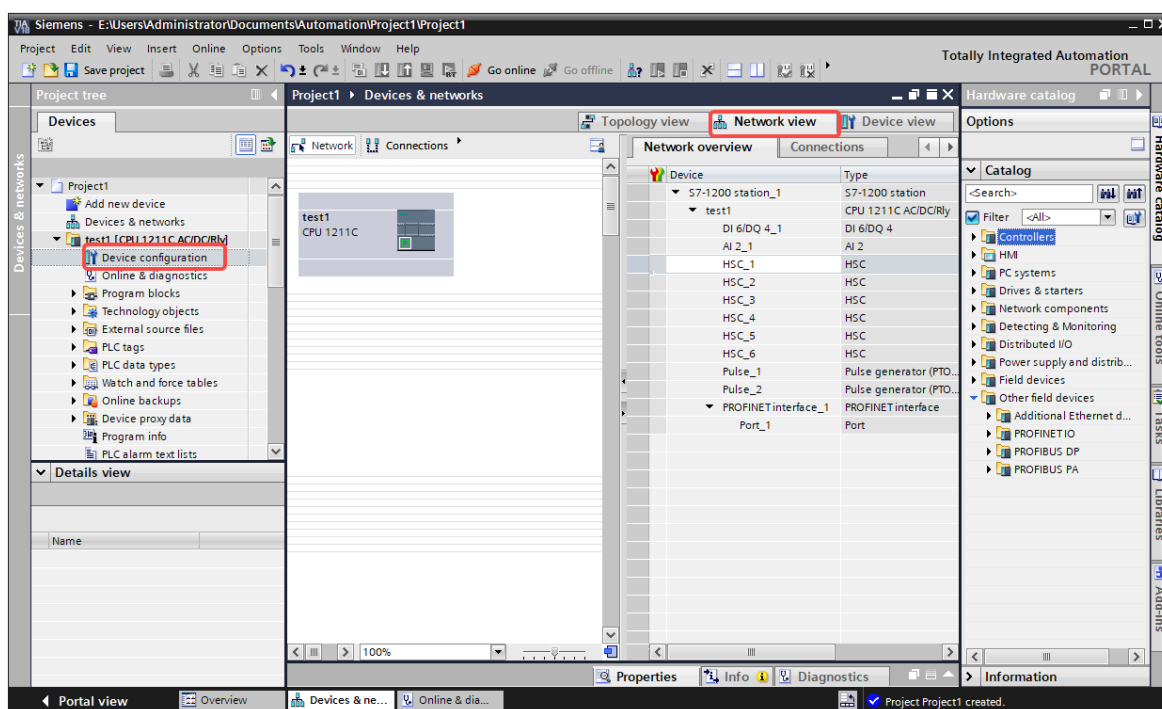


Figure 5-7 Switching to network view

- b. On the "Devices & networks" interface, select the PLC's Ethernet port, switch to "Properties" > "General" > "Ethernet addresses" for this port, and set the IP address and subnet mask for the PLC master.

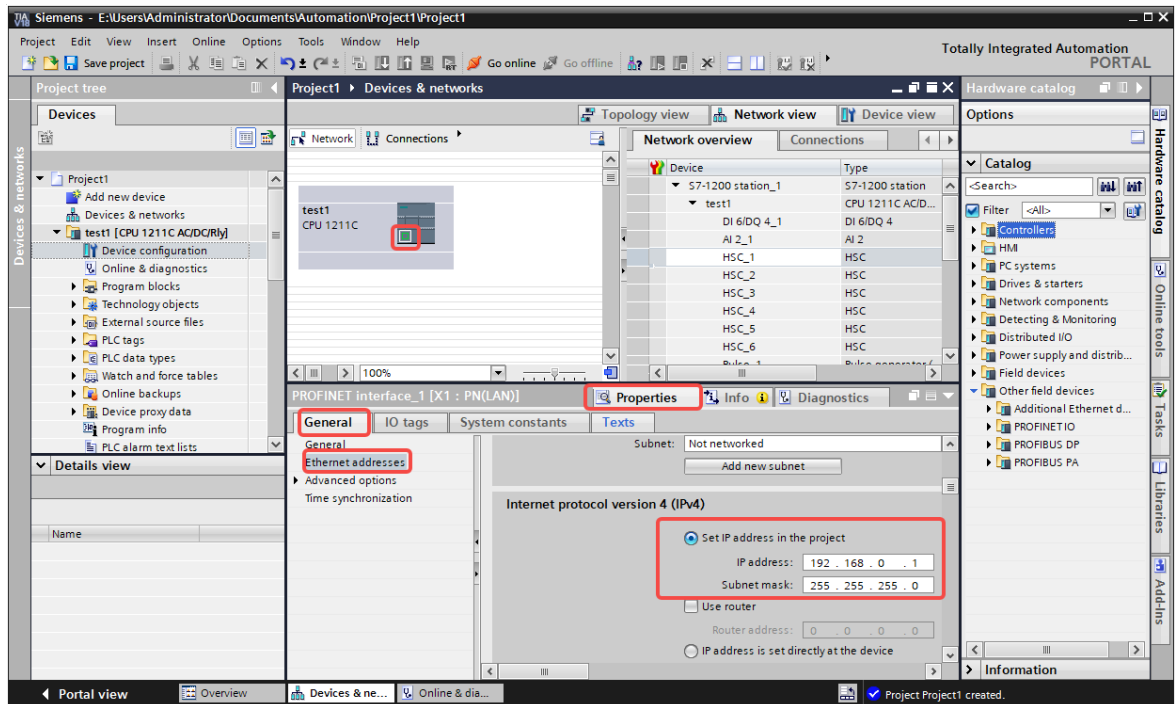


Figure 5-8 Setting the IP address of the PLC

- c. In the "Hardware catalog" on the right side, locate "Other field devices" > "PROFINET IO" > "Drives" > "INOVANCE" > "IRCB50X" > "IRCB501" > "IRCB501", double-click "IRCB501" to add the IRCB501 slave.

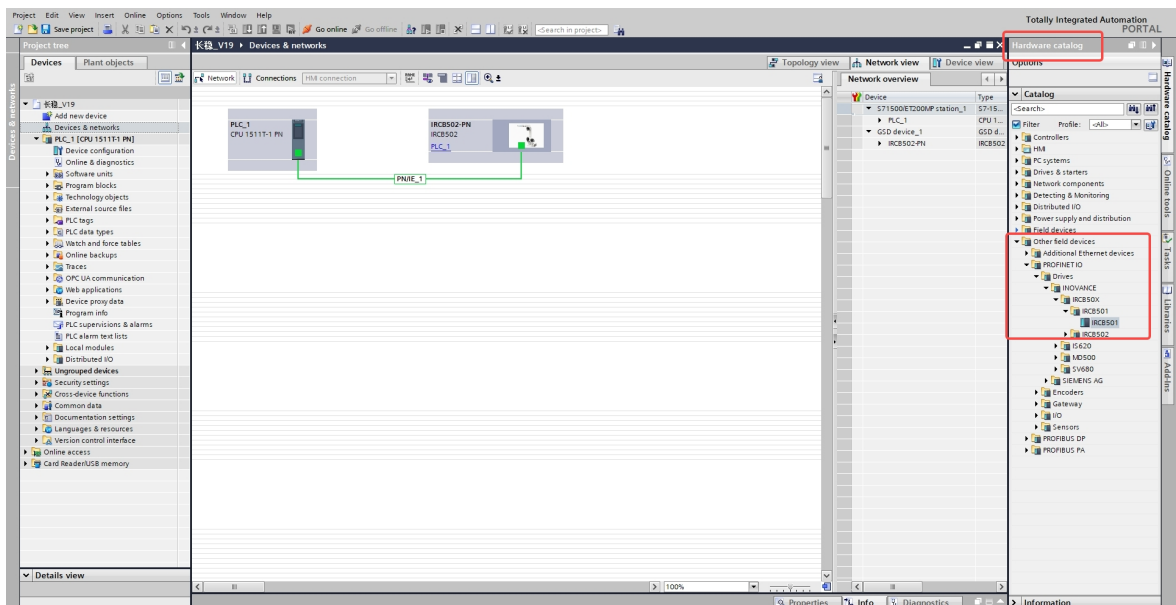


Figure 5-9 Adding the IRCB501 slave

- d. On the "Devices & networks" interface, click "Not assigned" and select the master system to which this slave needs to be connected.

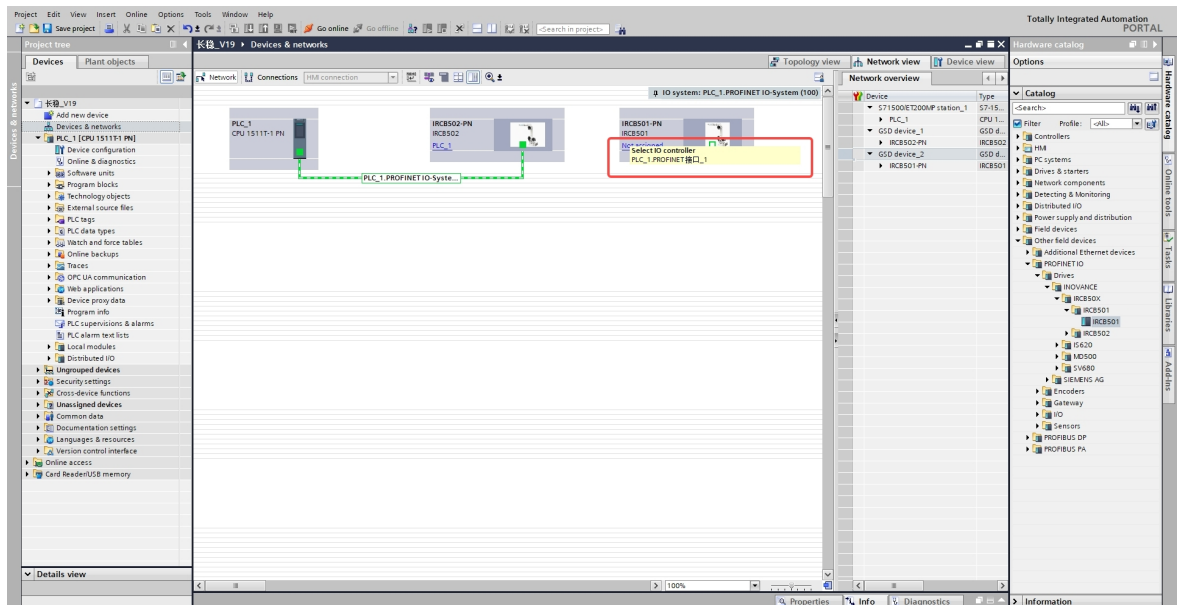


Figure 5-10 Assigning the master for the IRCB501 device

- e. On the "Devices & networks" interface, click the slave network port, and set the corresponding IP address under "Properties" > "General" > "Ethernet addresses" > "Set IP address in the project".

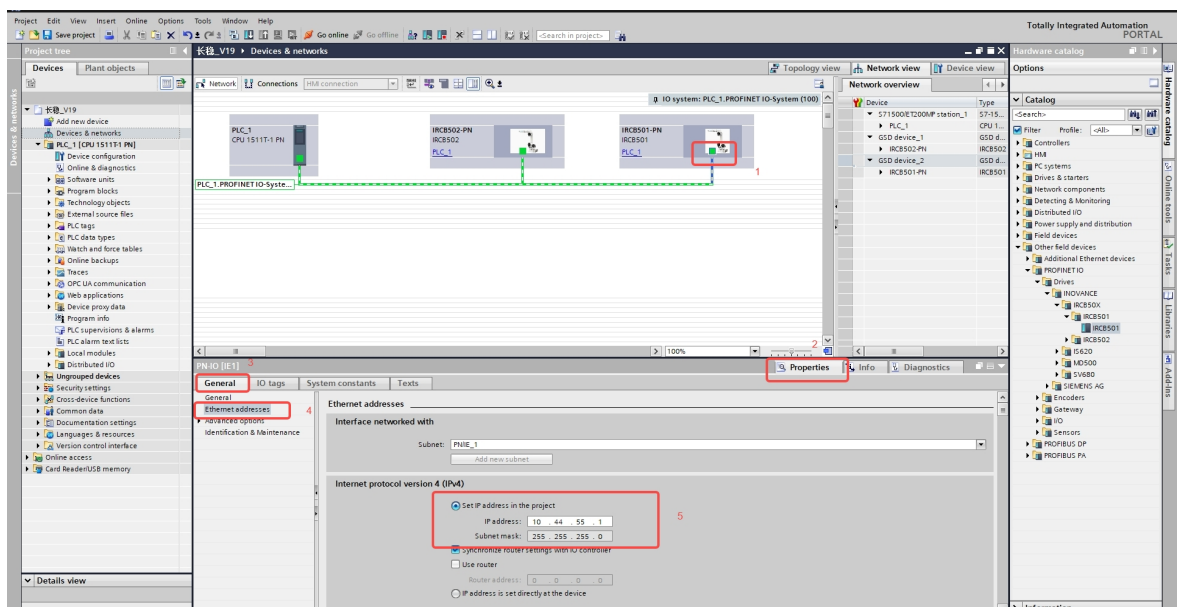


Figure 5-11 Setting the IP address of IRCB501

- f. On the "Device & networks" interface, scroll down to locate the "PROFINET" area, uncheck "Generate PROFINET device name automatically", and name the slave device in the "PROFINET device name" field. You can also select "Generate PROFINET device name automatically" to enable the system to generate the device name automatically.

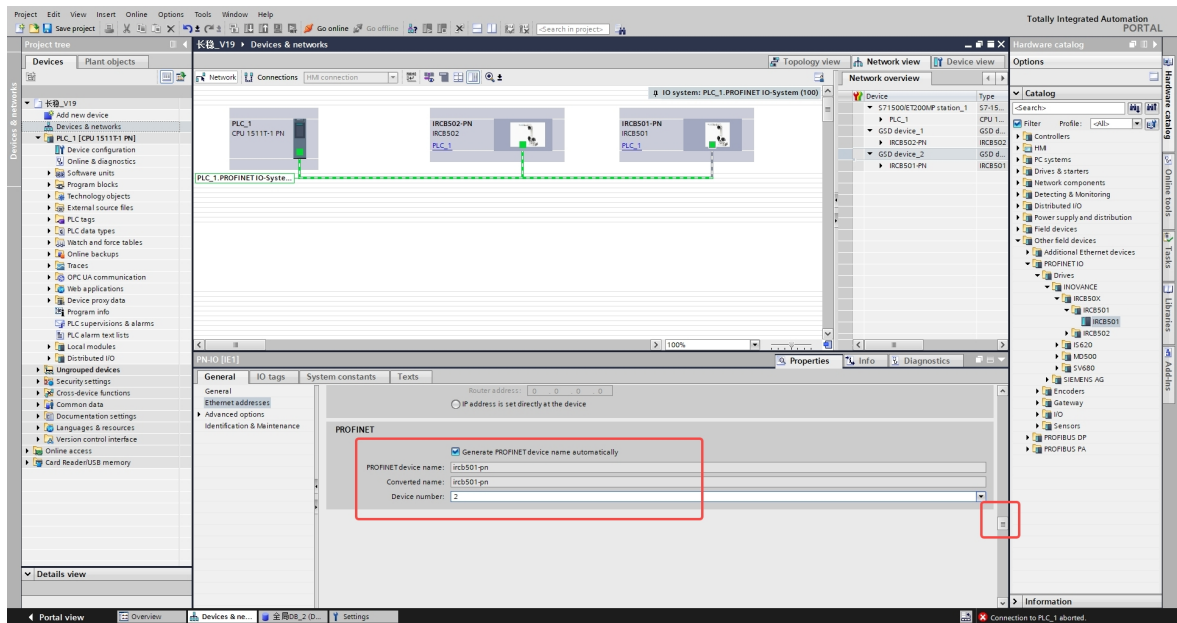


Figure 5-12 Setting PROFINET name for IRCB501

4. Configure the process data.

- Select the module in the "Device overview" list. Under the "Hardware catalog" > "Module" list, there are multiple modules for configuring the process data length of the slave.
- Drag and drop the corresponding module into slot 1 to configure process data of the corresponding length for the slave. After dragging and dropping, input and output address space of the corresponding size will be automatically allocated for this module.

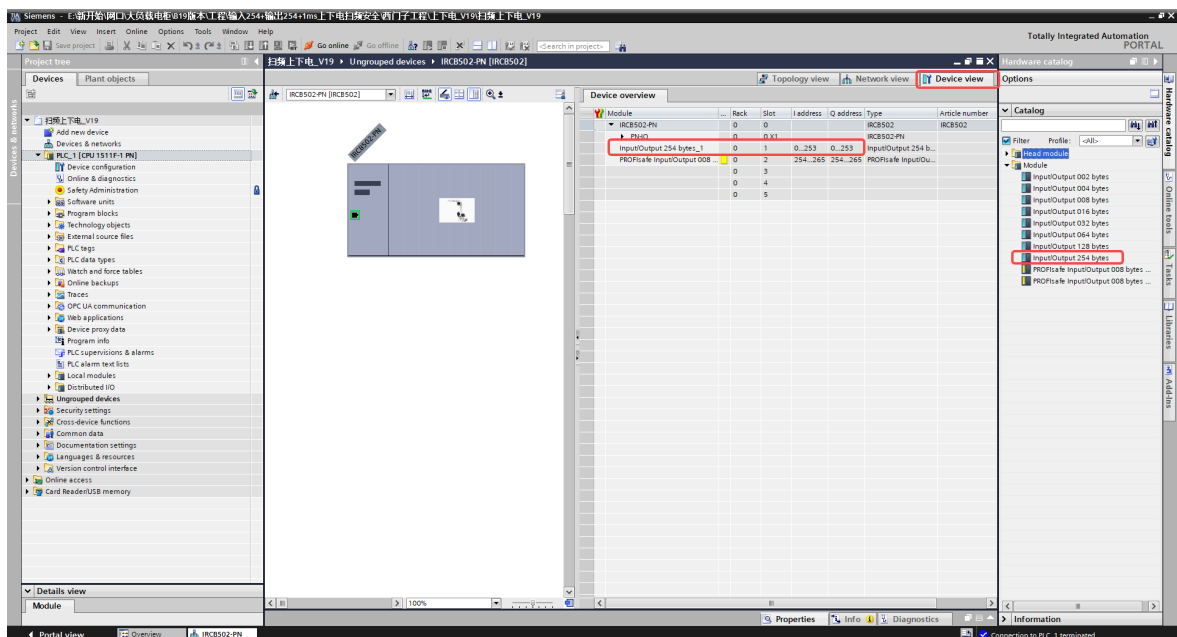


Figure 5-13 Configuring the process data

5. Download the project.

- After the configuration is complete, save the project and set the computer's IP address to be in the same subnet as the PLC (Note that this IP address cannot be the same as that of the slave in the configuration. You can set the PC to automatically obtain an IP address).
- On the "Extended download to device" interface, click "Download", select the appropriate interface, and then click "Start search".

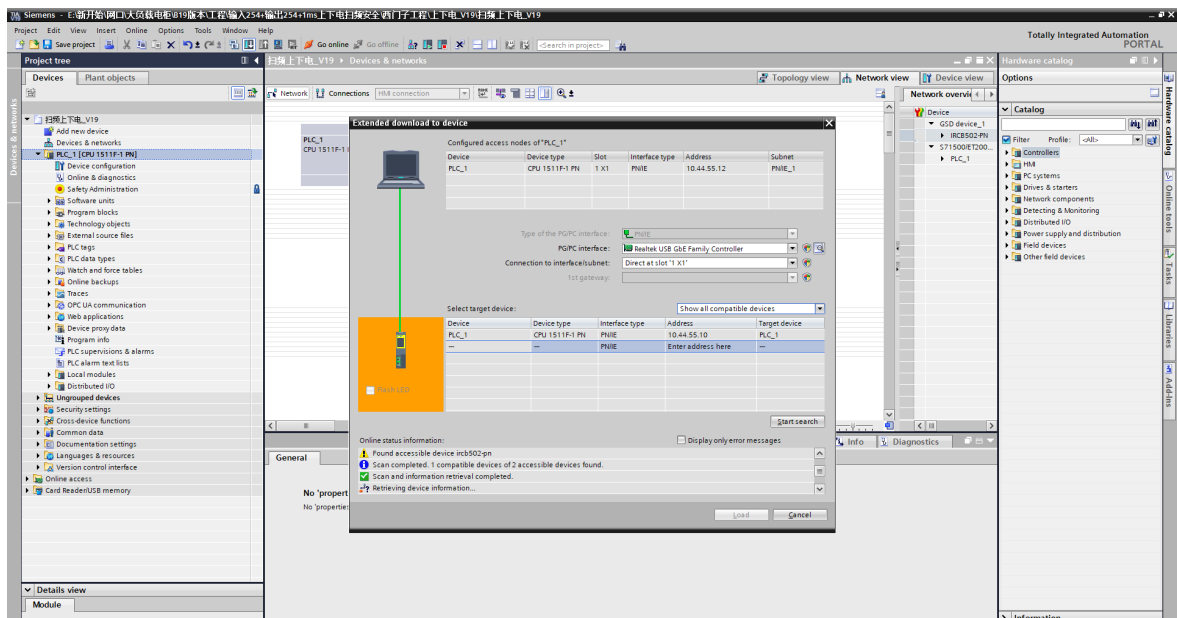


Figure 5-14 Downloading the project

6. Assign a device name.

- a. After completing the configuration file download, assign a device name to any slave node that has not been assigned one. Select a slave, and choose "Online" > "Assign device name".

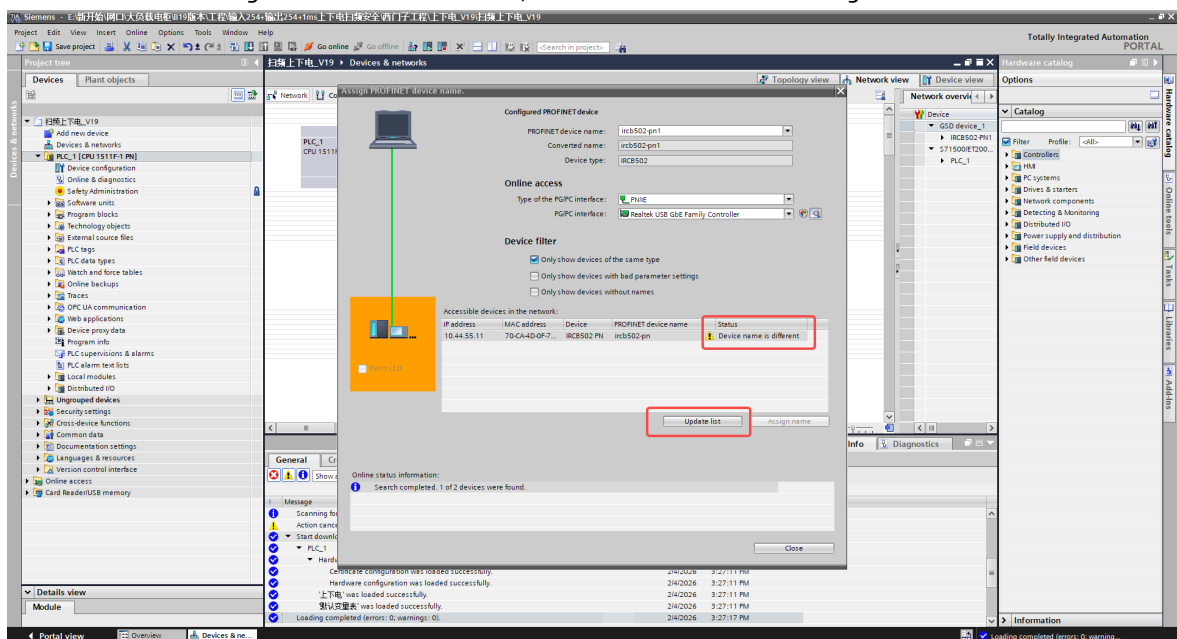


Figure 5-15 Assigning the device name

- b. Among the accessible nodes in the network, select the slave for which you want to assign a name (nodes are distinguished by MAC addresses) and then click "Assign name".
 - When the corresponding device status displays as "OK", the device name is written successfully, and it matches the "PROFINET device name" shown in the figure above.
 - After assigning the device name, close the interface or select another device from the "PROFINET device name" drop-down list to assign device names for other stations.
 - The slave will save the assigned name, and the master identifies each slave based on the device name. (The MAC address is not intuitive in use. The device name assigned is bound to the MAC address of a device for identification.)

4.2.4.1.2 MRP Function and Configuration

1. MRP overview

The standard Media Redundancy Protocol (MRP) is used, which meets the IEC 62439-2 standard. The typical reconstruction time is 200 ms and each ring network supports up to 50 devices.

MRP basic rules:

- All nodes in the ring network must support the MRP and have the MRP enabled.
- All devices must be interconnected through ports in the ring network.
- All devices in the ring network are in the same redundancy domain.
- A maximum of 50 devices can be connected in a ring network. More devices will prolong the reconstruction time of 200 ms.
- Only one device on the ring network can serve as the ring network manager, and other devices are all ring network clients.

2. MRP settings

- a. On the "Devices & networks" interface, click to switch to the "Network view" tab, select the PLC port, choose "Properties" > "General" > "Advanced options" > "Media redundancy", and set the "Media redundancy role" to "Manager". Then, check the parameters for "Ring port 1" and "Ring port 2".

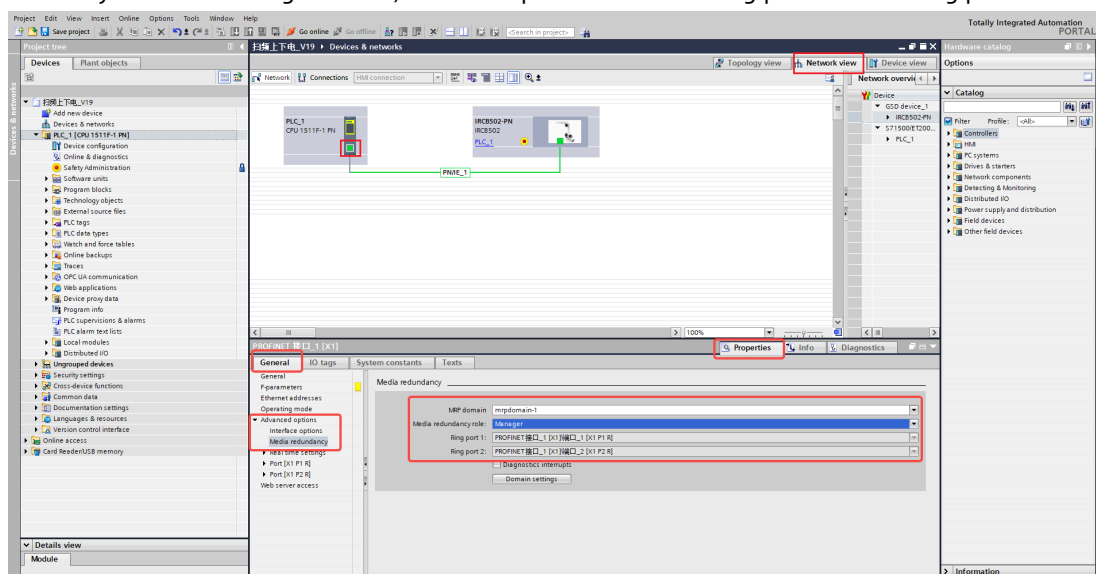


Figure 5-16 Ring network configuration for the master

- b. On the "Devices & networks" interface, click to switch to the "Device view" tab, select the I/O device port, choose "Properties" > "General" > "Advanced options" > "Media redundancy", and set the "Media redundancy role" to "Client". Then, check the parameters for "Ring port 1" and "Ring port 2".

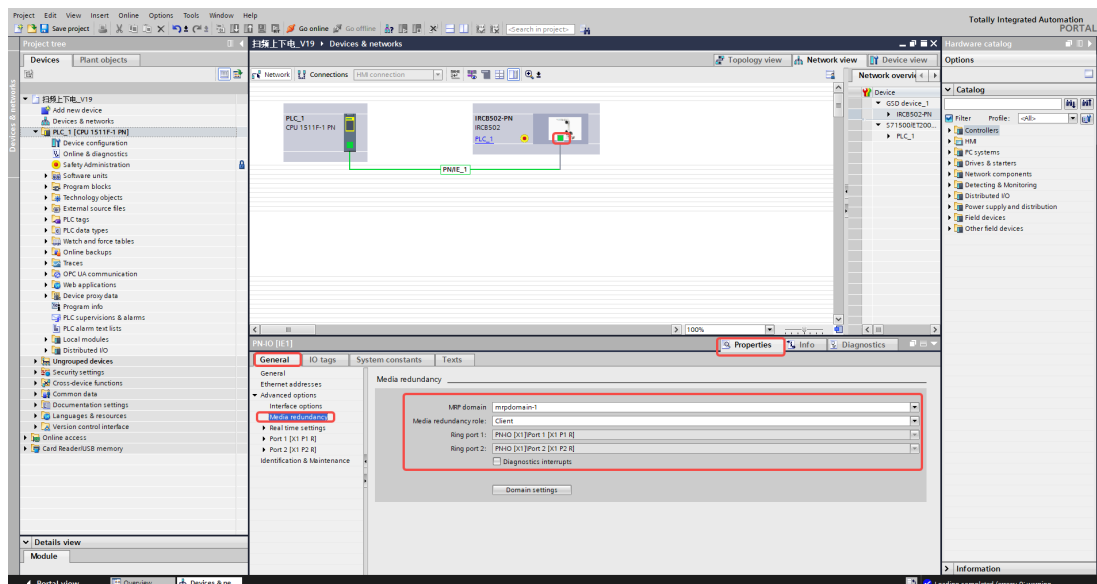


Figure 5-17 Ring network configuration for the slave

- MRP provides a typical reconstruction time of 200 ms and supports up to 50 devices in each ring network. To avoid interference to PROFINET communication during ring network reconstruction, set the PROFINET watchdog time of the I/O device to be greater than 200 ms.
- Select the I/O device port, "Properties" > "General" > "Advanced options" > "Real time settings" > "IO cycle", and adjust the "Update time" to an appropriate value so that the "Watchdog time" is greater than 200 ms.

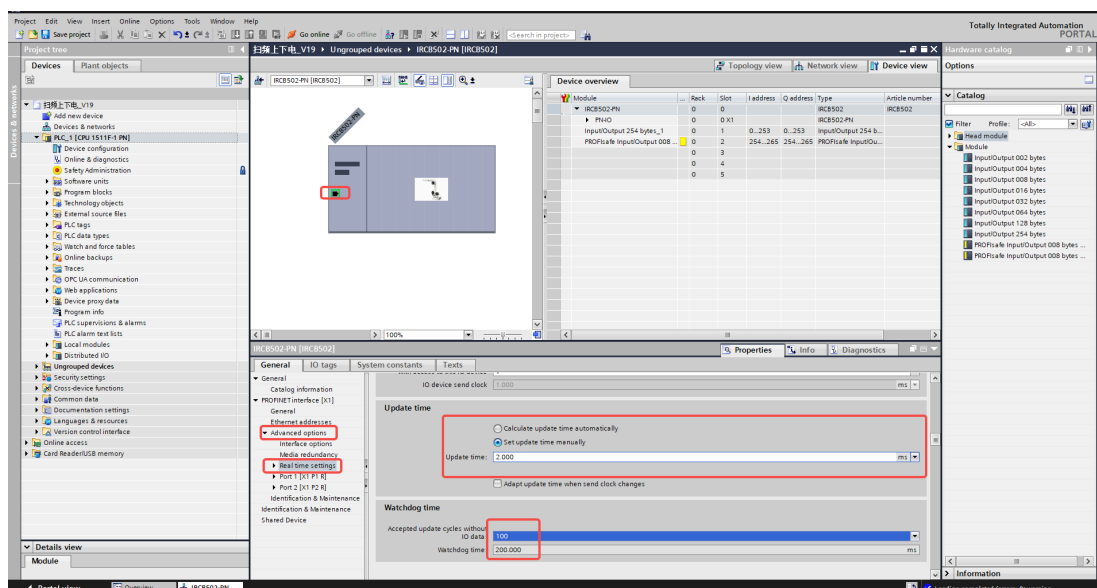


Figure 5-18 Slave watchdog time setting

4.2.4.2 Inovance Robot PROFINET Bus Configuration

4.2.4.2.1 PROFINET Expansion Card Activation

A maximum of 4 PROFINET expansion cards can be connected to an Inovance robot controller, but only one can be used at a time. The activation priority order is shown in [Figure 5-19](#). When the controller detects multiple PROFINET expansion cards, it activates the one with the highest priority.

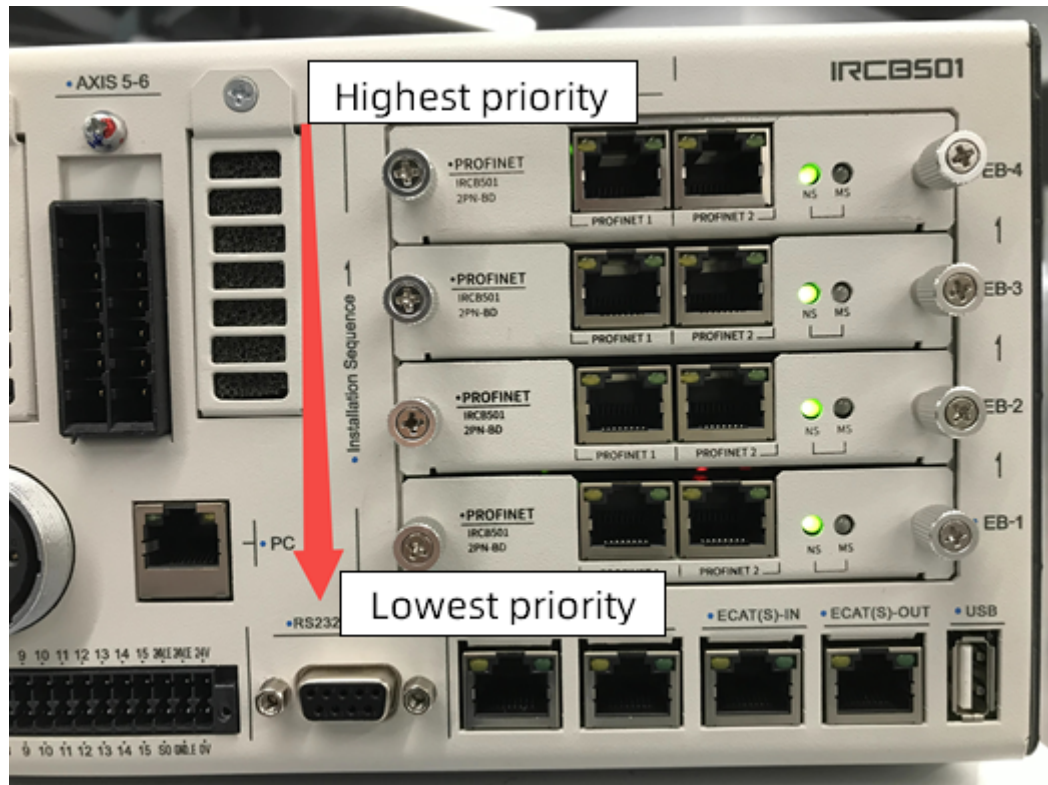


Figure 5-19 PROFINET expansion card activation priority

Follow the steps below to activate the expansion card:

1. Open the InoRobotLab software and connect to the target controller.
2. On the main interface of InoRobotLab software, click the  icon, select "Admin" for the user mode, enter the password "000000", and click "Login" to switch to administrator authority.

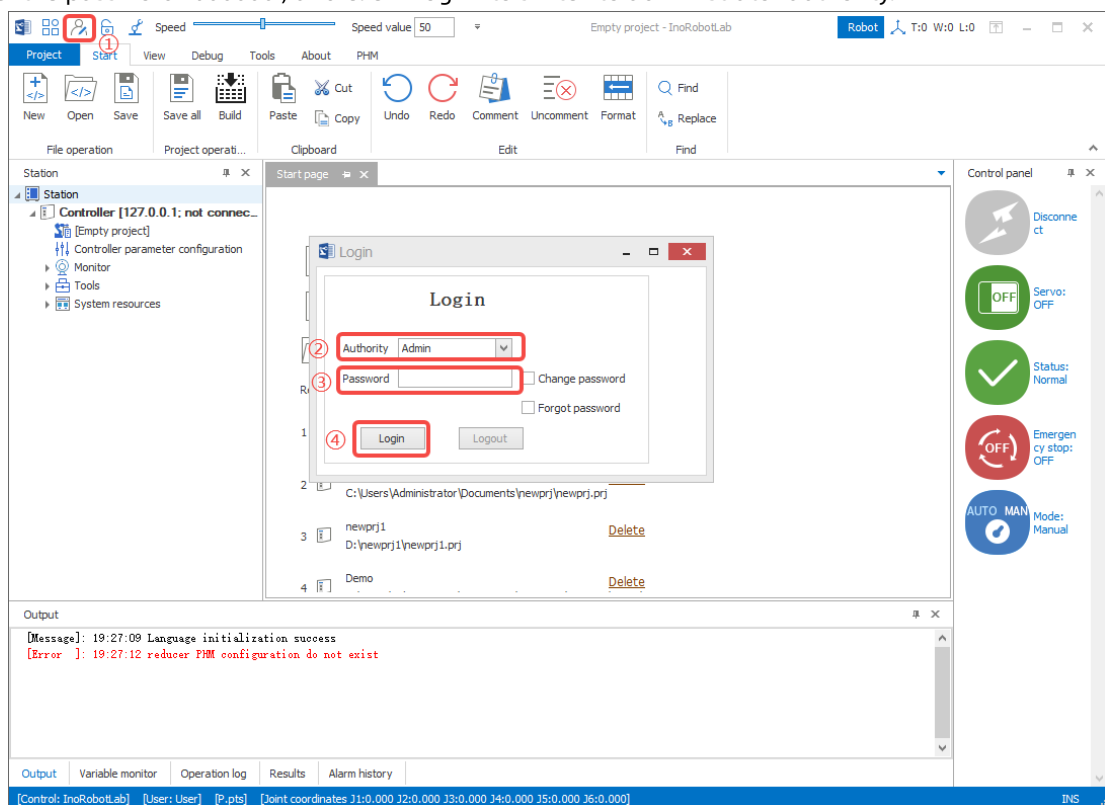


Figure 5-20 Switching the authority

3. In the navigation tree on the left side of the software interface, click "Controller parameter configuration" > "Peripheral configuration" > "Fieldbus slave configuration" > "Switch settings" > "Profinet", switch the "Profinet" button to "Open" and click "Save".

If the switch button remains in the open state and the InoRobotLab output window displays "PROFINET bus enabled, device in card slot [x] activated", it indicates the PROFINET card has been successfully activated. As shown in [Figure 5-21](#), after the PROFINET expansion card is activated, the NS indicator on the card will change from steady on to blinking.

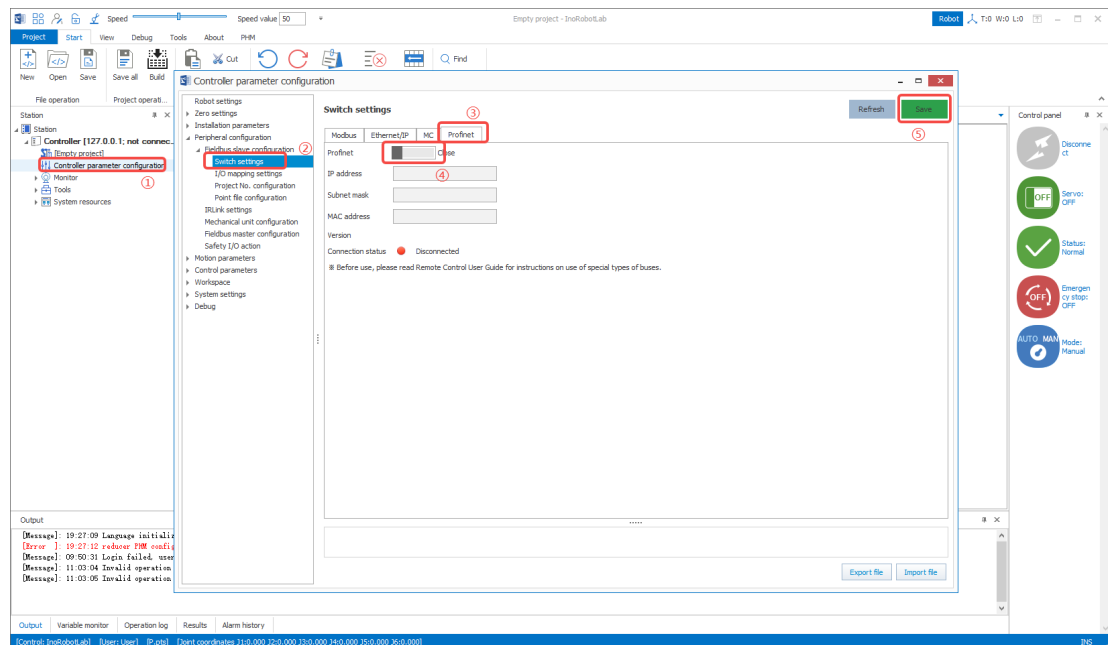


Figure 5-21 Activating the PROFINET expansion card

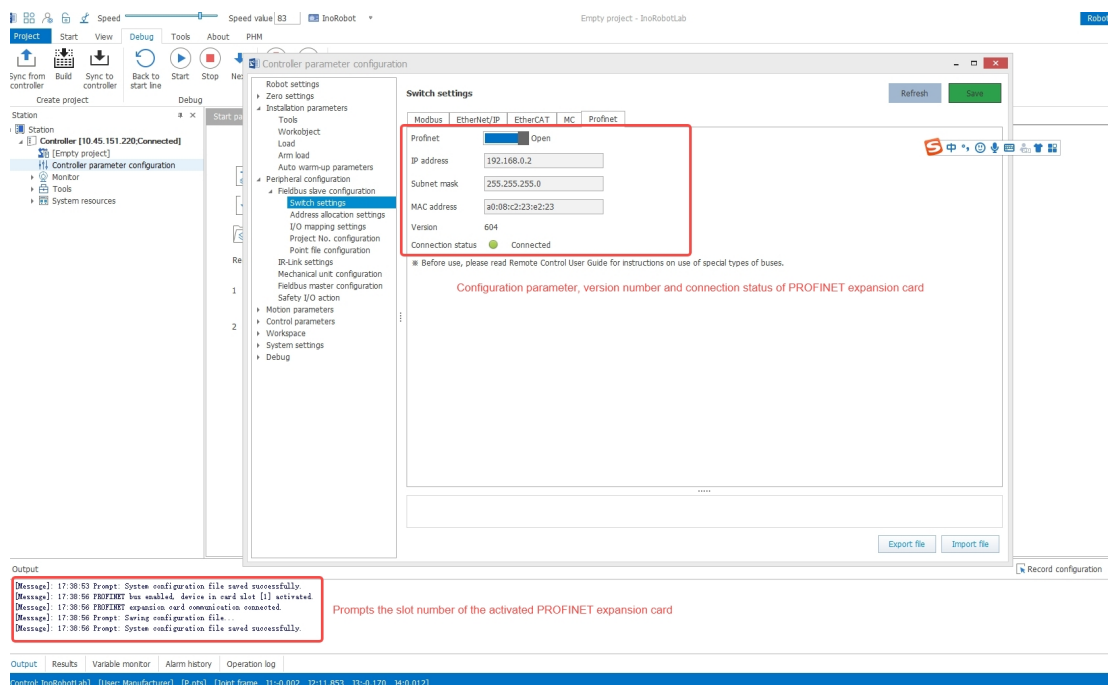


Figure 5-22 PROFINET expansion card activation success status

4.2.4.2.2 Address Allocation Configuration

Configure the length of process data sent and received by the robot controller to match the length of process data sent and received by the PLC master, ensuring the validity and integrity of PROFINET bus data.

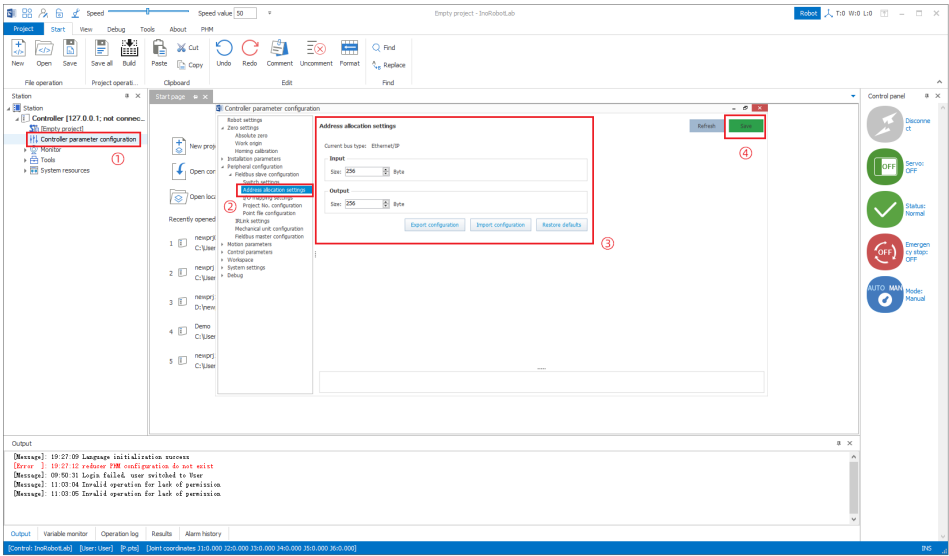


Figure 5-23 PROFINET Bus address allocation settings

1. On the main interface of the InoRobotLab software, click "Controller parameter configuration" > "Peripheral configuration" > "Fieldbus slave settings" > "Address allocation settings" to enter the address allocation settings interface, as shown in [Figure 5-23](#).
2. Select the Input/Output size in "Word", with an optional range of 1 word to 127 words (corresponding to 2 bytes to 254 bytes on the PLC side), then click "Save" to complete the configuration.



The robot controller will read and write bus data based on the input/output size configured in the current step. If the length entered here is inconsistent with the configuration in the process data configuration step in "Configuration Settings" section, the controller will generate a 0x3049 warning. The warning can be cleared when the length configuration matches that of the PLC.

4.2.5 Troubleshooting

4.2.5.1 PLC Diagnosis Information

For PLC fault diagnosis information, see the table below:

Table 6-1 PLC fault diagnosis information

Fault description	Fault cause	Solution
MAC address error	The MAC address of the PROFINET expansion card was not burned; the current value is the default address.	Burn the MAC address for the PROFINET expansion card.

Fault description	Fault cause	Solution
MCU not detected	PROFINET expansion card parallel port communication error or MCU chip crash	Check whether the MCU of the PROFINET expansion card is started, and check whether the parallel port communication is normal.
Inobus not connected	Handshake failure between the PROFINET expansion card and the controller	Check whether the Inobus link between the PROFINET expansion card and controller is properly connected.
Inobus data frame CRC error	PROFINET expansion card Inobus physical link unstable or interference present	Check the Inobus physical link of the PROFINET expansion card and eliminate interference.
Inobus data frame length verification error	The length of the given data in the Inobus standard frame data area of the PROFINET expansion card does not match the actual data length.	Check whether the set length of the data in the Inobus standard frame data area of the PROFINET expansion card matches the actual length.

You can locate and view the current error messages program through the diagnostic function of the PLC background. To view the detailed diagnosis information, choose "Online & diagnostics" > "Diagnostics buffer". As shown in the figure below, it shows the error "Safety program: F-I/O passivated" occurred in "Slot 2", and shows the solution in the "Help on event" box.

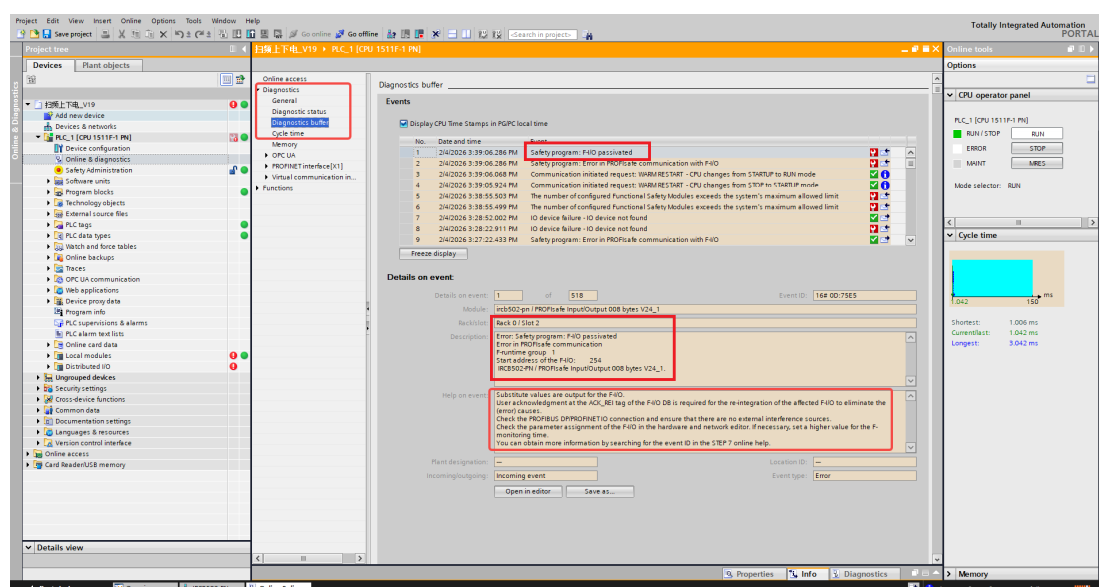


Figure 6-1 PLC background diagnosis

4.2.5.2 Robot Communication Fault Codes

When a PROFINET communication fault occurs, the robot controller generates a corresponding fault code. Diagnose the fault according to the table below.

Table 6-2 Expansion card fault code cross-reference table

Fault code	Description	Cause	Solution
0x3005	PROFINET expansion card missing	The PROFINET expansion card has lost communication with the controller.	Check in the following order: 1. Check that the PROFINET expansion card is securely connected to the card slot. 2. Check that the NS indicator light on the front panel of the PROFINET expansion card is solid green or flashing green. 3. Check that the PROFINET expansion card can be activated after the controller is powered on again.
0x3040	Illegal MAC address of PROFINET expansion card	The MAC address of the PROFINET expansion card was not burned, or an illegal MAC address was detected.	Burn a unique MAC address for the PROFINET expansion card.
0x3041	No network connection on PROFINET expansion card	PROFINET expansion card is not connected to the master.	Check the cable between the PROFINET expansion card and the master.
0x3042	PROFINET expansion card configuration parameter error	The configuration parameters for the PROFINET expansion card are incorrectly set.	Check the configuration parameter settings for the PROFINET expansion card.
0x3043	PROFINET expansion card disconnected from master	PROFINET expansion card is disconnected from master.	Check the cable between the PROFINET expansion card and the master.
0x3044	PROFINET expansion card chip not detected	No PROFINET expansion card slave chip is detected.	Reinstall the PROFINET expansion card or restart the controller.
0x3045	PROFINET bus master shutdown	The PROFINET master is detected in STOP state.	Check if the PROFINET master is in STOP state. If yes, switch it to RUN state to clear the alarm.
0x3046	PROFINET expansion card name not exist	No PROFINET expansion card device name is detected.	Assign a device name to the PROFINET expansion card through configuration software.
0x3047	PROFINET expansion card PDO length mismatch	A verification error has occurred in the length of the Inobus data frame.	Check that the set length of the data area of the Inobus standard frame matches the actual length.

Fault code	Description	Cause	Solution
0x3048	PROFINET expansion card Inobus CRC error	A CRC error has been detected in the received Inobus data frame.	Check and eliminate Inobus link interference.
0x3049	PROFINET expansion card master-slave data address length mismatch	The length of the PROFINET bus data configured on the host controller of the robot does not match the data length configured on the PLC.	Modify the bus communication data length at the robot or PLC end to ensure consistency between the two ends.

5 EtherCAT Bus Communication

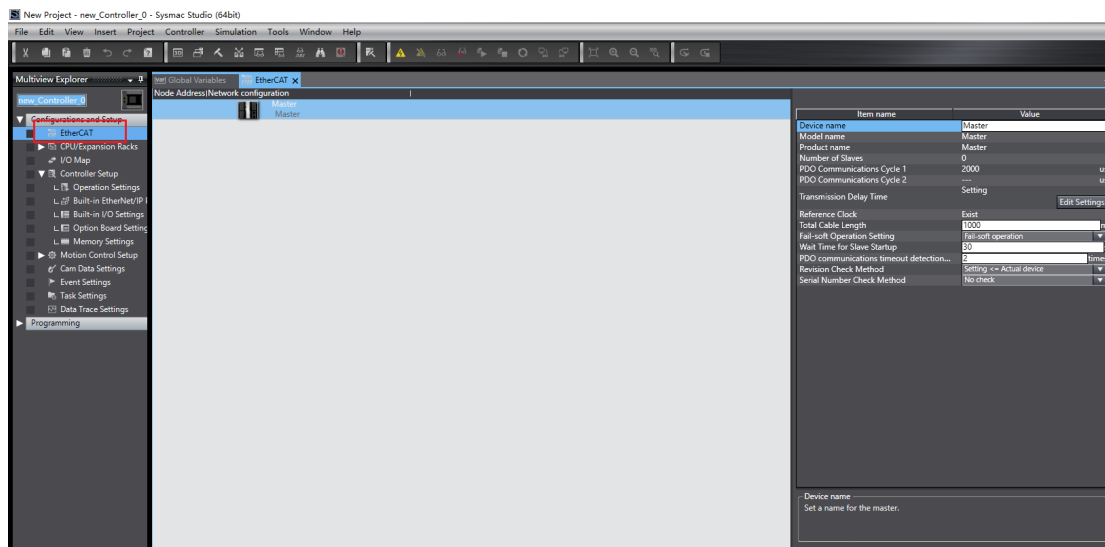
5.1 Communication Introduction

EtherCAT communication is a real-time industrial fieldbus communication protocol based on Ethernet, offering advantages such as high performance, flexible topology, low cost, high precision, and ease of application.

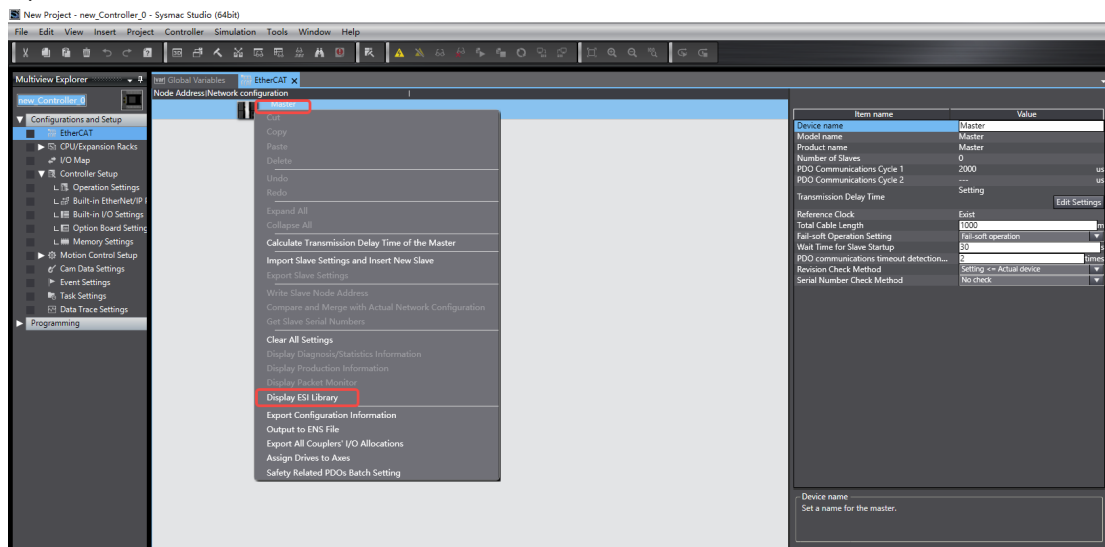
5.2 Sysmac Studio Software-related Configuration

5.2.1 Importing Inovance Robot EtherCAT Slave Description File

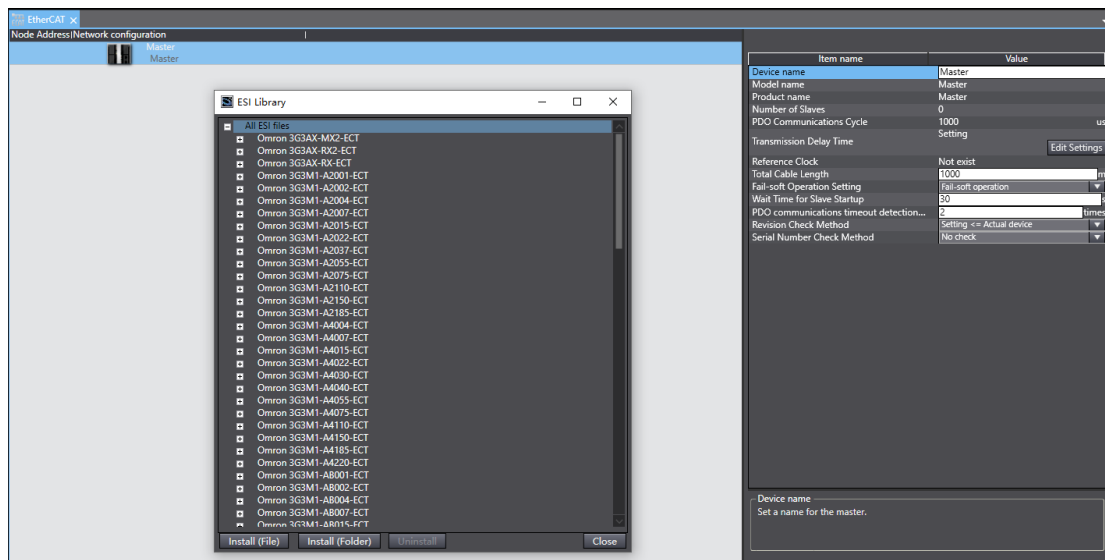
1. Take the Omron PLC NX-1P2 as an example. After creating a new PLC project, double-click "EtherCAT" on the left to enter the EtherCAT main interface.



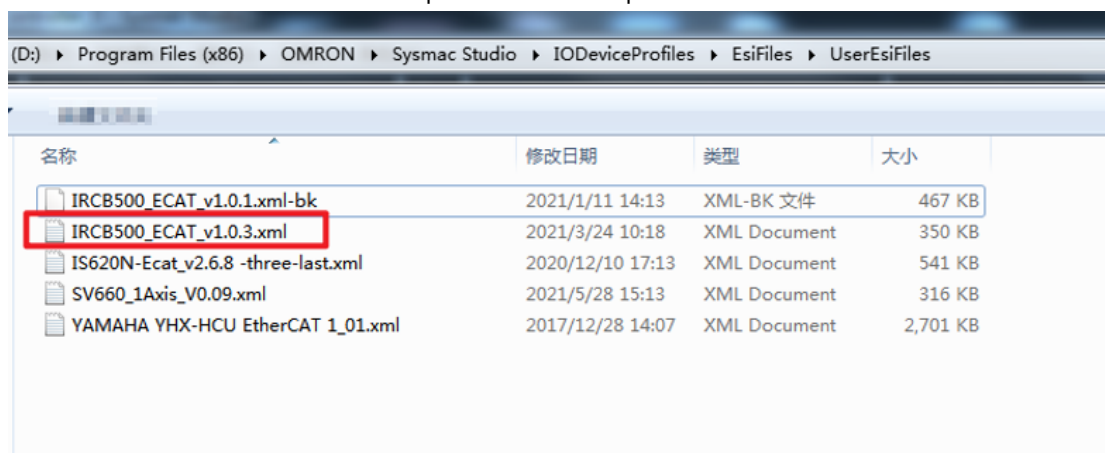
2. On the EtherCAT interface, locate the Master device, right-click on it, and click "Display ESI Library" in the pop-up menu bar.



3. In the "ESI Library" pop-up window, click "Install (file)" button.



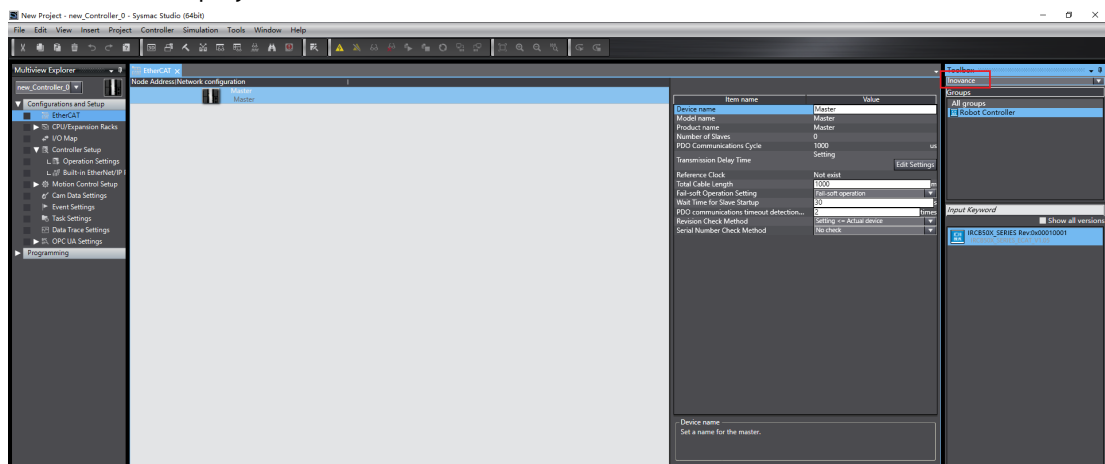
4. Select the Inovance robot EtherCAT description file to be imported.



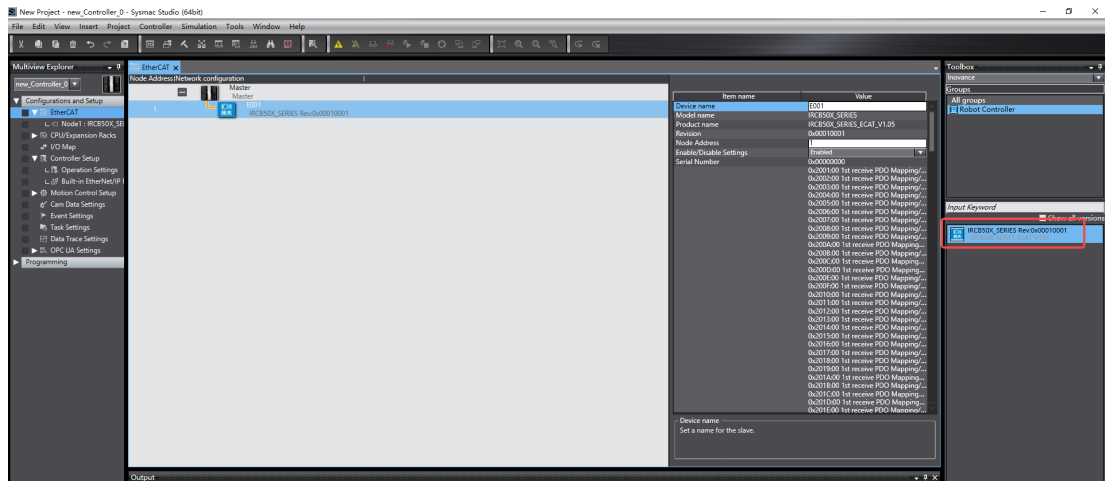
5. After completing the above operations, restart the software.

5.2.2 Adding Inovance Robot EtherCAT Slave to Network Configuration

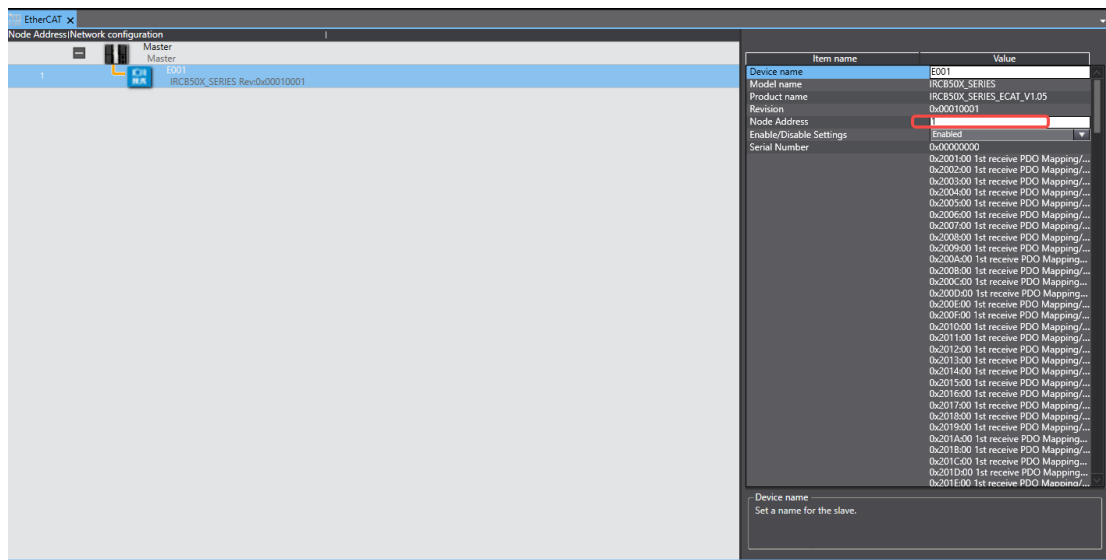
1. On the EtherCAT interface, select "Inovance" from the toolbox on the right side. All supported versions of this device will be displayed below.



2. Double-click the "InoIRCB50X" series controller on the EtherCAT interface to add the robot controller as a slave device.



3. Double-click the added slave "E001". In the "Node Address" section of the right navigation bar, set the node address (i.e., station alias) of the robot EtherCAT slave to a value greater than 0.

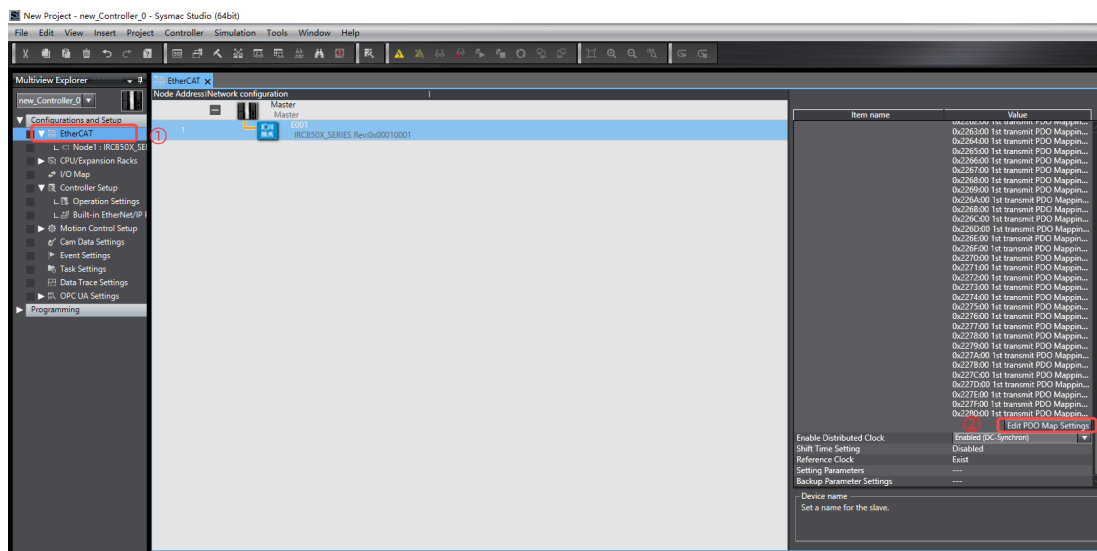


5.2.3 Editing PDO Mapping Settings

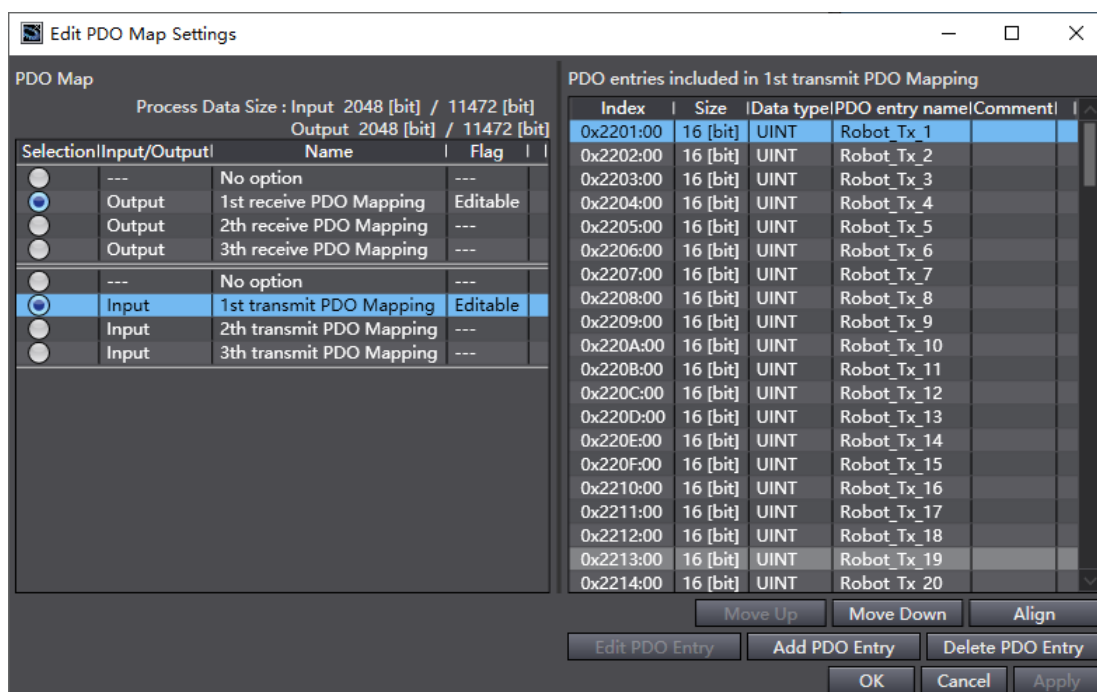
Inovance Robot EtherCAT provides three groups of PDO mapping configurations for user selection, supporting separate configuration of input and output PDOs.

- The default is the first group of mapping, with a fixed PDO length of 256 bytes of Uint type.
- The second group of PDO has a fixed length of 128 bytes of Uint type.
- The default length of the third group is 48 bytes (16 bytes of Byte type, 32 bytes of Uint type), and it supports editing.

1. On the "Multiview Explorer" interface on the left, click "Configuration and Setup" > "EtherCAT", then in the pop-up dialog box, click "Edit PDO Map Settings" to open the "Edit PDO Map Settings" interface.



2. On the "Edit PDO Map Settings" interface, select the appropriate PDO mapping according to the communication data length.

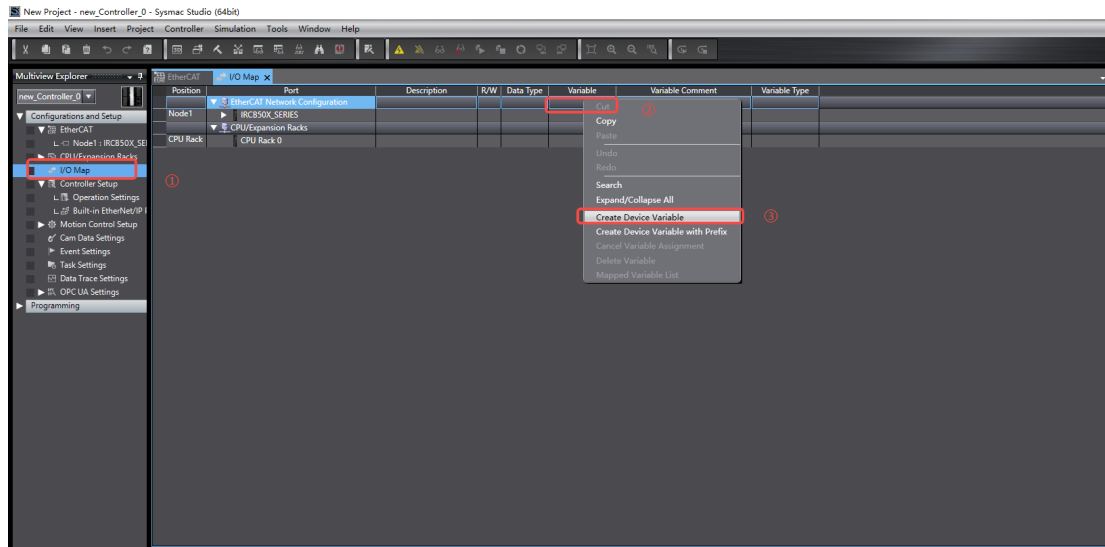


When editing the PDO mapping of group 3, the Omron PLC only supports the operations of deleting PDO entries, and does not support the operations of adding PDO entries or editing the data types of existing PDOs.

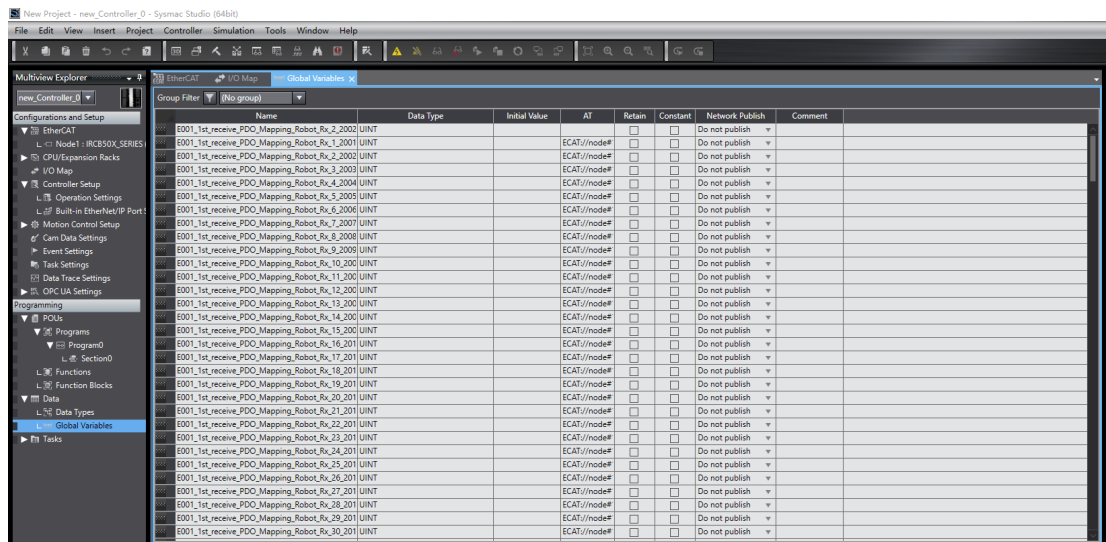
5.2.4 Variable Definitions and I/O Mapping Configuration

Global variable definition and mapping

1. On the main interface of Sysmac Studio software, double-click "I/O Map" on the left side to enter the "I/O Map" interface. Right-click at "Variable" and select "Create Device Variable" to automatically create global variables and complete I/O mapping.

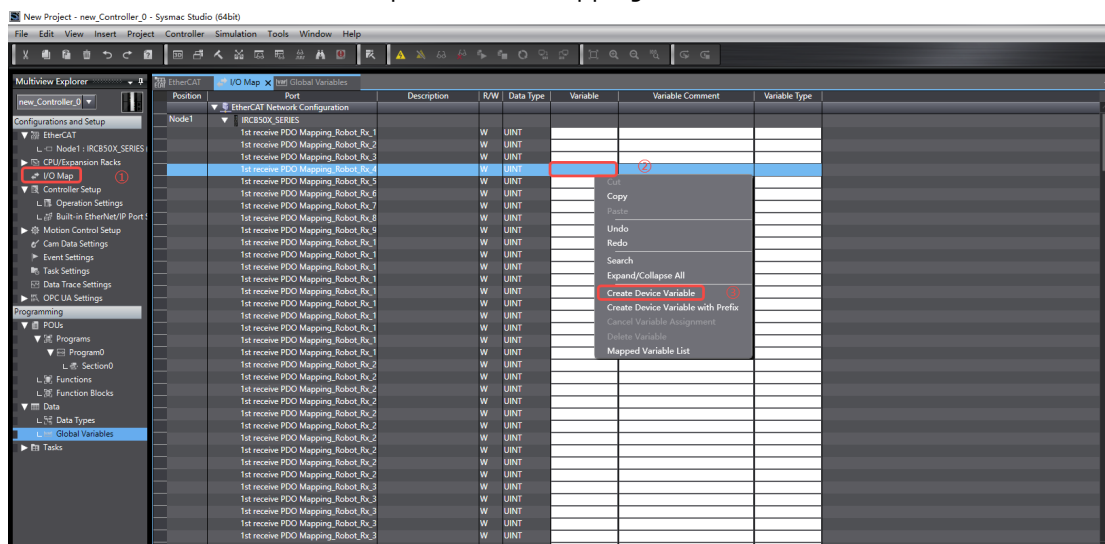


- On the main interface of Sysmac Studio software, double-click "Global Variables" on the left side to enter the "Global Variables" interface, where you can see the global variables automatically created by the system and associated with EtherCAT (you can also first define custom variables in the global variables page and then proceed with I/O mapping configuration).



Partial PDO variable definition and mapping

- On the main interface of Sysmac Studio software, double-click "I/O Map" on the left side to enter the "I/O Map" interface. Right-click in any "Variable" box and select "Create Device Variable" to create global variables for the current PDO and complete the I/O mapping.

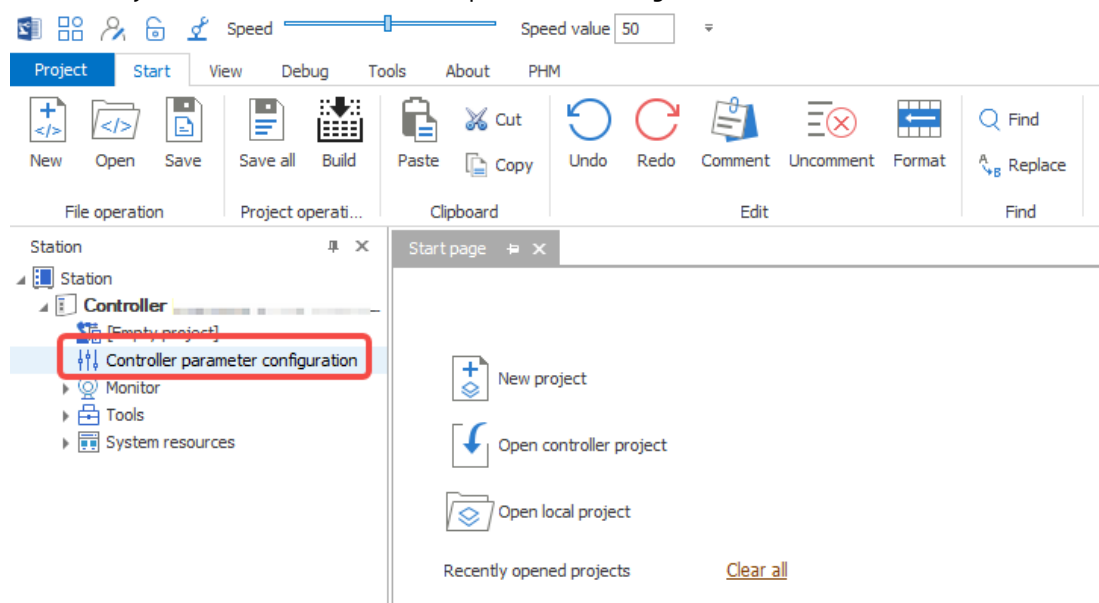


2. You can see the global variables automatically created by the system and associated with EtherCAT (you can also first define custom variables in the global variables page and then proceed with I/O mapping configuration).

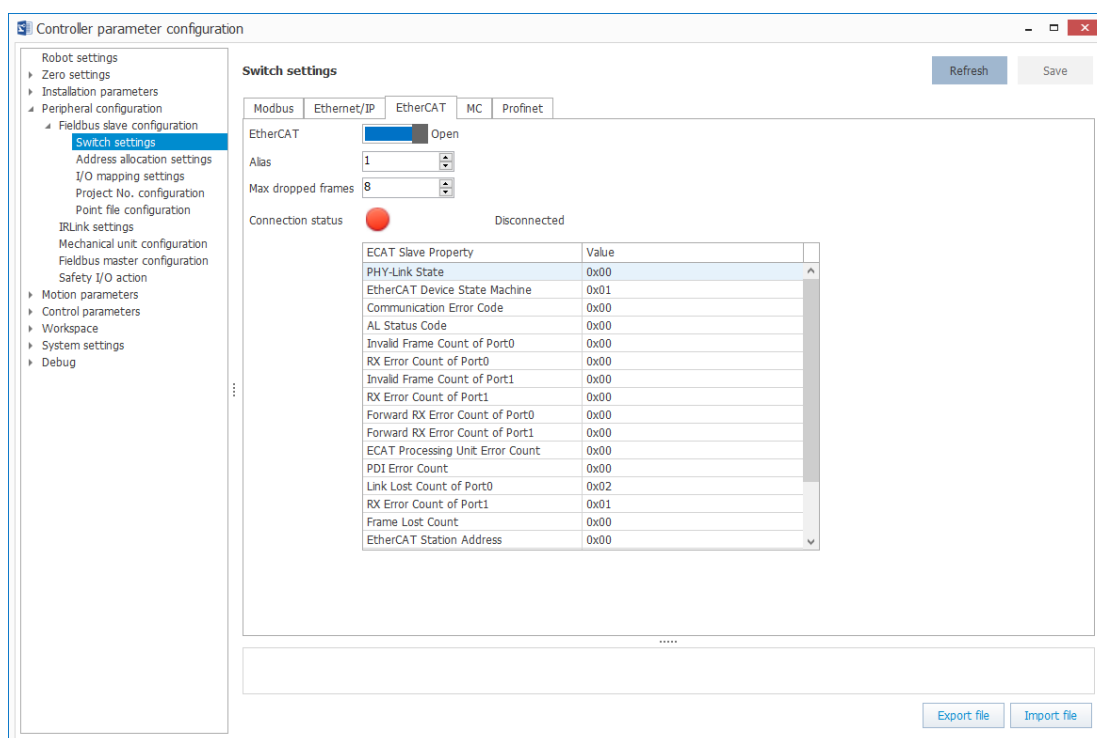
5.3 InoRobotLab Software-related Configuration

5.3.1 Fieldbus Activation and Configuration

1. On the main interface of the InoRobotLab software, double-click "Controller parameter configuration" in the left directory tree to enter the "Controller parameter configuration" interface.



2. On the "Controller Parameter Configuration" interface, click "Controller parameter configuration" > "Peripheral configuration" > "Fieldbus slave configuration" > "Switch settings" > "EtherCAT", toggle the switch to enable the EtherCAT slave, and set the station alias.
3. Click "Save" to enable the EtherCAT slave.

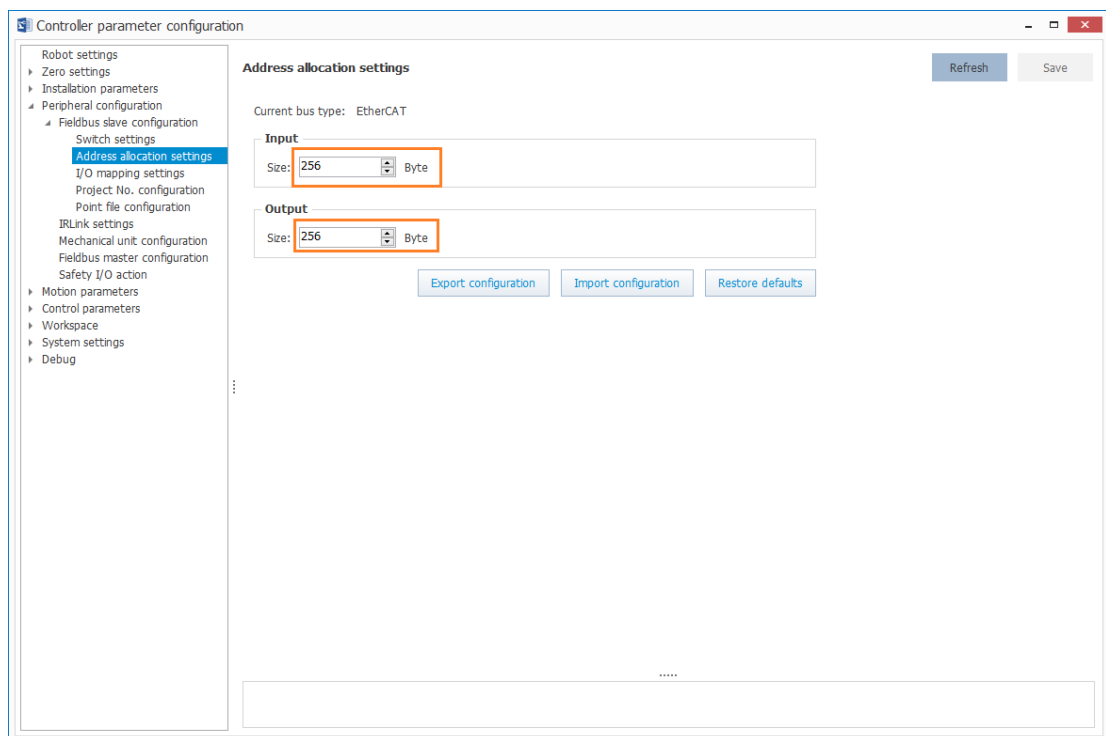




- Alias: Set the station alias (i.e., node address) to a value greater than 0.
- Max dropped frames: Keep the default value of 8 for the maximum lost frame count.

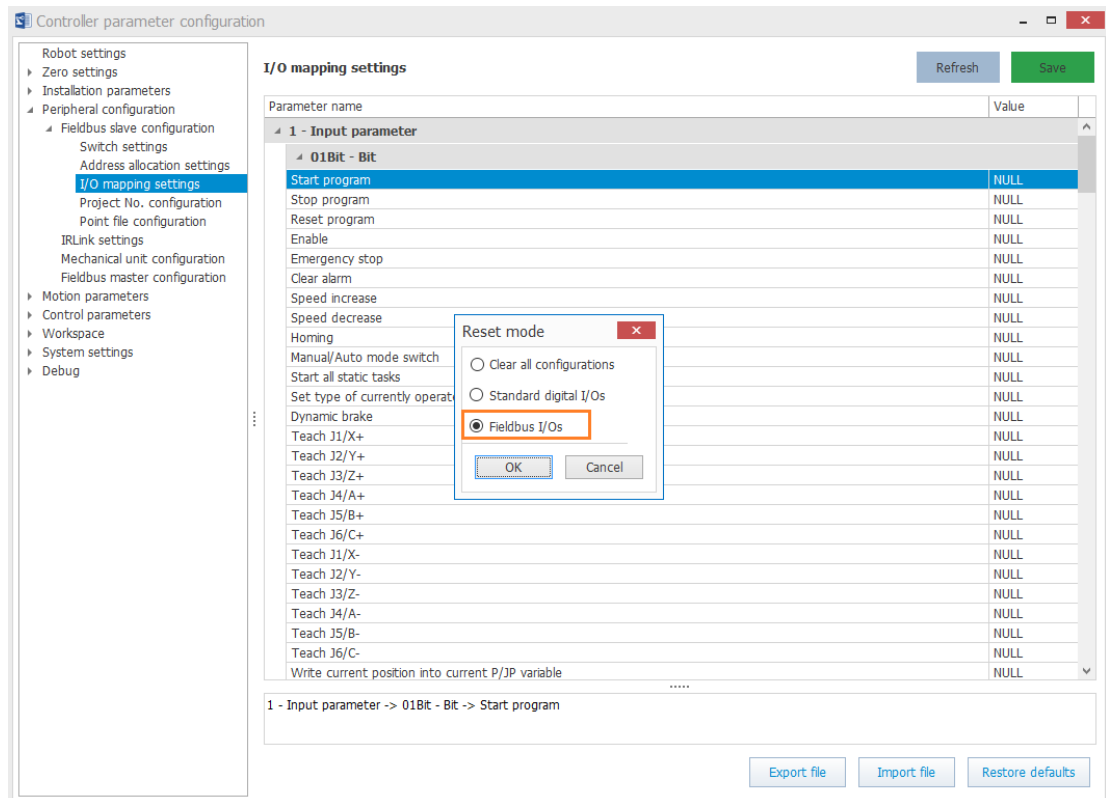
5.3.2 Bus Address Size Allocation Settings

1. On the main interface of the InoRobotLab software, double-click "Controller parameter configuration" in the left directory tree to enter the "Controller parameter configuration" interface.
2. On the "Controller parameter configuration" interface, click "Fieldbus slave configuration" > "Address allocation settings" to enter the "Address allocation settings" interface.
3. On the "Address allocation settings" interface, set the input/output size for the EtherCAT bus. The default configuration is 256 bytes, the maximum is 256 bytes, and the minimum is 2 bytes.



5.3.3 I/O Mapping Settings

1. On the main interface of the InoRobotLab software, double-click "Controller parameter configuration" in the left directory tree to enter the "Controller parameter configuration" interface.
2. On the "Controller parameter configuration" interface, click > "Fieldbus slave configuration" > "I/O mapping settings" to enter the "I/O mapping settings" interface.
3. On the "I/O mapping settings" interface, click "Restore defaults" in the lower right corner to pop up the "Reset mode" dialog box, then select the "Fieldbus I/Os".



The default fieldbus slave I/O configuration is as follows:

(1) Bus input address mapping table (basic remote control area)

Robot address	Function	Type
In512	Start program	Reset type
In513	Stop program	Reset type
In514	Program reset	Reset type
In515	Enable	Special type
In516	Emergency stop	Level type
In517	Clear alarm	Reset type
NULL	Speed increase	Reset type
NULL	Speed decrease	Reset type
In520	Homing	Reset type
In521	Switch to teaching mode	Reset type
In522	Switch to playback mode	Reset type
In523-In527	Reserved	-
In528	Teach J1/X+	Special type
In529	Teach J1/Y+	Special type

Robot address	Function	Type
In530	Teach J3/Z+	Special type
In531	Teach J4/A+	Special type
In532	Teach J5/B+	Special type
In533	Teach J6/C+	Special type
In534	Reserved	-
In535	Reserved	-
In536	Teach J1/X-	Special type
In537	Teach J2/Y-	Special type
In538	Teach J3/Z-	Special type
In539	Teach J4/A-	Special type
In540	Teach J5/B-	Special type
In541	Teach J6/C-	Special type
In542	Reserved	-
In543	Reserved	-
In544	Write the current position to the current P variable	Reset type
In545	Write the input point into the current P variable	Reset type
In546	Move directly to the position specified by the current P variable	Reset type
In547	OUT[0] control command	-
In548	OUT[1] control command	-
In549	OUT[2] control command	-
In550	OUT[3] control command	-
In551	OUT[4] control command	-
In552	OUT[5] control command	-
In553	OUT[6] control command	-
In554	OUT[7] control command	-

Robot address	Function	Type
In555	OUT[8] control command	-
In556	OUT[9] control command	-
In557	OUT[10] control command	-
In558	OUT[11] control command	-
In559	OUT[12] control command	-
In560	OUT[13] control command	-
In561	OUT[14] control command	-
In562	OUT[15] control command	-
In563-In719	Reserved area (can be used as a custom area)	-

(2) Bus output address mapping table (basic remote control area)

Robot address	Function
Out512	Program running state
Out513	Program stop state
Out514	Program reset successful state
Out515	Robot enable state
Out516	Robot emergency stop state
Out517	System fault state
Out518	System warning state
Out519	Servo fault state
Out520	Servo warning state
Out521	Safety door warning state
Out522	System startup completed state
Out523	Robot in motion state
Out524	Robot motion complete state
Out525	Bus communication heartbeat signal
Out526	P variable read successful
Out527	P variable write successful

Robot address	Function
Out528	IN[0] state
Out529	IN[1] state
Out530	IN[2] state
Out531	IN[3] state
Out532	IN[4] state
Out533	IN[5] state
Out534	IN[6] state
Out535	IN[7] state
Out536	IN[8] state
Out537	IN[9] state
Out538	IN[10] state
Out539	IN[11] state
Out540	IN[12] state
Out541	IN[13] state
Out542	IN[14] state
Out543	IN[15] state
Out544	OUT[0] state
Out545	OUT[1] state
Out546	OUT[2] state
Out547	OUT[3] state
Out548	OUT[4] state
Out549	OUT[5] state
Out550	OUT[6] state
Out551	OUT[7] state
Out552	OUT[8] state
Out553	OUT[9] state
Out554	OUT[10] state

Robot address	Function
Out555	OUT[11] state
Out556	OUT[12] state
Out557	OUT[13] state
Out558	OUT[14] state
Out559	OUT[15] state
Out560 to Out719	Reserved area (can be used as a custom area)

(3) Bus input address mapping table (extended remote control area)

No.	Function	Description
In720 to In735	Remote control command	-
In736 to In751	Remote control input parameter 1	-
In752 to In767	Remote control input parameter 2	-
In768 to In783	Remote control input parameter 3	-
...	...	...
In1680 to In1695	Remote control input parameter 60	-
...	Reserved	-
In1920 to In1935	Extended remote cycle command	-
In1936 to In1951	Extended remote cycle input parameter 1	-
In1952 to In1967	Extended remote cycle input parameter 2	-
...	Reserved	-
In2224 to In2239	Robot zero programming mode switch	-
In2240 to In2255	Reserved	-

No.	Function	Description
In2256 to In2271	Switch point file via I/O	• 0: Success • 1: Switching • 2: Point switching error
In2272 to In2287	Reserved	-
In2288 to In2303	Robot zero programming external permission settings	-
In2304 to In2319	Switch project via I/O	-
In2320 to In2335	Load number selection	-
In2336 to In2351	Frame selection	Added world frame types: • 1: Joint frame • 2: Base frame • 3: Flange frame • 4: Workobject frame • 5: World frame
In2352 to In2367	Tool frame No. selection	-
In2368 to In2383	Workobject frame No. selection	-
In2384 to In2399	Speed settings	-
In2400 to In2415	Direct motion mode settings • 0: MovJ • 1: MovL • 2: Jump • 3: JumpL • 4: MovABSJ	-
In2416 to In2431	Lower bits of the LH parameter of the jump motion	-
In2432 to In2447	Higher bits of the LH parameter of the jump motion	-
In2448 to In2463	Lower bits of the MH parameter of the jump motion	-
In2464 to In2497	Higher bits of the MH parameter of the jump motion	-

No.	Function	Description
In2480 to In2495	Lower bits of the RH parameter of the jump motion	-
In2496 to In2511	Higher bits of the RH parameter of the jump motion	-
In2512 to In2527	Write the index of the current P/JP variable	-
In2528 to In2543	Reserved	-
In2544 to In2559	Reserved	-

(4) Bus output address mapping table (extended remote control area)

No.	Function	Note
Out720 to Out735	Remote control state	-
Out736 to Out751	Remote control return parameter 1/Remote control error code	-
Out752 to Out767	Remote control return parameter 2/System fault code	-
Out768 to Out783	Remote control return parameter 3	-
...	...	...
Out1680 to Out1695	Remote control return parameter 60	-
...	Reserved	-
Out1920 to Out1935	Extended remote cycle state	-
Out1936 to Out1951	Extended remote cycle output parameter 1	-
Out1962 to Out1967	Extended remote cycle output parameter 2	-
Out1968 to Out1983	Robot zero programming mode status	-
Out1984 to Out1999	Robot zero programming external permission status	-
Out2000 to Out2015	X coordinate of J1 at current position - Low	Position of the active mechanical unit (robot by default)

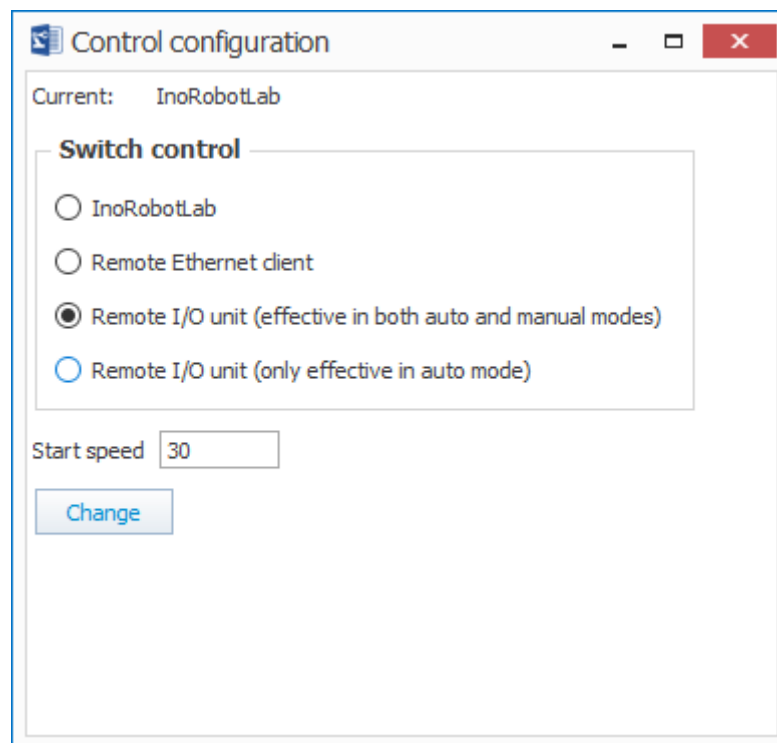
No.	Function	Note
Out2016 to Out2031	X coordinate of J1 at current position - High	
Out2032 to Out2047	Y coordinate of J2 at current position - Low	
Out2048 to Out2063	Y coordinate of J2 at current position - High	
Out2064 to Out2079	Z coordinate of J3 at current position - Low	
Out2080 to Out2095	Z coordinate of J3 at current position - High	
Out2096 to Out2111	A coordinate of J4 at current position - Low	
Out2112 to Out2127	A coordinate of J4 at current position - High	
Out2128 to Out2143	B coordinate of J5 at current position - Low	
Out2144 to Out2159	B coordinate of J5 at current position - High	
Out2160 to Out2175	C coordinate of J6 at current position - Low	
Out2176 to Out2191	C coordinate of J6 at current position - High	
...	...	...
Out2224 to Out2239	Collision alarm status	-
Out2240 to Out2255	Active P/JP variable index	-
Out2256 to Out2271	Position file switching status	-
Out2272 to Out2287	Current control	• 0: PC/TP • 2: Remote Ethernet client • 4: Remote I/O
Out2288 to Out2303	Active project No.	Range 1-256

No.	Function	Note
Out2304 to Out2319	Current project switchover state	• 0: Switchover successful • 1: Switchover in progress • 2: Project not obtained • 3: Invalid project name for remote I/O switchover • 4: The current project is running and needs to be stopped before switchover. • 5: Incoming path error (system error) • 6: Project compilation in progress, please wait • 7: Input value out of range (1-256)
Out2320 to Out2335	Current load number	-
Out2336 to Out2351	Current frame	-
Out2352 to Out2367	Current tool frame No.	-
Out2368 to Out2383	Current workobject frame No.	-
Out2384 to Out2399	Current speed	-
Out2400 to Out2415	Current fault code	-
Out2416 to Out2431	Current mode	-
Out2432 to Out2447	Current direct motion mode: • 0: MovJ • 1: MovL • 2: Jump • 3: JumpL	-
Out2448 to Out2463	Lower bits of the LH parameter of the current jump motion	-
Out2464 to Out2479	Higher bits of the LH parameter of the current jump motion	-
Out2480 to Out2495	Lower bits of the MH parameter of the current jump motion	-
Out2496 to Out2511	Higher bits of the MH parameter of the current jump motion	-

No.	Function	Note
Out2512 to Out2527	Lower bits of the RH parameter of the current jump motion	-
Out2528 to Out2543	Higher bits of the RH parameter of the current jump motion	-

5.3.4 Switching Control

For S03.21 version and later, the remote Modbus permission is no longer available and has been uniformly merged into the remote I/O unit. When using remote control, switch to the remote I/O unit control.



5.3.5 Fieldbus-to-Robot Bus Address Mapping

5.3.5.1 Fieldbus Input Address Mapping

- m is the start word address of the data transmitted by the EtherNet/IP master to the robot.
- p is the start word address of the data transmitted by the EtherCAT master to the robot, that is, the first object of RxPDO.

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
In512	4096 (coil address)	m (Bit0)	p (Bit0)
In513	4097 (coil address)	m (Bit1)	p (Bit1)
In514	4098 (coil address)	m (Bit2)	p (Bit2)
In515	4099 (coil address)	m (Bit3)	p (Bit3)

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
In516	4100 (coil address)	m (Bit4)	p (Bit4)
In517	4101 (coil address)	m (Bit5)	p (Bit5)
In518	4102 (coil address)	m (Bit6)	p (Bit6)
In519	4103 (coil address)	m (Bit7)	p (Bit7)
In520	4104 (coil address)	m (Bit8)	p (Bit8)
In521	4105 (coil address)	m (Bit9)	p (Bit9)
In522	4106 (coil address)	m (Bit10)	p (Bit10)
...	...	...	...
In527	4111 (coil address)	m (Bit15)	p (Bit15)
In528 to In543	4112-4127 (coil address)	m+1 (Bit0-Bit15)	p+1 (Bit0-Bit15)
In544 to In559	4128-4143 (coil address)	m+2 (Bit0-Bit15)	p+2 (Bit0-Bit15)
In560 to In575	4144-4159 (coil address)	m+3 (Bit0-Bit15)	p+3 (Bit0-Bit15)
In576 to In591	4160-4175 (coil address)	m+4 (Bit0-Bit15)	p+4 (Bit0-Bit15)
In592 to In607	4176-4191 (coil address)	m+5 (Bit0-Bit15)	p+5 (Bit0-Bit15)
In608 to In623	4192-4207 (coil address)	m+6 (Bit0-Bit15)	p+6 (Bit0-Bit15)
In624 to In639	4208-4223 (coil address)	m+7 (Bit0-Bit15)	p+7 (Bit0-Bit15)
In640 to In655	4224-4239 (coil address)	m+8 (Bit0-Bit15)	p+8 (Bit0-Bit15)
In656 to In671	4240-4255 (coil address)	m+9 (Bit0-Bit15)	p+9 (Bit0-Bit15)
In672 to In687	4256-4271 (coil address)	m+10 (Bit0-Bit15)	p+10 (Bit0-Bit15)
In688 to In703	4272-4287 (coil address)	m+11 (Bit0-Bit15)	p+11 (Bit0-Bit15)
In704 to In719	4288-4303 (coil address)	m+12 (Bit0-Bit15)	p+12 (Bit0-Bit15)
In720 to In735	34800 (register address)	m+13	p+13
In736 to In751	34801 (register address)	m+14	p+14
In752 to In767	34802 (register address)	m+15	p+15
In768 to In783	34803 (register address)	m+16	p+16
In784 to In799	34804 (register address)	m+17	p+17
In800 to In815	34805 (register address)	m+18	p+18

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
In816 to In831	34806 (register address)	m+19	p+19
In832 to In847	34807 (register address)	m+20	p+20
In848 to In863	34808 (register address)	m+21	p+21
In864 to In879	34809 (register address)	m+22	p+22
In880 to In895	34810 (register address)	m+23	p+23
In896 to In911	34811 (register address)	m+24	p+24
In912 to In927	34812 (register address)	m+25	p+25
In928 to In943	34813 (register address)	m+26	p+26
In944 to In959	34814 (register address)	m+27	p+27
In960 to In975	34815 (register address)	m+28	p+28
In976 to In991	34816 (register address)	m+29	p+29
In992 to In1007	34817 (register address)	m+30	p+30
In1008 to In1023	34818 (register address)	m+31	p+31
In1024 to In1039	34819 (register address)	m+32	p+32
In1040 to In1055	34820 (register address)	m+33	p+33
In1056 to In1071	34821 (register address)	m+34	p+34
In1072 to In1087	34822 (register address)	m+35	p+35
In1088 to In1103	34823 (register address)	m+36	p+36
In1104 to In1119	34824 (register address)	m+37	p+37
In1120 to In1135	34825 (register address)	m+38	p+38
In1136 to In1151	34826 (register address)	m+39	p+39
In1152 to In1167	34827 (register address)	m+40	p+40
In1168 to In1183	34828 (register address)	m+41	p+41
In1184 to In1199	34829 (register address)	m+42	p+42
In1200 to In1215	34830 (register address)	m+43	p+43
In1216 to In1231	34831 (register address)	m+44	p+44
In1232 to In1247	34832 (register address)	m+45	p+45

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
In1248 to In1263	34833 (register address)	m+46	p+46
In1264 to In1279	34834 (register address)	m+47	p+47
In1280 to In1295	34835 (register address)	m+48	p+48
In1296 to In1311	34836 (register address)	m+49	p+49
In1312 to In1327	34837 (register address)	m+50	p+50
In1328 to In1343	34838 (register address)	m+51	p+51
In1344 to In1359	34839 (register address)	m+52	p+52
In1360 to In1375	34840 (register address)	m+53	p+53
In1376 to In1391	34841 (register address)	m+54	p+54
In1392 to In1407	34842 (register address)	m+55	p+55
In1408 to In1423	34843 (register address)	m+56	p+56
In1424 to In1439	34844 (register address)	m+57	p+57
In1440 to In1455	34845 (register address)	m+58	p+58
In1456 to In1471	34846 (register address)	m+59	p+59
In1472 to In1487	34847 (register address)	m+60	p+60
In1488 to In1503	34848 (register address)	m+61	p+61
In1504 to In1519	34849 (register address)	m+62	p+62
In1520 to In1535	34850 (register address)	m+63	p+63
In1536 to In1551	34851 (register address)	m+64	p+64
In1552 to In1567	34852 (register address)	m+65	p+65
In1568 to In1583	34853 (register address)	m+66	p+66
In1584 to In1599	34854 (register address)	m+67	p+67
In1600 to In1615	34855 (register address)	m+68	p+68
In1616 to In1631	34856 (register address)	m+69	p+69
In1632-In1647	34857 (register address)	m+70	p+70
In1648 to In1663	34858 (register address)	m+71	p+71
In1664 to In1679	34859 (register address)	m+72	p+72

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
In1680 to In1695	34860 (register address)	m+73	p+73
In1696 to In1711	34861 (register address)	m+74	p+74
In1712 to In1727	34862 (register address)	m+75	p+75
In1728 to In1743	34863 (register address)	m+76	p+76
In1744 to In1759	34864 (register address)	m+77	p+77
In1760 to In1775	34865 (register address)	m+78	p+78
In1776 to In1791	34866 (register address)	m+79	p+79
In1792 to In1807	34867 (register address)	m+80	p+80
In1808 to In1823	34868 (register address)	m+81	p+81
In1824 to In1839	34869 (register address)	m+82	p+82
In1840 to In1855	34870 (register address)	m+83	p+83
In1856 to In1871	34871 (register address)	m+84	p+84
In1872 to In1887	34872 (register address)	m+85	p+85
In1888 to In1903	34873 (register address)	m+86	p+86
In1904 to In1919	34874 (register address)	m+87	p+87
In1920 to In1935	34875 (register address)	m+88	p+88
In1936 to In1951	34876 (register address)	m+89	p+89
In1952 to In1967	34877 (register address)	m+90	p+90
In1968 to In1983	34878 (register address)	m+91	p+91
In1984 to In1999	34879 (register address)	m+92	p+92
In2000 to In2015	34880 (register address)	m+93	p+93
In2016 to In2031	34881 (register address)	m+94	p+94
In2032 to In2047	34882 (register address)	m+95	p+95
In2048 to In2063	34883 (register address)	m+96	p+96
In2064 to In2079	34884 (register address)	m+97	p+97
In2080 to In2095	34885 (register address)	m+98	p+98
In2096 to In2111	34886 (register address)	m+99	p+99

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
In2112 to In2127	34887 (register address)	m+100	p+100
In2128 to In2143	34888 (register address)	m+101	p+101
In2144 to In2159	34889 (register address)	m+102	p+102
In2160 to In2175	34890 (register address)	m+103	p+103
In2176 to In2191	34891 (register address)	m+104	p+104
In2192 to In2207	34892 (register address)	m+105	p+105
In2208 to In2223	34893 (register address)	m+106	p+106
In2224 to In2239	34894 (register address)	m+107	p+107
In2240 to In2255	34895 (register address)	m+108	p+108
In2256 to In2271	34896 (register address)	m+109	p+109
In2272 to In2287	34897 (register address)	m+110	p+110
In2288 to In2303	34898 (register address)	m+111	p+111
In2304 to In2319	34899 (register address)	m+112	p+112
In2320 to In2335	34900 (register address)	m+113	p+113
In2336 to In2351	34901 (register address)	m+114	p+114
In2352 to In2367	34902 (register address)	m+115	p+115
In2368 to In2383	34903 (register address)	m+116	p+116
In2384 to In2399	34904 (register address)	m+117	p+117
In2400 to In2415	34905 (register address)	m+118	p+118
In2416 to In2431	34906 (register address)	m+119	p+119
In2432 to In2447	34907 (register address)	m+120	p+120
In2448 to In2463	34908 (register address)	m+121	p+121
In2464 to In2479	34909 (register address)	m+122	p+122
In2480 to In2495	34910 (register address)	m+123	p+123
In2496 to In2511	34911 (register address)	m+124	p+124
In2512 to In2527	34912 (register address)	m+125	p+125
In2528 to In2543	34913 (register address)	m+126	p+126

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
In2544 to In2559	34914 (register address)	m+127	p+127

5.3.5.2 Fieldbus Output Address Mapping

- n is the start word address of the data transmitted by the robot to the EtherNet/IP master.
- q is the start word address of the data transmitted by the robot to the EtherCAT master, that is, the first object of TxPDO.

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
Out512	0 (coil address)	n (Bit0)	q (Bit0)
Out513	1 (coil address)	n (Bit1)	q (Bit1)
Out514	2 (coil address)	n (Bit2)	q (Bit2)
Out515	3 (coil address)	n (Bit3)	q (Bit3)
Out516	4 (coil address)	n (Bit4)	q (Bit4)
Out517	5 (coil address)	n (Bit5)	q (Bit5)
Out518	6 (coil address)	n (Bit6)	q (Bit6)
Out519	7 (coil address)	n (Bit7)	q (Bit7)
Out520	8 (coil address)	n (Bit8)	q (Bit8)
Out521	9 (coil address)	n (Bit9)	q (Bit9)
Out522	10 (coil address)	n (Bit10)	q (Bit10)
...	...	...	...
Out527	15 (coil address)	n (Bit15)	q (Bit15)
Out528 to Out543	16-31 (coil address)	n+1 (Bit0-Bit15)	q+1 (Bit0-Bit15)
Out544 to Out559	32-47 (coil address)	n+2 (Bit0-Bit15)	q+2 (Bit0-Bit15)
Out560 to Out575	48-63 (coil address)	n+3 (Bit0-Bit15)	q+3 (Bit0-Bit15)
Out576 to Out591	64-79 (coil address)	n+4 (Bit0-Bit15)	q+4 (Bit0-Bit15)
Out592 to Out607	80-95 (coil address)	n+5 (Bit0-Bit15)	q+5 (Bit0-Bit15)
Out608 to Out623	96-111 (coil address)	n+6 (Bit0-Bit15)	q+6 (Bit0-Bit15)
Out624 to Out639	112-127 (coil address)	n+7 (Bit0-Bit15)	q+7 (Bit0-Bit15)
Out640 to Out655	128-143 (coil address)	n+8 (Bit0-Bit15)	q+8 (Bit0-Bit15)

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
Out656 to Out671	144-159 (coil address)	n+9 (Bit0-Bit15)	q+9 (Bit0-Bit15)
Out672 to Out687	160-175 (coil address)	n+10 (Bit0-Bit15)	q+10 (Bit0-Bit15)
Out688 to Out703	176-191 (coil address)	n+11 (Bit0-Bit15)	q+11 (Bit0-Bit15)
Out704 to Out719	192-207 (coil address)	n+12 (Bit0-Bit15)	q+12 (Bit0-Bit15)
Out720 to Out735	2048 (register address)	n+13	q+13
Out736 to Out751	2049 (register address)	n+14	q+14
Out752 to Out767	2050 (register address)	n+15	q+15
Out768 to Out783	2051 (register address)	n+16	q+16
Out784 to Out799	2052 (register address)	n+17	q+17
Out800 to Out815	2053 (register address)	n+18	q+18
Out816 to Out831	2054 (register address)	n+19	q+19
Out832 to Out847	2055 (register address)	n+20	q+20
Out848 to Out863	2056 (register address)	n+21	q+21
Out864 to Out879	2057 (register address)	n+22	q+22
Out880 to Out895	2058 (register address)	n+23	q+23
Out896 to Out911	2059 (register address)	n+24	q+24
Out912 to Out927	2060 (register address)	n+25	q+25
Out928 to Out943	2061 (register address)	n+26	q+26
Out944 to Out959	2062 (register address)	n+27	q+27
Out960 to Out975	2063 (register address)	n+28	q+28
Out976 to Out991	2064 (register address)	n+29	q+29
Out992 to Out1007	2065 (register address)	n+30	q+30
Out1008 to Out1023	2066 (register address)	n+31	q+31
Out1024 to Out1039	2067 (register address)	n+32	q+32
Out1040 to Out1055	2068 (register address)	n+33	q+33
Out1056 to Out1071	2069 (register address)	n+34	q+34
Out1072 to Out1087	2070 (register address)	n+35	q+35

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
Out1088 to Out1103	2071 (register address)	n+36	q+36
Out1104 to Out1119	2072 (register address)	n+37	q+37
Out1120 to Out1135	2073 (register address)	n+38	q+38
Out1136 to Out1151	2074 (register address)	n+39	q+39
Out1152 to Out1167	2075 (register address)	n+40	q+40
Out1168 to Out1183	2076 (register address)	n+41	q+41
Out1184 to Out1199	2077 (register address)	n+42	q+42
Out1200 to Out1215	2078 (register address)	n+43	q+43
Out1216 to Out1231	2079 (register address)	n+44	q+44
Out1232 to Out1247	2080 (register address)	n+45	q+45
Out1248 to Out1263	2081 (register address)	n+46	q+46
Out1264 to Out1279	2082 (register address)	n+47	q+47
Out1280 to Out1295	2083 (register address)	n+48	q+48
Out1296 to Out1311	2084 (register address)	n+49	q+49
Out1312 to Out1327	2085 (register address)	n+50	q+50
Out1328 to Out1343	2086 (register address)	n+51	q+51
Out1344 to Out1359	2087 (register address)	n+52	q+52
Out1360 to Out1375	2088 (register address)	n+53	q+53
Out1376 to Out1391	2089 (register address)	n+54	q+54
Out1392 to Out1407	2090 (register address)	n+55	q+55
Out1408 to Out1423	2091 (register address)	n+56	q+56
Out1424 to Out1439	2092 (register address)	n+57	q+57
Out1440 to Out1455	2093 (register address)	n+58	q+58
Out1456 to Out1471	2094 (register address)	n+59	q+59
Out1472 to Out1487	2095 (register address)	n+60	q+60
Out1488 to Out1503	2096 (register address)	n+61	q+61
Out1504 to Out1519	2097 (register address)	n+62	q+62

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
Out1520 to Out1535	2098 (register address)	n+63	q+63
Out1536 to Out1551	2099 (register address)	n+64	q+64
Out1552 to Out1567	2100 (register address)	n+65	q+65
Out1568 to Out1583	2101 (register address)	n+66	q+66
Out1584 to Out1599	2102 (register address)	n+67	q+67
Out1600 to Out1615	2103 (register address)	n+68	q+68
Out1616 to Out1631	2104 (register address)	n+69	q+69
Out1632 to Out1647	2105 (register address)	n+70	q+70
Out1648 to Out1663	2106 (register address)	n+71	q+71
Out1664 to Out1679	2107 (register address)	n+72	q+72
Out1680 to Out1695	2108 (register address)	n+73	q+73
Out1696 to Out1711	2109 (register address)	n+74	q+74
Out1712 to Out1727	2110 (register address)	n+75	q+75
Out1728 to Out1743	2111 (register address)	n+76	q+76
Out1744 to Out1759	2112 (register address)	n+77	q+77
Out1760 to Out1775	2113 (register address)	n+78	q+78
Out1776 to Out1791	2114 (register address)	n+79	q+79
Out1792 to Out1807	2115 (register address)	n+80	q+80
Out1808 to Out1823	2116 (register address)	n+81	q+81
Out1824 to Out1839	2117 (register address)	n+82	q+82
Out1840 to Out1855	2118 (register address)	n+83	q+83
Out1856 to Out1871	2119 (register address)	n+84	q+84
Out1872 to Out1887	2120 (register address)	n+85	q+85
Out1888 to Out1903	2121 (register address)	n+86	q+86
Out1904 to Out1919	2122 (register address)	n+87	q+87
Out1920 to Out1935	2123 (register address)	n+88	q+88
Out1936 to Out1951	2124 (register address)	n+89	q+89

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
Out1952 to Out1967	2125 (register address)	n+90	q+90
Out1968 to Out1983	2126 (register address)	n+91	q+91
Out1984 to Out1999	2127 (register address)	n+92	q+92
Out2000 to Out2015	2128 (register address)	n+93	q+93
Out2016 to Out2031	2129 (register address)	n+94	q+94
Out2032 to Out2047	2130 (register address)	n+95	q+95
Out2048 to Out2063	2131 (register address)	n+96	q+96
Out2064 to Out2079	2132 (register address)	n+97	q+97
Out2080 to Out2095	2133 (register address)	n+98	q+98
Out2096 to Out2111	2134 (register address)	n+99	q+99
Out2112 to Out2127	2135 (register address)	n+100	q+100
Out2128 to Out2143	2136 (register address)	n+101	q+101
Out2144 to Out2159	2137 (register address)	n+102	q+102
Out2160 to Out2175	2138 (register address)	n+103	q+103
Out2176 to Out2191	2139 (register address)	n+104	q+104
Out2192 to Out2207	2140 (register address)	n+105	q+105
Out2208 to Out2223	2141 (register address)	n+106	q+106
Out2224 to Out2239	2142 (register address)	n+107	q+107
Out2240 to Out2255	2143 (register address)	n+108	q+108
Out2256 to Out2271	2144 (register address)	n+109	q+109
Out2272 to Out2287	2145 (register address)	n+110	q+110
Out2288 to Out2303	2146 (register address)	n+111	q+111
Out2304 to Out2319	2147 (register address)	n+112	q+112
Out2320 to Out2335	2148 (register address)	n+113	q+113
Out2336 to Out2351	2149 (register address)	n+114	q+114
Out2352 to Out2367	2150 (register address)	n+115	q+115
Out2368 to Out2383	2151 (register address)	n+116	q+116

Robot address	Modbus address	EtherNet/IP address	EtherCAT address
Out2384 to Out2399	2152 (register address)	n+117	q+117
Out2400 to Out2415	2153 (register address)	n+118	q+118
Out2416 to Out2431	2154 (register address)	n+119	q+119
Out2432 to Out2447	2155 (register address)	n+120	q+120
Out2448 to Out2463	2156 (register address)	n+121	q+121
Out2464 to Out2479	2157 (register address)	n+122	q+122
Out2480 to Out2495	2158 (register address)	n+123	q+123
Out2496 to Out2511	2159 (register address)	n+124	q+124
Out2512 to Out2527	2160 (register address)	n+125	q+125
Out2528 to Out2543	2161 (register address)	n+126	q+126
Out2544 to Out2559	2162 (register address)	n+127	q+127

Service and Support

Should you encounter a safety accident during the use or operation of the product, or face challenges in operating and maintaining the equipment, which remain unresolved after the relevant documentation is consulted, we provide multiple channels to ensure prompt resolution:

- Channel #1: Contact service@inovance.com.
- Channel #2: Visit <https://www.inovance.com/global> to access document downloads, after-sales support, spare parts ordering, repair applications, and authenticity verification services.
- Channel #3: Download My Inovance app (<https://zshc-eu.inovance.com/download-pc/>) where you can access products info and documentation, and query product parameters.

We are committed to providing you with quick and professional technical support, and we look forward to your satisfaction and trust.



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